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UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

Form 10-K

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the Fiscal Year Ended December 31, 2023

or

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**
For the Transition Period from to

Commission File No. 001-32260

Westlake Corporation
(Exact name of registrant as specified in its charter)

Delaware	**76-0346924**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**

2801 Post Oak Boulevard, Suite 600
Houston, Texas 77056
(Address of principal executive offices, including zip code)
(713) 960-9111
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common Stock, $0.01 par value	WLK	The New York Stock Exchange
1.625% Senior Notes due 2029	WLK29	The New York Stock Exchange

Securities registered pursuant to Section 12(g) of the Act: None

Indicate by check mark if the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or Section 15(d) of the Exchange Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
		Emerging growth company	☐

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Act). Yes ☐ No ☒

The aggregate market value of the registrant's voting stock held by non-affiliates of the registrant on June 30, 2023, the end of the registrant's most recently completed second fiscal quarter, based on a closing price on June 30, 2023 of $119.47 on the New York Stock Exchange was approximately $3.9 billion.

There were 128,216,207 shares of the registrant's common stock outstanding as of February 14, 2024.

DOCUMENTS INCORPORATED BY REFERENCE:

Certain information required by Part II and Part III of this Form 10-K is incorporated by reference from the registrant's definitive Proxy Statement to be filed pursuant to Regulation 14A with respect to the registrant's 2024 Annual Meeting of Stockholders to be held on May 9, 2024.

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Explanatory Note

References in this Annual Report on Form 10-K (this "report") to "we," "our," "us" or like terms refer to Westlake Corporation ("Westlake" or the "Company").

Cautionary Statements about Forward-Looking Statements

The Private Securities Litigation Reform Act of 1995 provides safe harbor provisions for forward-looking information. Certain of the statements contained in this Form 10-K are forward-looking statements. All statements, other than statements of historical facts, included in this Form 10-K that address activities, events or developments that we expect, project, believe or anticipate will or may occur in the future are forward-looking statements. Forward-looking statements can be identified by the use of words such as "believes," "intends," "may," "should," "could," "anticipates," "expected" or comparable terminology, or by discussions of strategies or trends. Although we believe that the expectations reflected in such forward-looking statements are reasonable, we cannot give any assurances that these expectations will prove to be correct. Forward-looking statements relate to matters such as:

- the ultimate timing, outcome and results of integrating the operations of acquisitions and the ultimate outcome of our operating efficiencies applied to the products and services; the effects of the acquisition, including the combined company's future financial condition, results of operations, strategy and plans; and expected synergies and other benefits from the acquisition and our ability to realize such synergies and other benefits;
- future operating rates, margins, cash flows and demand for our products;
- industry market outlook, including the price of crude oil, natural gas, ethane, housing starts and repair and remodeling activity;
- macroeconomic outlook, including rising interest rates, inflation and possible recession;
- widespread outbreak of an illness or any other communicable disease, or any other public health crisis;
- production capacities;
- the impact of uncertainties and ongoing supply chain constraints caused by the Panama Canal reduced transit capacity and conflicts in the Middle East and between Russia and Ukraine;
- currency devaluation;
- our ability to borrow under our credit agreement;
- our ability to meet our liquidity needs;
- our ability to meet debt obligations under our debt instruments;
- our intended quarterly dividends;
- future capacity additions and expansions in the industries in which we compete;
- results of acquisitions;
- timing, funding and results of capital projects;
- pension plan obligations, funding requirements and investment policies;
- compliance with present and future environmental regulations and costs associated with environmentally related penalties, capital expenditures, remedial actions and proceedings, including any new laws, regulations or treaties that may come into force to limit or control carbon dioxide and other greenhouse gas emissions or to address other issues of climate change;
- recovery of losses under our insurance policies;
- effects of pending legal proceedings and settlements; and
- timing of and amount of capital expenditures.

We have based these statements on assumptions and analyses in light of our experience and perception of historical trends, current conditions, expected future developments and other factors we believe were appropriate in the circumstances when the statements were made. Forward-looking statements by their nature involve substantial risks and uncertainties that could significantly impact expected results, and actual future results could differ materially from those described in such statements. While it is not possible to identify all factors, we continue to face many risks and uncertainties. Among the factors that could cause actual future results to differ materially are the risks and uncertainties discussed under "Risk Factors" and those described from time to time in our other filings with the SEC including, but not limited to, the following:

- general economic and business conditions, including inflation, interest rates and possible recession;
- the cyclical nature of the chemical and building products industries;
- the availability, cost and volatility of raw materials and energy;
- uncertainties associated with the United States, European and worldwide economies, including those due to political tensions and conflict in the Middle East, Russia and Ukraine and elsewhere;
- uncertainties associated with pandemic infectious diseases;
- uncertainties associated with climate change;
- the potential impact on demand for ethylene, polyethylene and polyvinyl chloride due to initiatives such as recycling and customers seeking alternatives to polymers;
- current and potential governmental regulatory actions in the United States and other countries;
- industry production capacity and operating rates;
- the supply/demand balance for our products;
- competitive products and pricing pressures;
- instability in the credit and financial markets;
- access to capital markets;
- terrorist acts;
- operating interruptions (including leaks, explosions, fires, weather-related incidents, mechanical failure, unscheduled downtime, labor difficulties, transportation interruptions, spills and releases and other environmental risks);
- changes in laws or regulations, including trade policies;
- technological developments;
- information systems failures and cyberattacks;
- foreign currency exchange risks;
- our ability to implement our business strategies; and
- creditworthiness of our customers.

Many of such factors are beyond our ability to control or predict. Any of the factors, or a combination of these factors, could materially affect our future results of operations and the ultimate accuracy of the forward-looking statements. These forward-looking statements are not guarantees of our future performance, and our actual results and future developments may differ materially from those projected in the forward-looking statements. Management cautions against putting undue reliance on forward-looking statements or projecting any future results based on such statements or present or prior earnings levels. Every forward-looking statement speaks only as of the date of the particular statement, and we undertake no obligation to publicly update or revise any forward-looking statements.

Industry and Market Data

Industry and market data used throughout this Form 10-K were obtained through internal company research, surveys and studies conducted by unrelated third parties and publicly available industry and general publications. We have not independently verified market and industry data from external sources. While we believe internal company estimates are reliable and market definitions are appropriate, neither such estimates nor these definitions have been verified by any independent sources.

Production Capacity

Unless we state otherwise, annual production capacity estimates used throughout this Form 10-K represent rated capacity of the facilities at December 31, 2023. We calculated rated capacity by estimating the number of days in a typical year that a production unit of a plant is expected to operate, after allowing for downtime for regular maintenance, and multiplying that number by an amount equal to the unit's optimal daily output based on the design feedstock mix. Because the rated capacity of a production unit is an estimated amount, actual production volumes may be more or less than the rated capacity.

PART I

Item 1. Business

General

We are a vertically integrated global manufacturer and marketer of performance and essential materials and housing and infrastructure products that enhance the lives of people every day. Our products include some of the most widely used materials in the world, which are fundamental to many diverse consumer and industrial markets, including residential construction, flexible and rigid packaging, automotive products, healthcare products, water treatment, wind turbines, coatings as well as other durable and non-durable goods. We operate in two principal operating segments, Performance and Essential Materials and Housing and Infrastructure Products. Performance and Essential Materials includes Westlake North American Vinyls, Westlake North American Chlor-alkali & Derivatives, Westlake European & Asian Chlorovinyls, Westlake Olefins, Westlake Polyethylene and Westlake Epoxy. Housing and Infrastructure Products includes Westlake Royal Building Products, Westlake Pipe & Fittings, Westlake Global Compounds and Westlake Dimex. We are highly integrated along our materials chain with significant downstream integration from ethylene and chlor-alkali (chlorine and caustic soda) into vinyls, polyethylene and styrene monomer. We also have substantial downstream integration from polyvinyl chloride ("PVC") into our building products, PVC pipes and fittings and PVC compounds in our Housing and Infrastructure Products segment.

We began operations in 1986. Since 1986, we have grown rapidly into an integrated global producer of chemicals and building products. We achieved this growth by acquiring existing plants or constructing new plants and completing numerous capacity or production line expansions. We regularly consider acquisitions and other internal and external growth opportunities that would be consistent with, or complementary to, our overall business strategy.

In 2014, we formed Westlake Chemical Partners LP ("Westlake Partners") to operate, acquire and develop ethylene production facilities and related assets. Also in 2014, Westlake Partners completed an initial public offering of common units (the "Westlake Partners IPO"). As of February 14, 2024, Westlake Partners' assets consisted of a 22.8% limited partner interest in Westlake Chemical OpCo LP ("OpCo"), as well as the general partner interest in OpCo. Prior to the Westlake Partners IPO, OpCo's assets were wholly-owned by us. OpCo's assets include two ethylene production facilities at our olefins facility in Lake Charles, one ethylene production facility at our Calvert City site and a 200-mile common carrier ethylene pipeline that runs from Mont Belvieu, Texas to the Longview, Texas site, which includes our Longview polyethylene production facility. We retain a 77.2% limited partner interest in OpCo, a 40.1% limited partner interest in Westlake Partners (consisting of 14,122,230 common units), a general partner interest in Westlake Partners and incentive distribution rights. The operations of Westlake Partners are consolidated in our financial statements. We are party to certain agreements with Westlake Partners and OpCo whereby, among other things, OpCo sells us 95% of the ethylene it produces on a cost-plus basis that is expected to generate a fixed margin per pound of $0.10. We use this ethylene in the production processes of our Performance and Essential Materials segment. For more information, see "—Performance and Essential Materials Business" below.

On November 12, 2019, we completed the acquisition of an additional 34.8% of the membership interests in LACC, LLC ("LACC") from Lotte Chemical USA Corporation, a subsidiary of Lotte Chemical Corporation ("Lotte"), for approximately $817 million (the "Transaction"). Prior to the Transaction, we owned approximately 12% of the membership interests in LACC. On March 15, 2022, the Company completed the acquisition of an additional 3.2% membership interest in LACC from Lotte for approximately $89 million. As of December 31, 2023, we owned an aggregate 50% membership interest in LACC. The LACC ethylene plant has 2.2 billion pounds per year of ethylene production capacity and is adjacent to our chlor-alkali facility in Lake Charles. We receive our proportionate share of LACC's ethylene production on a cash-cost basis, which is expected to benefit our integrated downstream operations.

On June 20, 2021, we, through one of our wholly-owned subsidiaries, entered into an Equity Purchase Agreement (the "Boral Purchase Agreement") by and among Boral Building Products Inc., a Michigan corporation, Boral Stone Products LLC, a Delaware limited liability company, Boral Lifetile Inc., a California corporation, Boral Windows LLC, a Utah limited liability company, Boral Industries Inc., a California corporation ("Boral Industries"), and, solely for the limited purposes set forth therein, we and Boral Limited, an Australian corporation ("Boral"). Pursuant to the terms of the Boral Purchase Agreement, we agreed to acquire from Boral Industries all of the issued and outstanding equity interests of certain subsidiaries of Boral Industries engaged in Boral's North American building products businesses in roofing, siding, trim and shutters, decorative stone and windows (the "Boral Target Companies"). On October 1, 2021, we completed the acquisition of the Boral Target Companies. The total consideration was $2,140 million in an all-cash transaction. The assets acquired and liabilities assumed and the results of operations of this business are included in the Housing and Infrastructure Products segment.

On August 19, 2021, we completed the acquisition of, and acquired all of the equity interests in LASCO Fittings, Inc. ("LASCO"), a manufacturer of injected-molded PVC fittings that serve the plumbing, pool and spa, industrial, irrigation and retail markets in the United States from Aalberts U.S. Holding Corp. and Aalberts N.V. (the "LASCO Acquisition"). The total consideration was $277 million. The assets acquired and liabilities assumed and the results of operations of LASCO are included in the Housing and Infrastructure Products segment.

On September 10, 2021, we completed the acquisition of, and acquired all of the equity interests in, DX Acquisition Corp. ("Dimex"), a producer of various consumer products made from post-industrial-recycled polyvinyl chloride, polyethylene and thermoplastic elastomer materials, including, landscape edging; industrial, home and office matting; marine dock edging; and masonry joint controls The total consideration was $172 million. The assets acquired and liabilities assumed and the results of operations of Dimex are included in the Housing and Infrastructure Products segment.

On February 1, 2022, we completed the acquisition of the global epoxy business of Hexion Inc. ("Westlake Epoxy") for a total consideration of $1,207 million. The assets acquired and liabilities assumed and the results of operations of Westlake Epoxy are included in the Performance and Essential Materials segment. During the fourth quarter of 2023, the Westlake Epoxy business's sales volumes and prices, specifically base epoxy resins in Europe, continued to deteriorate. These lower sales volumes and prices were primarily driven by record exports at lower prices of bisphenol-A, epichlorohydrin and base epoxy resins (constituting the epoxy value chain) out of Asia into Europe and North America during the time when demand in the European market was contracting. In addition, Westlake Epoxy operations in Europe have experienced sustained high energy and power costs. These factors negatively impacted Westlake Epoxy financial results during 2023. Based on these developments, along with management's outlook for the Westlake Epoxy business over the foreseeable future, we determined, in the fourth quarter of 2023, that the carrying amount of long-lived assets of our base epoxy resin business in the Netherlands and all of the goodwill of the Westlake Epoxy business will not be recoverable. As a result of this assessment, a goodwill impairment charge of $128 million and a non-cash long-lived asset impairment charge related to Epoxy Netherlands base epoxy resin business assets of $347 million were recognized in the fourth quarter of 2023.

As a global manufacturer of products in the performance and essential materials and housing and infrastructure products businesses, we continue to build on our core strengths in delivering high-value, essential products for our customers and endeavor to produce and deliver these products in increasingly-sustainable ways. To further these objectives, we endeavor to reduce the environmental footprint of our operations and enhance the circularity in more of our products, including continuing to focus on recycling opportunities within our businesses, reducing waste at our facilities, incorporating more recycled content into our products, and seeking to incorporate renewable and bio-based materials. We established a target 20% reduction in our Scope 1 and Scope 2 CO_2 equivalent emissions intensity per ton of production by 2030 from a 2016 baseline. As of December 31, 2022, we had achieved a reduction of approximately 18% in such emissions intensity from our 2016 baseline.

Performance and Essential Materials Business

Products

Principal products in our integrated Performance and Essential Materials segment include ethylene, polyethylene, styrene, chlor-alkali (chlorine and caustic soda), chlorinated derivative products, ethylene dichloride ("EDC"), vinyl chloride monomer ("VCM") and PVC. We manage our integrated vinyls production chain to optimize product margins and capacity utilization.

We manufacture ethylene through three of the OpCo plants and our portion of LACC's production capacity located in Lake Charles and Calvert City. Chlor-alkali materials are produced at our three plants located in Lake Charles, two plants located in Germany and one plant each located in Calvert City, Plaquemine, Geismar, Natrium, Longview and Beauharnois. Our VCM is produced at our two plants in Lake Charles, two plants located in Germany and one plant each at Calvert City, Plaquemine and Geismar. Our PVC is produced at our four plants located in Germany and one plant each at Calvert City, Plaquemine, Geismar and Aberdeen. Polyethylene and associated products are produced at our two polyethylene plants in Lake Charles and three polyethylene plants and a specialty polyethylene wax plant at our Longview site. Our chlorinated derivative products are produced at our plants in Lake Charles and Natrium. Styrene monomer is produced at our plant located in our Lake Charles facility. Epoxy Specialty Resins are produced at two plants located in Germany, two plants in the United States, one plant in Spain and one plant in South Korea. Base Epoxy Resins and Intermediaries are produced at our plants in Pernis, the Netherlands and Deer Park, United States. Our other Asian manufacturing facilities are located near Shanghai, in China, and in Kaohsiung, Taiwan, through our 95%- and 60%-owned joint ventures, respectively, where we produce chlor-alkali, PVC and associated products. As of February 14, 2024, we (directly and through OpCo, our investment in LACC, and our 95%- and 60%-owned joint ventures in China and Taiwan, respectively) had approximately 43.3 billion pounds per year of aggregate production capacity at numerous manufacturing sites in North America, Europe and Asia in our Performance and Essential Materials segment.

The following table illustrates our Performance and Essential Materials segment production capacities at February 14, 2024 by principal product and the end uses of these products:

Product [1]	Annual Capacity [2]	End Uses	Principal Manufacturing Facilities [4] [5] [6]
	(Millions of pounds)		
Ethylene [3]	4,820	VCM, polyethylene, EDC, styrene, ethylene oxide/ethylene glycol	Calvert City, Kentucky Lake Charles, Louisiana
Chlorine	7,400	VCM, EDC, organic/inorganic chemicals, bleach and water treatment	Calvert City, Kentucky Geismar, Louisiana Lake Charles, Louisiana Plaquemine, Louisiana Natrium, West Virginia Gendorf and Knapsack, Germany
Caustic Soda	8,140	Pulp and paper, organic/inorganic chemicals, neutralization and alumina	Calvert City, Kentucky Geismar, Louisiana Lake Charles, Louisiana Plaquemine, Louisiana Natrium, West Virginia Gendorf and Knapsack, Germany
VCM	7,940	PVC, PVC Compounds	Calvert City, Kentucky Geismar, Louisiana Lake Charles, Louisiana Plaquemine, Louisiana Gendorf and Knapsack, Germany
Specialty PVC	980	Automotive sealants, cable sheathing, medical applications and other applications	Burghausen, Cologne, and Gendorf, Germany
Commodity PVC	6,820	Construction materials including pipe, siding, profiles for windows and doors, film and sheet for packaging and other applications	Calvert City, Kentucky Geismar, Louisiana Plaquemine, Louisiana Aberdeen, Mississippi Cologne and Knapsack, Germany
Low-Density Polyethylene ("LDPE")	1,500	High clarity packaging and bags, shrink films, food packaging, coated paper board, cup stock, paper folding cartons, lids, closures and general purpose molding	Lake Charles, Louisiana Longview, Texas
Linear Low-Density Polyethylene ("LLDPE")	1,070	Heavy-duty films and bags, general purpose liners	Lake Charles, Louisiana Longview, Texas
Chlorinated Derivative Materials	2,190	Coatings, flavorants, films, refrigerants, water treatment applications, chemicals and pharmaceutical production	Lake Charles, Louisiana Natrium, West Virginia
Styrene	570	Consumer disposables, packaging material, appliances, paints and coatings, resins and building materials	Lake Charles, Louisiana

Product [1]	Annual Capacity [2] (Millions of pounds)	End Uses	Principal Manufacturing Facilities [4] [5] [6]
Epoxy Specialty Resins	580	Protective Coatings and Adhesive Applications; Building and bridge construction, flooring, transportation, oil & gas Electrical Applications; Generators and bushings, transformers, medium and high-voltage switch gear components Composites Epoxy Resins; Wind energy, automotive, aerospace, construction, industrial applications	Lakeland, Florida Argo, Illinois Duisburg and Esslingen, Germany Onsan, South Korea Barbostro, Spain
Base Epoxy Resins and Intermediaries (BERI)	1,280	Electrocoat; Automotive, general industry Powder coatings; White goods, pipes for oil and gas transmission, general industry Heat Cured Coatings; Metal packaging and coil coated steel for construction and general industry	Deer Park, Texas Pernis, Rotterdam, The Netherlands

(1) EDC, a VCM intermediate product, is not included in the table.

(2) Includes capacity related to our 95%- and 60%-owned Asian joint ventures.

(3) Includes production capacity in Lake Charles and Calvert City owned by OpCo and our portion of LACC's production capacity in Lake Charles. For additional information on OpCo, see "*Ethylene*" below.

(4) Except as noted in notes (5) and (6) below, we own each of these facilities.

(5) We lease the land on which our Gendorf, Burghausen, Knapsack, Cologne and Esslingen, Germany facilities, Pernis, Rotterdam, The Netherlands facility, Longview and Deer Park, Texas facilities and Argo, Illinois facility are located. We also lease the Esslingen, Germany building.

(6) We lease a portion of the land on which our Aberdeen and Calvert City facilities are located.

Ethylene. Ethylene is the world's most widely used petrochemical in terms of volume. It is the key building block used to produce a large number of higher value-added chemicals including polyethylene, EDC, VCM and styrene. OpCo has the capacity to produce approximately 3.0 billion pounds of ethylene per year at our Lake Charles site, and we have the capability to consume all of OpCo's production that we purchase at Lake Charles to produce EDC, VCM, polyethylene and styrene monomer. In addition, we (through OpCo) produce ethylene co-products including chemical grade propylene, crude butadiene, pyrolysis gasoline and hydrogen. We (through OpCo) sell our entire output of these co-products to external customers. The ethylene from OpCo's facility in Calvert City and LACC is utilized to produce VCM at our facilities. We obtain the remainder of the ethylene we need for our business from third party purchases. The use of ethane feedstock by our ethylene plants enables us to enhance our low-cost materials chain integration.

Chlorine and Caustic Soda. We are the second-largest chlor-alkali producer in the world. We combine salt and electricity to produce chlorine and caustic soda, commonly referred to as chlor-alkali, at our Lake Charles, Plaquemine, Natrium, Calvert City, Geismar, Beauharnois, Longview (WA), Gendorf, Knapsack and Kaohsiung facilities. We use our chlorine production in our VCM and chlorinated derivative products plants. We currently have the capacity to supply all of our chlorine requirements internally. Any remaining chlorine is sold into the merchant chlorine market. Our caustic soda is sold to external customers who use it for, among other things, the production of pulp and paper, organic and inorganic chemicals and alumina.

VCM. VCM is used to produce PVC, solvents and PVC-related products. We use ethylene and chlorine to produce EDC, which is used in turn, to produce VCM. We have the capacity to produce approximately 6.3 billion pounds and 1.6 billion pounds of VCM per year at our North American and European facilities, respectively. The majority of our VCM is used internally in our PVC operations. VCM and EDC not used internally are sold externally.

PVC. PVC, the world's third most widely used plastic, is an attractive alternative to traditional materials such as glass, metal, wood, concrete and other plastic materials because of its versatility, durability and cost-competitiveness. PVC is produced from VCM, which is, in turn, made from chlorine and ethylene.

We are the second-largest PVC producer in the world. With the completion of our previously announced expansion projects at our Geismar and Burghausen plants in 2019, we have the capacity to produce approximately 6.8 billion pounds and 1.0 billion pounds of commodity and specialty PVC per year, respectively, at our various facilities globally. We use some of our PVC internally in the production of our building products, PVC pipes and fittings and PVC compounds in the Housing and Infrastructure Products segment. The remainder of our PVC is sold to downstream fabricators and the international markets.

Polyethylene. Polyethylene, the world's most widely consumed polymer, is used in the manufacture of a wide variety of film, coatings and molded product applications primarily used in packaging. Polyethylene is generally classified as either LDPE, LLDPE or high-density polyethylene ("HDPE"). The density correlates to the relative stiffness of the end-use products. The difference between LDPE and LLDPE is molecular, and products produced from LLDPE, in general, have higher strength properties than products produced from LDPE. LDPE exhibits better clarity and other physical properties and is used in end products such as bread bags, food wraps, milk carton coatings and food packaging. LLDPE is used for higher film strength applications such as stretch film and heavy-duty sacks. HDPE is used to manufacture products such as grocery, merchandise and trash bags, rigid plastic containers, plastic closures and pipe.

We are a leading producer of LDPE by capacity in North America and predominantly use the autoclave technology (versus tubular technology), which is capable of producing higher-margin specialty polyethylene products. In 2023, our annual capacity of approximately 1.5 billion pounds of LDPE was available in numerous formulations to meet the needs of our diverse customer base. We also have the capacity to produce approximately 1.1 billion pounds of LLDPE per year in various formulations. We produce LDPE and LLDPE at both the Lake Charles and Longview (TX) facilities. Our Lake Charles and Longview facilities also have the capability to produce HDPE. We sell polyethylene to external customers as a final product in pellet form.

Chlorinated Derivative Materials. Our chlorinated derivative products include ethyl chloride, perchloroethylene, trichloroethylene, tri-ethane® solvents, VersaTRANS® solvents, calcium hypochlorite, hydrochloric acid ("HCL") and pelletized caustic soda ("PELS"). We have the capacity to produce approximately 2.2 billion pounds of chlorinated derivative products per year, primarily at our Lake Charles, Natrium, Beauharnois and Longview (WA) facilities. The majority of our chlorinated derivative products are sold to external customers who use these products for, among other things, refrigerants, water treatment applications, chemicals and pharmaceutical production, food processing, steel pickling, solvent and cleaning chemicals and natural gas and oil production.

Styrene. Styrene is used to produce derivatives such as polystyrene, acrylonitrile butadiene styrene, unsaturated polyester and synthetic rubber. These derivatives are used in a number of applications including consumer disposables, food packaging, housewares, paints and coatings, building materials, tires and toys. We produce styrene at our Lake Charles plant, where we have the capacity to produce approximately 570 million pounds of styrene per year, all of which is sold to external customers.

Epoxy Specialty Resins. With the acquisition of the Westlake Epoxy business, Westlake is now one of the leading producers of epoxy specialty resins, modifiers and curing agents in Europe, the United States and Asia with a global reach to our end markets. Epoxy resins are the fundamental component of many types of materials and are often used in the automotive, construction, wind energy, aerospace and electronics industries due to their superior adhesion, strength and durability. Epoxy specialty resins are used for a variety of high-end coating applications that require superior adhesion, corrosion resistance and durability of epoxy such as protective coatings that are used for construction, flooring and transportation, among others. Epoxy-based surface coatings are among the most widely used industrial coatings due to their long service life and broad application functionality combined with overall economic efficiency. Our epoxy specialty resins are also used for electrical applications such as generators, bushings, transformers and medium and high voltage switch gear components. In composites, our specialty epoxy products are used either as replacements for traditional materials such as metal, wood and ceramics, or in applications where traditional materials do not meet demanding engineering specifications. We are also one of the leading producers of resins that are used in fiber reinforced composites. Composites are used in a wide variety of applications ranging from wind energy, automotive, aerospace, construction and industrial applications. We produce epoxy specialty resins at our Duisburg and Esslingen plants in Germany, Barbastro plant in Spain, Onsan plant in South Korea, and Lakeland and Argo plants in the United States. We have the total capacity to produce approximately 580 million pounds of epoxy specialty resins per year, all of which is sold to external customers.

Base Epoxy Resins and Intermediaries (BERI). We are a leading supplier of Liquid and Solid Epoxy Resin. These base epoxies are used in a wide variety of industrial coatings applications. In addition, we are a major producer of bisphenol-A ("BPA") and epichlorohydrin ("ECH"), key precursors in the downstream manufacturing of basic epoxy resins and specialty resins. We internally consume the majority of our BPA and ECH, which ensures consistent supply of our required intermediate materials. We produce base epoxies and intermediaries at our Deer Park, United States and Pernis, the Netherlands plants, where we have the capacity to produce approximately 1,280 million pounds per year.

Product and Application Development. Our product and application development activities are geared towards developing and enhancing products, processes and applications. Facilities where we perform such activities are located in the United States, Germany, China and the Netherlands.

Electricity. Our Lake Charles, Plaquemine and Natrium cogeneration assets have the capacity to generate electricity of approximately 845, 240 and 100 megawatts, respectively, per year.

Feedstocks

We are highly integrated along our materials production chain. We (through OpCo) produce most of the ethylene required to produce our polyethylene and styrene. Ethylene can be produced from either petroleum liquid feedstocks, such as naphtha, condensates and gas oils, or from natural gas liquid feedstocks, such as ethane, propane and butane. Both of OpCo's Lake Charles ethylene plants use ethane as the primary feedstock. Pursuant to a feedstock supply agreement between us and OpCo, OpCo receives ethane feedstock at our Lake Charles site through several pipelines from a variety of suppliers in Texas and Louisiana. We own a 50% interest in a 104-mile natural gas liquids pipeline from Mont Belvieu to our Lake Charles site. OpCo owns a 200-mile ethylene pipeline that runs from Mont Belvieu to our Longview (TX) site. Additionally, through OpCo, we produce most of the ethylene required at our Calvert City facility utilizing ethane feedstock. The LACC ethylene facility is located adjacent to our chlor-alkali facility in Lake Charles and has an ethylene production capacity of 2.2 billion pounds per year. During the third quarter of 2019, the LACC ethylene plant began its commercial operations. At December 31, 2023, we, through one of our subsidiaries, owned 50% of the membership interests in LACC. We receive our proportionate share in ethylene production on a cash-cost basis and primarily use it to produce VCM. In addition to ethylene supplied by OpCo and LACC, we also acquire ethylene from third parties in order to supply a portion of our ethylene requirements. In Germany, we have access to, and partially own, an ethylene pipeline.

We acquire butene and hexene to manufacture polyethylene and benzene to manufacture styrene. We receive butene and hexene at the Lake Charles site and hexene at the Longview (TX) site via rail car from several suppliers. We receive benzene via barges, ships and pipeline pursuant to short-term arrangements. We purchase butene and hexene pursuant to multi-year contracts.

The salt requirements for several of our larger chlor-alkali plants are supplied internally from salt domes we either own or lease and the salt is transported by pipelines we own. We purchase the salt required for our other chlor-alkali plants pursuant to long-term contracts. Electricity and steam for one of our Lake Charles facilities and our Plaquemine facility are produced by on-site cogeneration units. A portion of our Natrium facility's electricity requirements is produced by our on-site generation unit, and the remainder is purchased. We purchase electricity for our remaining facilities under long-term contracts. We purchase VCM for our Asian PVC plant on a contract and spot basis.

The raw materials that we primarily use to manufacture our epoxy products include chlorine, caustic soda, phenol and acetone, all of which are available from more than one source. Prices for our main feedstocks are generally driven by the underlying petrochemical benchmark prices and energy costs, which are subject to price fluctuations.

Sustainability

We continued our efforts to deliver high-value products in increasingly sustainable ways in 2023. For instance, having introduced lower-carbon caustic soda and PVC products since early 2021, in October 2022, we expanded our GreenVin® bio-attributed PVC, which is produced in Germany with renewable power (under European Guarantees of Origin) and renewable ethylene. The renewable ethylene is derived from biomass. GreenVin® bio-attributed PVC is both International Sustainability & Carbon Certification PLUS ("ISCC PLUS") and REDcert2 certified, using the mass balance approach. Bio-attributed GreenVin® PVC is approximately 90% less carbon-intensive compared with conventionally-produced Westlake PVC in Germany, based on a cradle-to-gate product carbon footprint study by Sustainable AG, tested by TÜV Rheinland, in accordance with the ISO 14067 standard, taking biogenic CO_2 fixations into account.

Another example of our efforts to improve the sustainability of our products is our incorporation of higher volumes of recycled materials in our polyethylene resin. Our customers use our one-pellet polyethylene compound, PIVOTAL®, a product line that contains a blend of post-consumer-recycled (PCR) and virgin polyethylene resins and has obtained GreenCircle certification, to achieve PCR content in their products ranging from 25% to 45%.

Our Westlake Epoxy business also continued to incorporate alternative, bio-based feedstocks. The Pernis, Netherlands site, for example, received certification by ISCC PLUS for our tracing and handling of bio-based materials in the production of epoxy products.

Marketing, Sales and Distribution

We have a dedicated sales force for our business, organized by product line and region that sells our products directly to our customers. In addition, we rely on distributors to market products to smaller customers. Our polyethylene customers are some of the nation's largest producers of film and flexible packaging. We and OpCo sell ethylene and ethylene co-products to external customers. OpCo's primary ethylene co-products are chemical grade propylene, crude butadiene, pyrolysis gasoline and hydrogen. We have storage agreements and exchange agreements that provide us and OpCo with access to customers who are not directly connected to the pipeline system that we own. OpCo ships crude butadiene and pyrolysis gasoline by rail or truck. Additionally, we transport our polyethylene and styrene by rail or truck. Further, styrene can be transported by barge or ship.

We use some of our PVC internally in the production of our building products, pipes and fittings and PVC compounds in the Housing and Infrastructure Products segment. The remainder of our PVC is sold to downstream fabricators and the international markets. We have the capacity to use a majority of our chlorine internally to produce VCM and EDC, most of which, in turn, is used to produce PVC. We also use our chlorine internally to produce chlorinated derivative products. We sell the remainder of our chlorine and substantially all of our caustic soda production to external customers. The majority of our products are shipped from production facilities directly to the customer via pipeline, truck, rail, barge and/or ship. The remaining products are shipped from production facilities to third party chemical terminals and warehouses until being sold to customers.

No single customer accounted for 10% or more of net sales for the Performance and Essential Materials segment in 2023.

Competition

The markets in which our Performance and Essential Materials businesses operate are highly competitive. Competition in the materials market is based on product availability, product quality and consistency, product performance, customer service and price. Our competitors in the ethylene, polyethylene and styrene markets are some of the world's largest chemical companies, including Chevron Phillips Chemical Company, Dow Inc., ExxonMobil Chemical Company, Formosa Plastics Corporation, LyondellBasell Industries, N.V., NOVA Chemicals Corporation and Sasol Limited. We compete in the chlor-alkali and PVC markets with other producers including Formosa Plastics Corporation, INOVYN ChlorVinyls Limited, KEM ONE Group SAS, Olin Corporation, Orbia Advance Corporation, Occidental Chemical Corporation, Shintech, Inc. and VYNOVA Group. Our competitors in the epoxy value chain include Olin Corporation, Nan Ya Plastics Corporation, the Spolchemie Group, LEUNA-Harze GmbH, Aditya Birla Chemicals (Thailand) Ltd., Huntsman Corporation, Swancor Holding Company Limited, Ningbo Bohui Chemical Technology Co., Ltd., Techstorm Advanced Material Co., Ltd., Shanghai Kangda Chemical New Material Group Co., Ltd., Evonik Industries AG, Allnex Management GmbH, Kukdo Chemical Co., Ltd., Kumhu Asiana Group and Chang Chun Plastics Co., Ltd.

Housing and Infrastructure Products Business

Our Brands and Products

We manufacture and sell housing and infrastructure products including residential PVC siding; PVC trim and mouldings; architectural stone veneer; windows; PVC decking; PVC films for various inflatables, wallcovering, tape and roofing applications; polymer composite and cement roof tiles; PVC pipe and fittings for various water, sewer, electrical and industrial applications; PVC compounds used in various housing, medical and automobile products; and a variety of consumer and commercial products such as landscape edging; industrial, home and office matting; marine dock edging; and masonry joint controls.

Our housing and infrastructure products consist of several product groups including: (i) exterior and interior building products, which includes siding, trim and mouldings, stone, roofing, windows and outdoor living products; (ii) PVC pipe, specialty PVC pipe and fittings; and (iii) PVC compounds. Many of our products are made from PVC, including products for water and sewer systems and home and light commercial applications for siding, trim and mouldings, outdoor living products, windows and PVC compounds.

Siding. Our siding products include insulated siding and vinyl siding and accessory products. Additionally, we offer premium siding products such as Celect® Cellular Composite Siding and, TruExterior® Siding. Our siding business is also a leader in non-wood shutters and siding accessories along with an array of specialty tools to aid installation. Our brands include Royal® Siding, Portsmouth® Shake and Shingle™, Foundry® Specialty Siding, TruExterior® Siding&Trim, Celect® Cellular Exteriors, Mid-America® Exteriors, Tapco Tools®, and many more.

Trim and Mouldings. We offer a wide variety of trim and moulding products, including exterior and interior products for homes, multi-family and light commercial structures that are used as substitute for wood and wood composite offerings. Our brands include Royal® Trim and Mouldings, TruExterior® Siding&Trim, and, Kleer Lumber®, among others.

Stone. We are a leader in the masonry stone veneer category, with both mortar applied products and mechanically fastened products that are used as a substitute for stone on exterior walls and accents as well as in interior applications such as fireplaces, kitchens and bathrooms. Our stone brands include Cultured Stone®, Eldorado Stone®, Versetta Stone®, StoneCraft Industries® and Dutch Quality Stone®, among others.

Roofing. Our DaVinci® Roofscapes is a premium composite roofing. Additional product offerings include concrete and clay roof tiles and stone coated steel roofing. Our other major roofing name brands include NewPoint™, Concrete Roof Tile, US Tile® Clay Roofing Products, Unified Steel™, and Stone Coated Roofing among others.

Windows. We are a regional fabricator of vinyl windows in the South and Southeast markets of the United States. Our brands include Legacy Collection™, the Krestmark® Collection, and the Magnolia Collection™.

Outdoor Living Products. Our outdoor living products include Zuri™ Premium Decking and Kindred® Outdoor kitchens and fire bowls.

PVC Pipe. We manufacture and sell pipe ranging from ½ inch to 36 inches in diameter, in gasketed, solvent welded, and restrained joint configurations.Our pipe products are used in residential water and sewer applications; municipal potable water and sewer infrastructure; plumbing and industrial applications, including drain, waste & vent ("DWV"); electrical duct and conduit; turf irrigation, water well and other major water transport market segments. We manufacture and market pipe for water, sewer, irrigation and conduit pipe products under the Westlake Pipe & Fittings brand name.

Specialty PVC Pipe. Our specialty PVC pipe includes the Certa-Lok® pipe and Certa-Lok® CLIC™ joining systems, which provide restrained joints with rapid assembly, designed for use in potable water, sewer, fire protection, agriculture, well-casing, electrical conduit and other piping system applications in the residential and various infrastructure markets. Other specialty products include a system for high rise DWV installations that incorporates low smoke and flame properties. Our molecularly-oriented PVC ("PVCO") pipe, AquaMax™, is produced with less PVC than conventional pipe improving our environmental footprint while delivering higher performance. We also manufacture and market specialty pipe under the Certa-Set®, Certa-Flo®, Certa-Com®, Yelomine®, Fluid-Tite®, Kwik-Set®, Sure-Fit®, Cobra Lock™, and Kor-Flo® brand names, among others.

Fittings. Our fittings products include a range of injection molded and custom fabricated fittings including: injection mold DWV fittings for residential, low-rise and high-rise commercial installations; molded gasketed and solvent weld sewer fittings up to 15 inches, molded gasketed municipal pressure fittings and molded fittings for the pool, spa, industrial markets and electrical assemblies; and fabricated custom fittings up to 36 inches for municipal and plumbing installations. We manufacture and market specialty fittings under the Westlake Pipe & Fittings brand name.

*PVC Compound*s. PVC compounds are custom blended formulations made by combining PVC resin with functional additives. They offer specific end-use properties based on customer-determined specifications and are used to produce rigid and flexible PVC applications. Our compounds are processed via extrusion, injection molding, blow molding, calendering. Our primary markets are building materials (housing), pipe and fittings, industrial materials, automotive, healthcare, telecommunications and consumer goods. Flexible PVC compounds are used for, but not limited to, the following applications: wire and cable, flooring, roofing, wall coverings, window and door trims, gaskets, industrial applications, automotive interior and exterior trims, footwear and medical applications. Rigid extrusion PVC compounds are commonly used for pipe, window and door profiles, fencing, decking, railing, siding and trim. Injection molding and blow molding PVC compounds are commonly used for pipe fittings, electrical components, industrial applications and packaging for consumer goods. We manufacture and market custom PVC compounds under the Westlake Global Compounds and Nakan brand names.

Recycled Products. Westlake Dimex is a producer of post-industrial-recycled PVC, PE and thermoplastic elastomer (TPE) compounds in addition to various consumer and professional products made from recycled PVC, PE and TPE materials. These products include landscape edging; industrial, home and office matting; marine dock edging; and masonry control joints.

Raw Materials and Suppliers

Our North American PVC facilities within the Performance and Essential Materials segment supply most of the PVC required by building products for our housing exteriors and PVC pipes and fittings plants. Our raw materials for stone, roofing and accessories, windows, shutters, and specialty tool products are externally purchased. PVC required for the PVC compounds plants is either internally sourced from our North American and Asian facilities within the Performance and Essential Materials segment or externally purchased based on the location of the plants. The remaining raw materials required, including pigments, fillers, stabilizers, and other ingredients, are purchased under short-term contracts, long-term contracts and in the spot market based on prevailing market prices.

Manufacturing

We operate 58 manufacturing locations primarily in the United States and Canada where we produce siding, trim and mouldings, stone, roofing, windows, outdoor living products, PVC pipes, specialty PVC pipes and fittings. In addition, we have 12 manufacturing locations across the world where we produce PVC compounds, including locations in North America, Europe and Asia. The following table illustrates the properties owned and leased by the Housing and Infrastructure Products business:

	Manufacturing Facilities	
	Owned	**Leased**
North America	45	18
Europe	2	2
Asia	3	—

Marketing, Sales and Distribution

We sell a majority of our siding, trim and mouldings, stone, roofing, windows and outdoor living products, PVC pipe, specialty PVC pipe and fittings through a combination of our internal sales force and some manufacturer's representatives. In North America, we operate 38 leased and 6 owned distribution centers and warehouses that service and supply our products to local customers, contractors and distributors. We also engage in advertising programs primarily directed at trade professionals and homeowners that are intended to develop awareness and interest in our products. In addition, we display our products at trade shows. Our 12 PVC Compounds facilities across the world sell through a combination of our internal sales force and distributors.

No single customer accounted for 10% or more of net sales for the Housing and Infrastructure Products segment in 2023.

Sustainability

We endeavor to continue to focus on manufacturing products in a more sustainable way and developing products with improved sustainable features. For instance, we expanded production of Westlake molecular-oriented polyvinyl chloride (PVCO) pipe, which offers both enhanced water flow and a lower-carbon footprint due to its lighter weight compared to traditional Westlake PVC pipe. Westlake PVCO pipe provides an approximately 10% increase in internal flow area compared with pipes with the same outside diameter, and approximately 40% less PVC pipe compound compared to traditional Westlake PVC pipe used for distribution of potable water, sewer and other systems. Less PVC pipe compound per unit length means a lighter weight product that provides secondary benefits for shipping the product.

We also enhanced circularity of more of our products as discussed in our latest 2022 Environmental Social and Governance Report, which is available at our website at www.westlake.com. For instance, we produced more final products made using in-house generated regrind, post-industrial recycled materials, and/or post-consumer recycled material. Several of our stone veneer products produced at two of our facilities incorporated approximately 45% (Versetta Stone® and ProStone®) to 55% (Cultured Stone®) recycled content. Following our acquisition of Westlake Dimex, we started to utilize increasing amounts of in-house generated PVC plastic regrind purchased from other Westlake businesses, in addition to purchasing post-industrial recycled materials from third parties. Unless specifically stated herein, documents and information on our website are not incorporated by reference in this Form 10-K.

Competition

The markets in which our housing and infrastructure businesses operate are highly competitive. Competition in the housing and infrastructure products market is based on product quality, product innovation, customer service, product consistency, on-time delivery and price. We compete in the housing and infrastructure products market with other producers and fabricators including Associated Materials LLC., CertainTeed Corporation, Cornerstone Building Brands, Inc., Diamond Plastics Corporation, IPEX Inc., JM Eagle Inc., Trex Company, Inc. and The Azek Company and with producers of PVC compounds including GEON Performance Solutions and Teknor Apex Company, Inc.

Seasonality

Though we generally experience demand for our products throughout the year, our sales, particularly of housing products have historically experienced some seasonality. We have typically experienced moderately higher levels of sales of our residential products in the first half of the year due to inventory restocking and improved weather for construction. Our sales are affected by the individual decisions of distributors and dealers on the levels of inventory they carry, their views on product demand, their financial condition and the manner in which they choose to manage inventory risk. Our sales are also generally impacted by the number of days in a quarter or a year that contractors and other professionals are able to install our products. This can vary dramatically based on, among other things, weather events such as rain, snow and extreme temperatures. We have generally experienced lower levels of sales of our housing products in the fourth quarter due to adverse weather conditions in certain markets, which typically reduces the construction and renovation activity during the winter season.

Environmental

As is common in our industry, we are subject to environmental laws and regulations related to the use, storage, handling, generation, transportation, emission, discharge, disposal and remediation of, and exposure to, hazardous and non-hazardous substances and wastes in all of the countries in which we do business. National, state or provincial and local standards regulating air, water and land quality affect substantially all of our manufacturing locations around the world. Compliance with such laws and regulations has required and will continue to require capital expenditures and increase operating costs.

It is our policy to comply with all environmental, health and safety requirements and to provide safe and environmentally sound workplaces for our employees. In some cases, compliance can be achieved only by incurring capital expenditures. In 2023, we made capital expenditures of $54 million related to environmental compliance. We estimate that we will make capital expenditures of approximately $56 million in 2024 and $74 million in 2025, respectively, related to environmental compliance. The expected 2024 and 2025 capital expenditures are relatively higher than the amounts we have spent related to environmental compliance in recent years in large part due to capital expenditures related to previously existing and new Environmental Protection Agency (the "EPA") regulations and corrective actions required by the EPA to resolve the flare enforcement matter discussed below. The remainder of the 2024 and 2025 estimated expenditures are related to equipment replacement and upgrades. We anticipate that stringent environmental regulations will continue to be imposed on us and the industry in general.

From time to time, we receive notices or inquiries from government entities regarding alleged violations of environmental laws and regulations pertaining to, among other things, the disposal, emission and storage of chemical substances, including hazardous wastes. Pursuant to Item 103 of the SEC's Regulation S-K, the following environmental matters involve a governmental authority as a party to the proceedings and potential monetary sanctions that we believe could exceed $1 million (which is less than one percent of our current assets on a consolidated basis as of December 31, 2023):

- To resolve alleged violations associated with exceedances of discharge limits under the Natrium facility's National Pollutant Discharge Elimination System ("NPDES") permit effective August 2020, we have entered into enforcement negotiations with the West Virginia Department of Environmental Protection ("WVDEP"). The resolution of this matter may involve a penalty in excess of $1 million.

- For several years, the EPA has been conducting an enforcement initiative against petroleum refineries and petrochemical plants with respect to emissions from flares. On April 21, 2014, we received a Clean Air Act Section 114 Information Request from the EPA which sought information regarding flares at the Calvert City facility and certain Lake Charles facilities. The EPA informed us that the information provided led the EPA to believe that some of the flares were out of compliance with applicable standards. In June 2022, the Department of Justice announced that the Company, the EPA and state environmental agencies had reached agreement on a consent decree resolving this matter. The consent decree was entered by the court and became effective in October 2022. The consent decree requires us to install flare gas recovery units, implement fence line monitoring, install and operate flare monitoring and control equipment to meet certain performance standards, and pay a civil penalty of $1 million. We estimate that the total cost to implement the requirements under the decree will be approximately $110 million, which includes capital expenditures associated with installation of the flare gas recovery units and monitoring and control equipment we are required to install at our Calvert City and Lake Charles facilities. The substantial majority of the capital expenditures and other costs required to comply with the consent decree were incurred in 2021, 2022 or 2023. We have paid all penalties required under the consent decree and have planned, budgeted for and scheduled all compliance requirements going forward. While the final consent decree does provide for stipulated penalties if certain requirements are not met, we do not believe that any stipulated penalties, if incurred and assessed, will have a material adverse effect on our financial condition, results of operations or cash flows.
- Westlake owns and operates salt solution-mining caverns at the Sulphur Brine Dome in Sulphur, Louisiana. The Louisiana Department of Natural Resources issued Compliance Order No. IMD 2022-027 and several supplements to that order, the latest in October 2023, in response to pressure anomaly events in two of our brine caverns. The brine caverns were not active, operating wells but under ongoing, post-operational monitoring requirements. The compliance order and supplements thereto have required us to undertake various activities related to response planning, monitoring, investigation and mitigation of potential impacts in the event of future cavern pressure anomalies or failures. Since December 2022, continuous brine injection has maintained cavern pressure while we continue to pursue active monitoring, studies, groundwater monitoring, modeling and other activities under the compliance order and supplements. In September 2023, the Office of Conservation issued an emergency declaration as a conservative step and to ensure that the full suite of powers and resources are available to the government in its response and management of the evolving circumstances at the Sulphur Dome. Capital costs and expenditures required to comply with the compliance order and supplements have been and will continue to be incurred throughout 2024. In response to the most recent supplement to the compliance order, the Company reserved approximately $13 million in connection with groundwater monitoring wells and monitoring required by the supplement. At this time, the Company is unable to estimate the impact that other ongoing expenditures or future injunctive relief ordered by the government could have on the Company's financial condition, results of operations or cash flows.

Also see our discussion of our environmental matters contained in Item 1A, "Risk Factors" below, Item 3, "Legal Proceedings" below and Note 22 to our consolidated financial statements included in Item 8 of this Form 10-K.

Human Capital

Westlake is committed to acting in a safe, ethical, environmentally, and socially responsible manner. We have been a public company for 19 years, but we still think of our employees as members of our extended family. We value the integrity, creativity, dedication and diversity of ideas that our employees bring to work every day.

Diversity, Equity and Inclusion (DEI)

As a global company, we recognize the diversity of our employees, customers and communities, and believe in creating an inclusive and equitable environment that represents a broad spectrum of backgrounds and cultures. A significant portion of our management team and our Board of Directors comes from diverse backgrounds, and we are focused on hiring and retaining diverse and talented employees. Our Board of Directors has charged the Compensation Committee with oversight responsibility of the Company's DEI efforts.

As an Asian American and Pacific Islander ("AAPI")-controlled business, we feel a special commitment to ensuring that Westlake continues to offer opportunities for employees of all backgrounds and experiences. As of December 31, 2023, approximately 36% of our employees in the United States and Canada self-identified as Black, Indigenous, Hispanic, or AAPI. Although we do not collect race and ethnicity data of our workforces in Latin America, Europe, and Asia, we know that we are a diverse, multinational company.

Training and Professional Development

As part of our retention and promotion efforts, we invest in internal leadership development. Westlake regularly provides its employees with a blend of live, virtual, and digital training opportunities, including safety training, technical courses, compliance training relating to company policies, business and professional development training, and professional growth classes. In conjunction with technical training, we believe that the non-technical training helps to create well-rounded, and highly-driven, employees. In addition, we periodically conduct employee surveys to gauge employee engagement and identify areas for additional focus.

Headcount

As of December 31, 2023, we had approximately 15,520 employees in the following areas:

Category	Number
Performance and Essential Materials segment	6,830
Housing and Infrastructure Products segment	8,130
Corporate and other	560

Our employees are distributed across 19 countries. As of December 31, 2023, approximately 68% were employed in the United States. Approximatly 27% of our employees are represented by labor unions, including works councils in Europe, and our collective bargaining agreements in Europe, North America and Asia expire at various times through 2026. There were no strikes, lockouts or work stoppages in 2023, and we believe that our relationship with our employees and unions is open and positive.

Attracting, developing and retaining talented people is crucial to executing our strategy. Our ability to recruit and retain such talent depends on a number of factors, including compensation and benefits, career opportunities and work environment. We provide our employees with competitive compensation packages, development programs that enable continued learning and growth, and comprehensive and competitive benefit packages worldwide. Our compensation and benefits arrangements generally are tailored to local markets of operation.

Health and Safety

The health and safety of our employees and our operations is our highest priority. Our health and safety programs are designed around global standards with appropriate variations addressing the multiple jurisdictions and regulations, specific hazards and unique working environments of our manufacturing, research and development, and administrative office locations. We require each of our locations to perform regular safety audits to ensure proper safety policies, program procedures, analyses and training are in place.

Our focus on safety starts at the top with our Board of Directors. Safety is a key performance indicator that is reported on and discussed at every Board meeting. Our Health, Safety and Environment ("HSE") team plays a pivotal role in overseeing all related policy protections, risk identification and other aspects of our business.

Several of our U.S. manufacturing sites have been recognized by the U.S. Occupational Safety & Health Administration's ("OSHA") Voluntary Protection Program ("VPP") for their low injury rates, employee engagement and safety programs. Most of our manufacturing sites in Germany satisfy the Deutsche Industrie Norm ISO 45001 certification program, which is comparable to VPP.

With respect to the COVID-19 pandemic, we adopted safety guidelines and practices, including safety and health training for existing and new employees, remote working, health screening of employees, contact tracing for reported exposure, enforcing self-isolation after exposure, provisions for mask wearing, modifications to the in-office work environment, social distancing, increased sanitation stations and increased cleaning of offices and workstations.

Intellectual Property and Technology

We rely upon both trademark and service mark protection to protect our brands, and have registered or applied to register many of these on a world-wide basis. We have over 1,650 active and pending trademark registrations worldwide for our various business segments. We also rely on a combination of patents and un-patented proprietary know-how and trade secrets to preserve our competitive technology position in the market. We have over 1,300 issued patents and pending-patent applications in the United States and several other countries. While some of our production capacity operates under licenses from third parties, other parts of our production operate under technology that was developed internally. Consequently, we offer our own independently developed technology for licensing when the opportunity makes sense on a commercial basis. Although the Company considers its patents, licenses, trademarks and trade secrets in the aggregate to constitute a valuable asset that provides competitive advantage to us, we do not regard any one of our businesses as being materially dependent upon any single or group of related patents, trademarks, licenses, or trade secrets.

Available Information

Our website address is www.westlake.com. Our website content is available for information purposes only. It should not be relied upon for investment purposes, nor is it incorporated by reference in this Form 10-K. We make available on this website under "Investor Relations/Financials," free of charge, our proxy statements, annual reports on Form 10-K, quarterly reports on Form 10-Q, current reports on Form 8-K and amendments to those materials as soon as reasonably practicable after we electronically file those materials with, or furnish those materials to, the SEC. The SEC also maintains a website at www.sec.gov that contains reports, proxy statements and other information regarding SEC registrants, including us.

We intend to satisfy the requirement under Item 5.05 of Form 8-K to disclose any amendments to our Code of Ethics and any waiver from a provision of our Code of Ethics by posting such information on our website at www.westlake.com under "Investor Relations/Governance."

Item 1A. Risk Factors

We are subject to various risks and uncertainties in the course of our business. The following summarizes the risks and uncertainties that we consider to be material and that may materially and adversely affect our business, financial condition, results of operations or cash flows and the market value of our securities. Investors should consider these matters, in addition to the other information we have provided in this Annual Report on Form 10-K and the documents we incorporate by reference.

Risk Factor Summary

Risks Related to Our Business, Industry and Operations

- Cyclicality in the petrochemical industry has in the past, and may in the future, result in reduced operating margins or operating losses.
- We sell commodity products in highly competitive markets and face significant competition and price pressure.
- Our operations depend on the availability and costs of raw materials, energy and utilities, and volatility in costs of raw materials, energy and utilities and supply chain constraints may result in increased operating expenses and adversely affect our results of operations and cash flows.
- External factors beyond our control can cause fluctuations in demand for our products and in our prices and margins, which may negatively affect our results of operations and cash flows.
- The housing market may continue its recent decline or decline further, and any such continuation or decline in the homebuilding industry may adversely affect our operating results.
- We operate internationally and are subject to related risks, including exchange rate fluctuations, exchange controls, political risk and other risks relating to international operations.
- Our inability to compete successfully may reduce our operating profits.
- Our production facilities process some volatile and hazardous materials that subject us to operating and litigation risks that could adversely affect our operating results.
- We rely on a limited number of outside suppliers for specified feedstocks and services.
- We rely heavily on third party transportation, which subjects us to risks and costs that we cannot control. Such risks and costs may adversely affect our operations.

- We may pursue acquisitions, dispositions, joint ventures or other transactions that may impact our results of operations and financial condition. We may have difficulties integrating the operations of recently acquired businesses, such as Westlake Epoxy, and future acquired businesses.
- Capital projects are subject to risks, including delays and cost overruns, which could have an adverse impact on our financial condition and results of operations.
- Public and investor sentiment towards climate change and other environmental, social and governance ("ESG") matters could adversely affect our cost of capital and the price of our common stock.
- Our participation in joint ventures and similar arrangements exposes us to a number of risks, including risks of shared control.
- Our operations could be adversely affected by labor relations.
- We have certain material pension and other post-retirement employment benefit ("OPEB") obligations. Future funding obligations related to these obligations could restrict cash available for our operations, capital expenditures or other requirements or require us to borrow additional funds.
- If our goodwill or other long-lived assets become impaired in the future, we may be required to record non-cash charges to earnings, which could be significant.
- Failure to adequately protect critical data and technology systems could materially affect our operations.
- Fluctuations in foreign currency exchange and interest rates could affect our consolidated financial results.
- Our property insurance has only partial coverage for acts of terrorism and, in the event of terrorist attack, we could lose net sales and our facilities. Our casualty and other insurance policies may be subject to coverage limitations.
- The impact and effects of public health crises, pandemics and epidemics could adversely affect our business, financial condition and results of operations.

Legal, Governmental and Regulatory Risks

- Our operations and assets are subject to extensive environmental, health and safety laws and regulations.
- Our operations and assets are subject to climate-related risks and uncertainties.
- We are subject to operational and financial risks and liabilities associated with the implementation of and efforts to achieve our carbon emission reduction goals.

Risks Related to Our Indebtedness

- Our level of debt could adversely affect our ability to operate our business.
- To service our indebtedness and fund our capital requirements, we will require a significant amount of cash. Our ability to generate cash depends on many factors beyond our control.
- The Credit Agreement and the indenture governing certain of our senior notes impose significant operating and financial restrictions, which may prevent us from capitalizing on business opportunities and taking some actions.

Risks Related to Taxes

- A change in tax laws, treaties or regulations, or their interpretation or application, could have a negative impact on our business and results of operations.
- We depend in part on distributions from Westlake Partners to generate cash for our operations, capital expenditures, debt service and other uses. Westlake Partners' tax treatment depends on its status as a partnership for federal income tax purposes, and it not being subject to a material amount of entity-level taxation. If the Internal Revenue Service ("IRS") were to treat Westlake Partners as a corporation for federal income tax purposes, or if Westlake Partners became subject to entity-level taxation for state tax purposes, its cash available for distribution would be substantially reduced, which would also likely cause a substantial reduction in the value of its common units that we hold.

Risks Related to the Ownership of Our Securities

- We will be controlled by our principal stockholder and its affiliates as long as they own a majority of our common stock, and our other stockholders will be unable to affect the outcome of stockholder voting during that time. Our interests may conflict with those of the principal stockholder and its affiliates, and we may not be able to resolve these conflicts on terms possible in arms-length transactions.

Risk Related to Business, Industry and Operations

Cyclicality in the petrochemical industry has in the past, and may in the future, result in reduced operating margins or operating losses.

Our historical operating results reflect the cyclical and volatile nature of the petrochemical industry. The industry is mature and capital intensive. Margins in this industry are sensitive to supply and demand balances both domestically and internationally, which historically have been cyclical. The cycles are generally characterized by periods of tight supply, leading to high operating rates and margins, followed by periods of oversupply primarily resulting from excess new capacity additions, leading to reduced operating rates and lower margins.

Moreover, profitability in the petrochemical industry is affected by the worldwide level of demand along with vigorous price competition which may intensify due to, among other things, new industry capacity. In general, weak economic conditions reduce demand and put pressure on margins. It is not possible to predict accurately the supply and demand balances, market conditions and other factors that will affect industry operating margins in the future.

New capacity additions, principally of ethylene, polyethylene, chlorine, caustic soda and PVC in North America, Asia and the Middle East and in the epoxy value chain in Asia, a number of which have been recently completed, may lead to periods of over-supply and lower profitability. Additionally, new entrants to the market, including when customers backward integrate into products we supply, can further exacerbate supply and demand imbalances. As a result, our Performance and Essential Materials ("PEM") segment operating margins may be negatively impacted.

The impact of COVID-19 and the conflicts in the Middle East and between Russia and Ukraine may lead to further supply chain constraints, supply and demand shifts, workforce availability issues and increased uncertainty in general economic and business conditions, including inflationary pressures, persistent high interest rates and possible recession.

We sell most of our commodity products in highly competitive markets and face significant competition and price pressure.

We sell most of our commodity products in highly competitive markets. Competition in commodity markets is based primarily on price and to a lesser extent on performance, product quality, product deliverability and customer service. As a result, we generally are not able to protect our market position for most of these products by product differentiation and may not be able to pass on cost increases to our customers. Accordingly, increases in raw material and other costs may not necessarily correlate with changes in prices for these products, either in the direction of the price change or in magnitude. Specifically, timing differences in pricing between raw material prices, which may change daily, and contract product prices, which in many cases are negotiated monthly or less often, sometimes with an additional lag in effective dates for increases, have had and may continue to have a negative effect on profitability. Significant volatility in raw material costs tends to place pressure on product margins as sales price increases could lag behind raw material cost increases. Conversely, when raw material costs decrease, customers could seek relief in the form of lower sales prices.

Our operations depend on the availability of raw materials, energy and utilities, and volatility in costs of raw materials, energy and utilities may result in increased operating expenses and adversely affect our results of operations and cash flows.

Our operations depend on the continued supply of raw materials and reliable energy. The availability of natural gas and electricity can be affected by numerous events such as weather (e.g., hurricanes and periods of considerable heat or cold, like Winter Storm Uri in 2021), pipeline and other logistics interruptions, electrical grid outages, cybersecurity incidents, intermittent electricity generation (particularly from wind and solar), hostilities and sanctions arising from geopolitical tensions, human error, and supply and demand imbalances for raw materials and electricity.

Significant variations in the costs and availability of raw materials and energy may negatively affect our results of operations. These costs have risen significantly in the past due primarily to oil and natural gas cost increases. We purchase significant amounts of ethane feedstock, natural gas, ethylene and salt to produce several basic chemicals. We also purchase significant amounts of electricity to supply the energy required in our production processes. The cost of these raw materials and energy, in the aggregate, represents a substantial portion of our operating expenses. The prices of raw materials and energy generally follow price trends of, and vary with market conditions for, crude oil and natural gas, which are highly volatile and cyclical, as well as the ability of domestic producers to export natural gas liquids, ethane and ethylene. Changes to regulatory policies applicable to the German energy sector for industrial users have contributed to higher prices for industrial users of energy in the past and may continue to do so in the future. Our results of operations have been and could in the future be significantly affected by increases in these costs.

Price increases increase our working capital needs and, accordingly, can adversely affect our liquidity and cash flows. In addition, because we utilize the first-in, first-out ("FIFO") method of inventory accounting, during periods of falling raw material prices and declining sales prices, our results of operations for a particular reporting period could be negatively impacted as the lower sales prices would be reflected in operating income more quickly than the corresponding drop in feedstock costs. We use derivative instruments (including commodity swaps and options) in an attempt to reduce price volatility risk on some feedstock commodities. In the future, we may decide not to hedge any of our raw material costs or any hedges we enter into may not have successful results. Also, our hedging activities involve credit risk associated with our hedging counterparties, and a deterioration in the financial markets could adversely affect our hedging counterparties and their abilities to fulfill their obligations to us.

Lower prices of crude oil, such as those experienced from the third quarter of 2014 through 2020, led to a reduction in the cost advantage for natural gas liquids-based ethylene crackers in North America, such as ours, as compared to naphtha-based ethylene crackers. As a result, our margins and cash flows were negatively impacted. Lower crude oil and natural gas prices could lead to a reduction in hydraulic fracturing in the United States, which could reduce the availability of feedstock and increase prices of feedstock for our operations. Higher natural gas prices could also adversely affect our ability to export products that we produce in the United States. In addition to the impact that this has on our exports from the United States, reduced competitiveness of U.S. producers also has in the past increased the availability of chemicals in North America, as U.S. production that would otherwise have been sold overseas was instead offered for sale domestically, resulting in excess supply and lower prices in North America. We could also face the threat of imported products from countries that have a cost advantage. Furthermore, additional export storage facilities for natural gas liquids, ethane and ethylene may lead to higher exports of such products from the United States or greater restrictions on hydraulic fracturing could restrict the availability of our raw materials in the United States, thereby increasing our costs.

External factors beyond our control can cause fluctuations in demand for our products and in our prices and margins, which may negatively affect our results of operations and cash flows.

External factors beyond our control can cause volatility in raw material prices, demand for our products, product prices and volumes and deterioration in operating margins. These factors can also magnify the impact of economic cycles on our business and results of operations. Examples of external factors include:

- general economic and business conditions, including in North America, Europe and Asia, including inflation, persistent high interest rates and possible recession;
- new capacity additions in North America, Europe, Asia and the Middle East;
- the level of business activity in the industries that use our products;
- competitor action;
- technological innovations;
- currency fluctuations;
- the impact of supply chain constraints and workforce availability;
- international events and circumstances;
- pandemics, such as COVID-19 pandemic, and other public health threats and efforts to contain their transmission;
- war, sabotage, terrorism and civil unrest, including the conflicts between Russia and Ukraine and in the Middle East;
- governmental regulation, including in the United States, Europe and Asia;
- public attitude towards climate change and safety, health and the environment;
- perceptions of our products by potential buyers of our products, as well as the public generally, and related changes in behavior, including with respect to recycling;
- severe weather and natural disasters;
- long-term impacts of climate change, including rising sea levels and changes in weather patterns, such as drought and flooding; and
- credit worthiness of customers and vendors.

A number of our products are highly dependent on durable goods markets, such as housing and construction, which are themselves particularly cyclical. Weakness in the U.S. residential housing market and economic weakness in North America, Europe, Asia and the Middle East could have an adverse effect on demand and margins for our products. Further, persistent high interest rates, inflationary pressures and the possibility of recession can have an unfavorable impact on the demand for housing and our products.

We may reduce production at or idle a facility for an extended period of time or exit a business because of high raw material prices, an oversupply of a particular product and/or a lack of demand for that particular product, which makes production uneconomical. Temporary outages sometimes last for several quarters or, in certain cases, longer, and cause us to incur costs, including the expenses of maintaining and restarting these facilities. Factors such as increases in raw material costs or lower demand in the future may cause us to further reduce operating rates, idle facilities or exit uncompetitive businesses.

A lower level of economic activity in the United States, Europe or globally could result in a decline in demand for our products, which could adversely affect our net sales and margins and limit our future growth prospects. In addition, these risks could cause increased instability in the financial and insurance markets and could adversely affect our ability to access capital and to obtain insurance coverage that we consider adequate or is otherwise required by our contracts with third parties.

The housing market may continue its recent decline or decline further, and any such continuation or decline in the homebuilding industry may adversely affect our operating results.

We cannot predict whether and to what extent the housing market in the United States will grow, particularly if interest rates for mortgage loans remain elevated or continue to rise. The U.S. housing market remained strong throughout the COVID-19 pandemic, but began softening at the end of the second quarter of 2022 and continued to decline throughout 2023 primarily due to inflationary pricing, rapidly rising interest rates for mortgage loans and increasing construction costs. Other factors that might impact the homebuilding industry include uncertainty in domestic and international financial, credit and consumer lending markets amid slow economic growth or recessionary conditions in various regions or industries around the world, including as a result of the conflicts in the Middle East and between Russia and Ukraine, higher interest rates, tight lending standards and practices for mortgage loans that limit consumers' ability to qualify for mortgage financing to purchase a home, higher home prices, population declines or slower rates of population growth or U.S. Federal Reserve policy changes.

If there is limited economic growth, a decline in employment and consumer income, a general change in consumer behavior and/or tightening of mortgage lending standards, practices and regulation, or if interest rates for mortgage loans or home prices rise, there could be a corresponding adverse effect on our financial condition, results of operations or cash flows, including, but not limited to, the amount of revenues or profits we generate in our Housing and Infrastructure Products segment.

We operate internationally and are subject to related risks, including exchange rate fluctuations, exchange controls, political risk and other risks relating to international operations.

We operate internationally and are subject to the risks of doing business on a global basis. These risks include, but are not limited to, fluctuations in currency exchange rates, currency devaluations, imposition or the threat of trade barriers (which could, among other things, negatively impact our ability to export our products outside of the United States), imposition or the threat of tariffs and duties (which could, among other things, lead to lower demand for our products outside of the United States), inflationary pressures and possibility of recession, restrictions on the transfer of funds, changes in law and regulatory requirements, involvement in judicial proceedings in unfavorable jurisdictions, economic instability and disruptions, political unrest and epidemics. U.S. foreign trade policies could lead to the imposition of additional trade barriers and tariffs on us in foreign jurisdictions. Our operating results could be negatively affected by any of these risks.

Our inability to compete successfully may reduce our operating profits.

The industries in which we operate are highly competitive. Historically, there have been a number of mergers, acquisitions, spin-offs and joint ventures in the industry in which the PEM business operates. This restructuring activity has resulted in fewer but more competitive PEM producers, many of which are larger than we are and have greater financial resources than we do. Among our PEM competitors are some of the world's largest chemical companies and chemical industry joint ventures. Competition within the petrochemical industry and in the manufacturing of housing and infrastructure products is affected by a variety of factors, including:

- product price;
- balance of product supply/demand;
- material, technology and process innovation;
- technical support and customer service;

- quality;
- reliability of raw material and utility supply;
- availability of potential substitute materials; and
- product performance.

Changes in the competitive environment could have a material adverse effect on our business and our operations. These changes could include:

- the emergence of new domestic and international competitors;
- the rate of capacity additions by competitors;
- the additions of export storage facilities for natural gas liquids, ethane and ethylene;
- changes in customer base due to mergers;
- the intensification of price competition in our markets;
- the introduction of new or substitute products by competitors; and
- the technological innovations of competitors.

Our production facilities process some volatile and hazardous materials that subject us to operating risks that could adversely affect our operating results.

We have manufacturing sites in North America, Europe and Asia. Our operations are subject to the usual hazards associated with chemical, plastics, housing and infrastructure products manufacturing and the related use, storage, transportation and disposal of feedstocks, products and wastes, and litigation arising as a result of such hazards, including:

- pipeline leaks and ruptures;
- explosions;
- fires;
- severe weather and natural disasters;
- mechanical failure;
- unscheduled downtime;
- labor difficulties;
- transportation interruptions;
- transportation accidents involving our products;
- remediation complications;
- chemical spills, discharges or releases of toxic or hazardous substances or gases;
- other environmental risks;
- sabotage;
- terrorist attacks; and
- political unrest.

All of these hazards can cause personal injury and loss of life, catastrophic damage to or destruction of property and equipment and environmental damage, and may result in a suspension of operations and the imposition of civil or criminal penalties. We are from time to time subject to environmental claims brought by governmental entities or third parties. A loss or shutdown over an extended period of operations at any one of our chemical manufacturing facilities would have a material adverse effect on us. We maintain property, business interruption and casualty insurance that we believe is in accordance with customary industry practices, but we cannot be fully insured against all potential hazards incident to our business, including losses resulting from wars or terrorist acts, among other things. In addition, certain policies may be subject to coverage limitations, which may affect the extent of any recovery thereunder. As a result of market conditions and past claims, premiums and deductibles for certain insurance policies can increase substantially and, in some instances, certain insurance may become unavailable or available only for reduced amounts of coverage. If we were to incur a significant liability for which we were not fully insured, or if an insurer was unwilling or unable to meet its obligations, it could have a material adverse effect on our financial condition, results of operations or cash flows.

We are exposed to significant losses from products liability, personal injury and other claims relating to the products we manufacture. Additionally, individuals currently seek, and likely will continue to seek, damages for alleged personal injury or property damage due to alleged exposure to chemicals at our facilities or to chemicals otherwise owned, controlled or manufactured by us. We are also subject to present and future claims with respect to workplace exposure, workers' compensation and other matters. Any such claims, whether with or without merit, could be time consuming, expensive to defend and could divert management's attention and resources. We maintain and expect to continue to maintain insurance for products liability, workplace exposure, workers' compensation and other claims, but the amount and scope of such insurance may not be adequate or available to cover a claim that is successfully asserted against us. In addition, such insurance could become more expensive and difficult to maintain and may not be available to us on commercially reasonable terms or at all. The results of any future litigation or claims are inherently unpredictable, but such outcomes could have a material adverse effect on our financial condition, results of operations or cash flows.

We rely on a limited number of outside suppliers for specified feedstocks and services.

We obtain a significant portion of our raw materials from a few key suppliers. If any of these suppliers is unable to meet its obligations under any present or future supply agreements, we may be forced to pay higher prices to obtain the necessary raw materials. Any interruption of supply or any price increase of raw materials could have a material adverse effect on our business and results of operations. A vendor may choose, subject to existing contracts, to modify its relationship due to general economic concerns or concerns relating to the vendor or us, at any time. Any significant change in the terms that we have with our key suppliers, or any significant additional requirements from our suppliers that we provide them additional security in the form of prepayments or with letters of credits, could materially adversely affect our financial condition, results of operations or cash flows.

We rely heavily on third party transportation, which subjects us to risks and costs that we cannot control. Such risks and costs may materially adversely affect our operations.

We rely heavily on railroads, barges, pipelines, ships, trucks and other shipping companies to transport raw materials to the manufacturing facilities used by our businesses and to ship finished products to customers. These transport operations are subject to various hazards and risks, including extreme weather conditions, work stoppages and operating hazards (including pipeline leaks and ruptures and storage tank leaks), as well as interstate transportation regulations. In addition, the methods of transportation we utilize, including shipping chlorine and other chemicals by railroad, may be subject to additional, more stringent and more costly regulations in the future. If we are delayed or unable to ship finished products or unable to obtain raw materials as a result of any such new regulations or public policy changes related to transportation safety, or these transportation companies fail to operate properly, or if there were significant changes in the cost of these services due to new or additional regulations, or otherwise, we may not be able to arrange efficient alternatives and timely means to obtain raw materials or ship goods, which could result in a material adverse effect on our business and results of operations.

We may pursue acquisitions, dispositions, joint ventures or other transactions that may impact our results of operations and financial condition. We may have difficulties integrating the operations of acquired businesses.

We seek opportunities to maximize efficiency and create stockholder value through various transactions. These transactions may include domestic and international business combinations, purchases or sales of assets or contractual arrangements or joint ventures. In this regard, we regularly consider acquisition opportunities that would be consistent or complementary to our existing business strategies. To the extent permitted under our credit facility, the indenture governing our senior notes and other debt agreements, some of these transactions may be financed by additional borrowings by us. Although we would pursue these transactions because we expect them to yield longer-term benefits if the efficiencies and synergies we expect are realized, they could adversely affect our results of operations in the short term because of the costs associated with such transactions and because they may divert management's attention from existing business operations. These transactions may not yield the business benefits, synergies or financial benefits anticipated by management. Integration of acquired operations could lead to restructuring charges or other costs.

Our ability to realize the anticipated benefits of acquisitions will depend, to a large extent, on our ability to integrate our business with the acquired businesses. The combination of such independent businesses is a complex, costly and time-consuming process. As a result, we have devoted, and will continue to devote significant management attention and resources to integrating business practices and operations. The integration process may disrupt the businesses and, if implemented ineffectively or if impacted by unforeseen negative economic or market conditions or other factors, we may not realize the full anticipated benefits of the acquisitions. Our failure to meet the challenges involved in integrating such businesses could adversely affect our results of operations.

In addition, the overall integration of the businesses may result in material unanticipated problems, expenses, liabilities, competitive responses, loss of customer relationships, or diversion of management's attention. Even if the operations of our businesses acquired are integrated successfully, we may not realize the full benefits of the acquisitions, including the synergies, cost savings or sales or growth opportunities that we expect. These benefits may not be achieved within the anticipated time frame, or at all. Furthermore, additional unanticipated costs may be incurred in the integration of the businesses. All of these factors could decrease or delay the expected benefits of the acquisitions and negatively impact us.

If we are unable to integrate or to successfully manage businesses that we may acquire in the future, our business, financial condition and results of operations could be adversely affected. We may not be able to realize the operating efficiencies, synergies, cost savings or other benefits expected from acquisitions for a number of reasons, including the following:

- we may fail to integrate the businesses we acquire into a cohesive, efficient enterprise;
- our resources, including management resources, are limited and may be strained if we engage in a large acquisition or significant number of acquisitions, and acquisitions may divert our management's attention from initiating or carrying out programs to save costs or enhance revenues; and
- our failure to retain key employees and contracts of the businesses we acquire.

Capital projects are subject to risks, including delays and cost overruns, which could have an adverse impact on our financial condition and results of operations.

We have capital expansion plans for our facilities. Expansion projects may be subject to delays or cost overruns, including delays or cost overruns resulting from any one or more of the following:

- unexpectedly long delivery times for, or shortages of, key equipment, parts or materials;
- shortages of skilled labor and other personnel necessary to perform the work;
- delays and performance issues;
- failures or delays of third-party equipment vendors or service providers;
- unforeseen increases in the cost of equipment, labor and raw materials;
- work stoppages and other labor disputes;
- unanticipated actual or purported change orders;
- disputes with contractors and suppliers;
- design and engineering problems;
- latent damages or deterioration to equipment and machinery in excess of engineering estimates and assumptions;
- financial or other difficulties of our contractors and suppliers;
- sabotage;
- terrorist attacks;
- interference from adverse weather conditions; and
- difficulties in obtaining necessary permits or in meeting permit conditions.

Significant cost overruns or delays could materially affect our financial condition and results of operations. Additionally, actual capital expenditures could materially exceed our planned capital expenditures.

Public and investor sentiment towards climate change and other environmental, social and governance ("ESG") matters could adversely affect our cost of capital and the price of our common stock.

There have been intensifying efforts within the investment community (including investment advisors, investment fund managers, sovereign wealth funds, public pension funds, universities and individual investors) to promote the divestment of, or limit investment in, the stock of companies in the petrochemical industry. There has also been pressure on lenders and other financial services companies to limit or curtail financing of companies in the industry. Because we operate within the petrochemical industry, if these efforts continue or expand, our stock price and our ability to raise capital may be negatively impacted.

Members of the investment community are increasing their focus on ESG practices and disclosures by public companies, including practices and disclosures related to climate change and sustainability, DEI initiatives and heightened governance standards. As a result, we may continue to face increasing pressure regarding our ESG disclosures and practices. Additionally, members of the investment community may screen companies such as ours for ESG disclosures and performance before investing in our stock. Over the past few years, there has also been an acceleration in investor demand for ESG investing opportunities, and many large institutional investors have committed to increasing the percentage of their portfolios that are allocated towards ESG investments. With respect to any of these investors, our ESG disclosures and efforts may not satisfy the investor requirements or their requirements may not be made known to us. If we or our securities are unable to meet the ESG standards or investment criteria set by these investors and funds, we may lose investors or investors may allocate a portion of their capital away from us, our cost of capital may increase, and our stock price may be negatively impacted.

Our participation in joint ventures and similar arrangements exposes us to a number of risks, including risks of shared control.

We are party to several joint ventures and similar arrangements, including an investment, together with Lotte Chemical USA Corporation ("Lotte"), in a joint venture, LACC, LLC ("LACC"), to build and operate an ethylene facility. Our participation in joint ventures and similar arrangements, by their nature, requires us to share control with unaffiliated third parties. If there are differences in views among joint venture participants in how to operate a joint venture that result in delayed decisions or the failure to make decisions, or our joint venture partners do not fulfill their obligations, the affected joint venture may not be able to operate according to its business plan and fulfill its obligations. In that case, we may be required to write down the value of our investment in a joint venture, increase the level of financial or other commitments to the joint venture or, if we have contractual agreements with the joint venture, our operations may be materially adversely affected. Any of the foregoing could have a material adverse effect on our financial condition, results of operations or cash flows.

Our operations could be adversely affected by labor relations.

The vast majority of our employees in Europe and Asia, and some of our employees in North America, are represented by labor unions and works councils. Our operations may be adversely affected by strikes, work stoppages and other labor disputes.

We have certain material pension and other post-retirement employment benefit ("OPEB") obligations. Future funding obligations related to these obligations could restrict cash available for our operations, capital expenditures or other requirements or require us to borrow additional funds.

We have U.S. and non-U.S. defined benefit pension plans covering certain current and former employees. Certain non-U.S. defined benefit plans associated with our European operations have not been funded and we are not obligated to fund those plans under applicable law. As of December 31, 2023, the projected benefit obligations for our pension and OPEB plans were $1,123 million and $44 million, respectively. The fair value of pension investment assets was $818 million as of December 31, 2023. The total underfunded status of the pension obligations calculated on a projected benefit obligation basis as of December 31, 2023 was $305 million, including the Westlake Defined Benefit Plan and the Vinnolit Pension Plan (locally known as 'Grund- und Zusatzversorgung' in Germany), which were underfunded by $47 million and $142 million, respectively, on an individual plan basis.

The unfunded OPEB obligations as of December 31, 2023 were $44 million. We will require future operating cash flows to fund our pension and OPEB obligations, which could restrict available cash for our operations, capital expenditures and other requirements. We may also not generate sufficient cash to satisfy these obligations, which could require us to seek funding from other sources, including through additional borrowings, which could materially increase our outstanding debt or debt service requirements.

If our goodwill or other long-lived assets become impaired in the future, we may be required to record non-cash charges to earnings, which could be significant.

Under the accounting principles generally accepted in the United States ("GAAP"), we review goodwill for impairment on an annual basis or more frequently if events or circumstances indicate that the carrying amount may not be recoverable. Long-lived assets, including tangible assets and intangible assets with finite lives, are reviewed if events or circumstances indicate that their carrying amount may not be recoverable. Our balance sheet includes significant goodwill and long-lived assets. In the fourth quarter of 2023, we recorded a goodwill impairment charge in the PEM segment of $128 million related to Westlake Epoxy and a non-cash long-lived asset impairment charge of $347 million related to our base epoxy resin business in the Netherlands.

The process of impairment testing for our goodwill and long-lived assets involves a number of judgments and estimates made by management including the fair values of assets and liabilities, future cash flows, our interpretation of current economic indicators and market conditions, overall economic conditions and our strategic operational plans with regards to our business units. If the judgments and estimates used in our analysis are not realized or change due to future external factors, then actual results may not be consistent with these judgments and estimates, and our goodwill and other long-lived assets may become further impaired in future periods. If our goodwill and long-lived assets are determined to be impaired in the future, we may be required to record further non-cash charges to earnings during the period in which the impairment is determined, which could be significant and have an adverse effect on our financial condition and results of operations.

Failure to adequately protect critical data and technology systems could materially affect our operations.

We are increasingly dependent on digital technologies and services to conduct our business. We use these technologies for internal and operational purposes, including data storage, processing, and transmission, as well as in our interactions with our business associates, such as customers and suppliers.

Information technology system failures, network disruptions and breaches of data security due to internal or external factors including cyber-attacks could have material adverse impacts on our business—and the business of our subsidiaries and affiliates that we support—or cause disruptions to our operations. Such disruptions could include, but are not limited to, delaying or cancelling customer orders, impeding the manufacture or shipment of products or causing standard business processes to become ineffective, or the unintentional or malicious disclosure of proprietary, confidential or other sensitive information. Cyber-attacks could include, but are not limited to, ransomware attacks, malicious software, attempts to gain unauthorized access to our systems or data or other electronic security breaches that could lead to disruptions in critical systems, unauthorized release, corruption or loss of data including protected information such as personal information of our employees, interruptions in communication, loss of our intellectual property or theft of our sensitive or proprietary technology, loss or damage to our data delivery systems, or other cybersecurity and infrastructure systems, including our property and equipment.

We have taken steps to address these risks by implementing network security and internal control measures, including employee training, comprehensive monitoring of our networks and systems, maintenance of backup and protective systems and disaster recovery and incident response plans. However, we also rely on our business associates with whom we may share data and digital services to defend their digital technologies, systems, and services against attack. As a result, there is a risk that an incident could originate from our business associates, as well.

Our employees, systems, networks, products, facilities and services remain potentially vulnerable to sophisticated cyber-assault, including the additional cybersecurity risks posed by the increased use of remote networking technologies and services, and, as such, there can be no assurance that a system failure, network disruption or data security breach will not have a material adverse effect on our business, financial condition, operating results or cash flows. In addition, laws and regulations governing cybersecurity, data privacy, and the unauthorized disclosure of confidential or protected information pose increasingly complex compliance challenges, and failure to comply with these laws could result in penalties and legal liability.

Fluctuations in foreign currency exchange and interest rates could affect our consolidated financial results.

We earn revenues, pay expenses, own assets and incur liabilities in countries using currencies other than the U.S. dollar. Because our consolidated financial statements are presented in U.S. dollars, we must translate revenues and expenses into U.S. dollars at the average exchange rate during each reporting period, as well as assets and liabilities into U.S. dollars at exchange rates in effect at the end of each reporting period. Therefore, increases or decreases in the value of the U.S. dollar against other major currencies will affect our net sales, operating income and the value of balance sheet items denominated in foreign currencies. Because of the geographic diversity of our operations, weaknesses in various currencies might occur in one or many of such currencies over time. From time to time, we may use derivative financial instruments to further reduce our net exposure to currency exchange rate fluctuations. However, fluctuations in foreign currency exchange rates, such as the strengthening of the U.S. dollar against major currencies, including, in particular, the Euro and the Canadian dollar, could nevertheless materially adversely affect our financial results.

In addition, we are exposed to volatility in interest rates. When appropriate, we may use derivative financial instruments to reduce our exposure to interest rate risks. However, our financial risk management program may not be successful in reducing the risks inherent in exposures to interest rate fluctuations.

Our property insurance has only partial coverage for acts of terrorism and, in the event of terrorist attack, we could lose net sales and our facilities.

Our insurance carriers maintain certain exclusions for losses from terrorism from our property insurance policies. While separate terrorism insurance coverage is available, premiums for full coverage are very expensive, especially for chemical facilities, and the policies are subject to high deductibles. Available terrorism coverage typically excludes coverage for losses from acts of war and from acts of foreign governments as well as nuclear, biological and chemical attacks. We have determined that it is not economically prudent to obtain full terrorism insurance, especially given the significant risks that are not covered by such insurance. Where feasible we have secured some limited terrorism insurance coverage on our property where insurers have included it in their overall programs. In the event of a terrorist attack impacting one or more of our facilities, we could lose the net sales from the facilities and the facilities themselves, and could become liable for any contamination or for personal or property damage due to exposure to hazardous materials caused by any catastrophic release that may result from a terrorist attack.

The impact and effects of public health crises, pandemics and epidemics could adversely affect our business, financial condition and results of operations.

Public health crises, pandemics and epidemics, such as the COVID-19 pandemic, could materially adversely affect our business, financial condition and results of operations. Such events have resulted in and could again result in authorities implementing numerous measures to try to contain the disease, such as travel bans and restrictions, quarantines, shelter-in-place orders and shutdowns, among others. Such events have had and could again have widespread adverse impacts on the global economy, many of our facilities and on our employees, customers and suppliers. These and similar events have caused and may again cause supply chain constraints and disruptions and workforce availability issues as well.

Legal, Governmental and Regulatory Risks

Our operations and assets are subject to extensive environmental, health and safety laws and regulations.

We use large quantities of hazardous substances and generate hazardous wastes and emissions in our manufacturing operations. Due to the associated quantities of hazardous substances and wastes, our industry is highly regulated and monitored by various environmental regulatory authorities such as the EPA, federal or state analogs in other countries and the European Union, which promulgated the Industrial Emission Directive ("IED"). As such, we are subject to extensive international, national, state and local laws, regulations and directives pertaining to pollution and protection of the environment, health and safety, which govern, among other things, emissions to the air, discharges onto land or waters, the maintenance of safe conditions in the workplace, the remediation of contaminated sites, and the generation, handling, storage, transportation, treatment and disposal of waste materials. Some of these laws, regulations and directives are subject to varying and conflicting interpretations. Many of these laws, regulations and directives provide for substantial fines and potential criminal sanctions for violations and require the installation of costly pollution control equipment or operational changes to limit pollution emissions or reduce the likelihood or impact of hazardous substance releases, whether permitted or not. For example, all of our petrochemical facilities in the United States and Europe may require improvements to comply with certain changes in process safety management requirements.

New laws, rules and regulations as well as changes to laws, rules and regulations may also affect us. For example, on April 17, 2012, the EPA promulgated maximum achievable control technology standards for major sources and generally available control technology standards for area sources of PVC production. The rule sets emission limits and work practice standards for total organic air toxics and for three specific air toxics: vinyl chloride, chlorinated di-benzo dioxins and furans and hydrogen chloride and includes requirements to demonstrate initial and continuous compliance with the emission standards. In June 2012, the EPA received petitions for reconsideration of the rule. On November 9, 2020, the EPA proposed rule amendments to address issues raised in the petitions for reconsideration. While this rule is the subject of legal challenge and EPA reconsideration, the rule has not been stayed. Although we cannot predict the outcome or timing of the legal challenges or EPA reconsideration, the EPA's proposed rule amendments could require us to incur further capital expenditures, or increase our operating costs, to levels higher than what we have previously estimated.

In April 2023, the EPA proposed amendments to new source performance standards for the synthetic organic chemical manufacturing industry and amendments to the national emissions standards for hazardous air pollutants for the synthetic organic chemical manufacturing industry and group I & II polymers and resins industry. These proposed amendments, among other things, would impose tougher emissions limits, additional leak detection and repair obligations, certain performance standard for the operation of flares at applicable facilities, and new fenceline air monitoring for several chemicals. Although we cannot predict the outcome or timing of EPA's final rule amendments, the amendments as proposed would require us to incur further capital expenditures and increase operating costs.

On May 6, 2022, the EPA finalized rules amending (i) the national emission standards for hazardous air pollutants ("NESHAPs") for mercury emissions from mercury cell chlor-alkali plants and (ii) the 2003 NESHAPs for mercury cell chlor-alkali plants residual risk and technology review. Among other things, the amendments require improvements in work practices to reduce fugitive mercury emissions and work practice standards for the cell rooms and instrumental monitoring of cell room fugitive emissions, modify regulatory provisions regarding startup, shutdown, and malfunctions, and add standards for fugitive chlorine emissions from mercury cell chlor-alkali plants, which are not currently regulated under the NESHAP. The final rules also include a requirement to cease all mercury emissions from the operation of mercury cell chlor-alkali facilities by May 6, 2025. We operate a mercury cell production unit at our Natrium facility. Compliance with the final rules resulted in additional restrictions on our operations, increased compliance costs and will result in the cessation of operation of the mercury cell production unit.

Our operations produce greenhouse gas ("GHG") emissions, which have been the subject of increased scrutiny and regulation. In December 2015, the United States joined the international community at the 21st Conference of the Parties of the United Nations Framework Convention on Climate Change in Paris, France. The resulting Paris Agreement calls for the parties to undertake "ambitious efforts" to limit the average global temperature and to conserve and enhance sinks and reservoirs of greenhouse gases. The United States signed the Paris Agreement in April 2016, and the Paris Agreement went into effect in November 2016. In November 2019, the United States submitted formal notification to the United Nations that it intended to withdraw from the Paris Agreement. The withdrawal took effect in November 2020. However, President Biden signed an executive order on January 20, 2021 for reentry of the United States into the Paris Agreement and on February 19, 2021, President Biden formally rejoined the Paris Agreement. As part of rejoining the Paris Agreement, President Biden announced that the United States would commit to a 50 to 52 percent reduction from 2005 levels of GHG emissions by 2030 and set the goal of reaching net-zero GHG emissions by 2050. To measure progress towards this target, the Paris Agreement requires the parties to complete a global stocktake, assessing members' collective efforts and achievements in reducing GHG emissions and adapting to the impacts of climate change, every five years. On December 13, 2023, the 28th annual United Nations Climate Change Conference ("COP 28"), which was held in Dubai, issued its first global stocktake, which calls on parties, including the United States, to contribute to the transitioning away from fossil fuels, reduce methane emissions, and increase renewable energy capacity, among other things, to achieve net zero emissions by 2050. Legislation to regulate GHG emissions has periodically been introduced in the United States Congress, and such legislation may be proposed or adopted in the future. There has been a wide-ranging policy debate regarding the impact of these gases and possible means for their regulation. Some of the proposals would require industries to meet stringent new standards that would require substantial reductions in carbon emissions. The adoption and implementation of any international, federal or state legislation or regulations that restrict emissions of GHGs could result in increased compliance costs or additional operating restrictions.

Various jurisdictions have considered or adopted laws and regulations on GHG emissions, with the general aim of reducing such emissions. The EPA currently requires certain industrial facilities to report their GHG emissions, and to obtain permits with stringent control requirements before constructing or modifying new facilities with significant GHG emissions. In the European Union, the Emissions Trading Scheme obligates certain emitters to obtain GHG emission allowances to comply with a cap and trade system for GHG emissions. In addition, the European Union has committed to reduce domestic GHG emissions by at least 57% below the 1990 level by 2030. Effective January 2023, the European Union expanded its sustainability-related reporting with its Corporate Sustainability Reporting Directive, which requires the reporting of Scope 1, 2, and 3 emissions and climate-related financial risks, in addition to a broad array of sustainability-related information. The SEC proposed similar climate-related disclosures in March 2022 and, in September of 2023, California passed climate-related disclosure mandates that are broader than the SEC's proposed rules. As our chemical manufacturing processes result in GHG emissions, these and other GHG laws and regulations could affect our costs of doing business.

Similarly, the Toxic Substances Control Act ("TSCA") imposes reporting, record-keeping and testing requirements, and restrictions relating to the production, handling, and use of chemical substances. The TSCA reform legislation enacted in June 2016 expanded the EPA's authority to review and regulate new and existing chemicals. Under the reform legislation, the EPA is required to, among other things, undertake rule making within statutory time frames related to: (1) chemical risk evaluation, designation and management; (2) reporting of mercury supply, use and trade; and (3) management of persistent, bioaccumulative, and toxic chemical substances ("PBTs"). In response to this mandate, the EPA issued rules establishing the EPA's process and criteria for identifying high priority chemicals for risk evaluation and setting the EPA's approach for determining whether these high priority chemicals present an unreasonable risk to health or the environment. Pursuant to its rules, the EPA designated certain chemical substances as high priority for risk evaluation. We manufacture several of these chemical substances. On December 14, 2023, the EPA announced that it will prioritize vinyl chloride (used to make PVC) for evaluation and potential regulation under TSCA as a High Priority Substance. In November 2023, the European Chemicals Agency ("ECHA") sent its investigation results of the risks from PVC and PVC additives to the European Commission for further consideration. The EPA has proposed risk management rules which would phase out the manufacturing, processing and distribution of TCE completely, PCE for consumer and most industrial and commercial uses, and asbestos for commercial use. Under the risk management rule process established by the TSCA, the EPA has also issued several Test Orders for chemical substances that we manufacture or import, including EDC. Although we cannot predict with certainty the extent of our future liabilities and costs at this time, we do not anticipate that the Test Order requirements or risk evaluation and management rules of these chemical substances will have a material adverse effect on our business, financial condition, operating results or cash flows.

In addition, the TSCA inventory reset rule required industry reporting of chemicals manufactured or processed in the United States over a 10-year period ending in 2016. This reporting is used by the EPA to identify which chemicals are active or inactive on the TSCA Inventory. Beginning in 2019, chemical manufacturers and processors are required to notify and obtain approval by the EPA before reintroducing inactive chemicals into commerce. A final mercury reporting rule published in June 2018 requires manufacturers, including manufacturers who use mercury in a manufacturing process, to report information about their mercury supply, use and trade. The first periodic reporting deadline under the mercury reporting rule was July 1, 2019. The EPA used the information collected to develop an inventory of mercury and mercury-added products as well as mercury-use manufacturing processes. The EPA also recommended actions and rule amendments based on the collected information. We cannot predict the timing or content of these actions or amendments, or their ultimate cost to, or impact on us.

On June 28, 2021, the EPA proposed reporting and recordkeeping requirements for Per- and Polyfluoroalkyl Substances ("PFAS") under TSCA. The EPA issued its final rule on September 28, 2023, which requires manufacturers, and importers, that have manufactured or imported PFAS chemicals since January 1, 2011, to electronically report information regarding PFAS uses, production volumes, disposal, exposures, and hazards. On September 6, 2022, the EPA proposed listing perfluorooctanoic acid ("PFOA") and perfluorooctane sulfonic acid (PFOS) as CERCLA hazardous substances. PFAS chemicals have come under increased scrutiny by federal, state, and local governments. For example, many states have banned the use of PFAS in certain consumer products and set Maximum Contaminant Levels for PFAS in drinking water. On March 14, 2023, the EPA announced a proposed National Primary Drinking Water Regulation for six PFAS, including PFOA, PFOS, perfluorononanoic acid (PFNA), hexafluoropropylene oxide dimer acid (HFPO-DA, commonly known as GenX Chemicals), perfluorohexane sulfonic acid (PFHxS), and perfluorobutane sulfonic acid (PFBS). On February 7, 2023, the ECHA published proposed restrictions on PFAS. We are unable to predict the impact these requirements and concepts may have on our future costs of compliance.

Under the IED, European Union member state governments are expected to adopt rules and implement environmental permitting programs relating to air, water and waste for industrial facilities. In this context, concepts such as the "best available technique" are being explored. Future implementation of these concepts may result in technical modifications in our European facilities. In addition, under the Environmental Liability Directive, European Union member states can require the remediation of soil and groundwater contamination in certain circumstances, under the "polluter pays principle." The European Chemical Agency is collecting information on PVC and its additives to determine whether further regulation is warranted. We are unable to predict the impact these requirements and concepts may have on our future costs of compliance.

Local, state, federal and foreign governments have increasingly proposed or implemented restrictions on certain plastic-based products, including single-use plastics and plastic food packaging. Plastics have also faced increased public scrutiny due to negative coverage of plastic waste in the environment. On January 12, 2023, the EPA published a tentative denial of the 2014 Center for Biological Diversity Petition to regulate discarded PVC as hazardous waste under the Resource Conservation and Recovery Act. Pursuant to a consent decree entered into by the EPA and the Center for Biological Diversity, the EPA has until April 12, 2024 to issue its final determination. We are unable to predict the EPA's final decision in this matter or its impact on us. However, increased regulation on the use of plastics could cause reduced demand for our polyethylene products, which could adversely affect our business, operating results and financial condition.

These rules or future new, amended or proposed laws or rules could increase our costs or reduce our production, which could have a material adverse effect on our business, financial condition, operating results or cash flows. In addition, we cannot accurately predict future developments, such as increasingly strict environmental and safety laws or regulations, and inspection and enforcement policies, as well as resulting higher compliance costs, which might affect the handling, manufacture, use, emission, disposal or remediation of products, other materials or hazardous and non-hazardous waste, and we cannot predict with certainty the extent of our future liabilities and costs under environmental, health and safety laws and regulations. These liabilities and costs may be material.

We also may face liability for alleged personal injury or property damage due to exposure to chemicals or other hazardous substances at our facilities or to chemicals that we otherwise manufacture, handle or own. Although these types of claims have not historically had a material impact on our operations, a significant increase in the success of these types of claims could have a material adverse effect on our business, financial condition, operating results or cash flows.

Environmental laws may have a significant effect on the nature and scope of, and responsibility for, cleanup of contamination at our current and former operating facilities, the costs of transportation and storage of raw materials and finished products, the costs of reducing emissions and the costs of the storage and disposal of wastewater. The U.S. Comprehensive Environmental Response, Compensation, and Liability Act ("CERCLA"), similar state laws and certain European directives impose joint and several liability for the costs of remedial investigations and actions on the entities that generated waste, arranged for disposal of the waste, transported to or selected the disposal sites, and the past and present owners and operators of such sites. All such potentially responsible parties (or any one of them, including us) may be required to bear all of such costs regardless of fault, legality of the original disposal or ownership of the disposal site. In addition, CERCLA, similar state laws and certain European directives could impose liability for damages to natural resources caused by contamination.

Although we seek to take preventive action, our operations are inherently subject to accidental spills, discharges or other releases of hazardous substances that may make us liable to governmental entities or private parties. This may involve contamination associated with our current and former facilities, facilities to which we sent wastes or by-products for treatment or disposal and other contamination. Accidental discharges may occur in the future, future action may be taken in connection with past discharges, governmental agencies may assess damages or penalties against us in connection with any past or future contamination, or third parties may assert claims against us for damages allegedly arising out of any past or future contamination. In addition, we may be liable for existing contamination related to certain of our facilities for which, in some cases, we believe third parties are liable in the event such third parties fail to perform their obligations.

Our operations and assets are subject to climate-related risks and uncertainties.

We are subject to increasing climate-related risks and uncertainties, many of which are outside of our control. Climate change may result in more frequent severe weather events, potential changes in precipitation patterns, flooding, sea level rise, and variability in weather patterns, which can disrupt our operations as well as those of our customers, partners and suppliers. Climate change may result in heightened hurricane activity in the Gulf of Mexico and other weather and natural disaster hazards that pose a risk to our facilities, particularly those in Louisiana. The transition to lower greenhouse gas emissions technology, the effects of carbon pricing, changes in public sentiment, regulations, taxes, public mandates or requirements, increases in climate-related lawsuits, insurance premiums and implementation of more robust disaster recovery and business continuity plans may increase costs to maintain or resume our operations, which could in turn negatively impact our business and results of operations.

We are subject to operational and financial risks and liabilities associated with the implementation of and efforts to achieve our carbon emission reduction goals.

We have publicly announced a GHG emissions reduction target for 2030. We believe that our expected allocation of growth capital into lower-carbon projects is consistent with such targets; however, our analysis and plan for execution requires us to make a number of assumptions. These goals and underlying assumptions involve risks and uncertainties and are not guarantees. Should one or more of our underlying assumptions prove incorrect, our actual results and ability to achieve our GHG emissions reduction goal could differ materially from our expectations. For example, certain lower-carbon projects that we plan to make investments in are currently at various stages of progress, evaluation or approval.

Developing and implementing plans for compliance with voluntary climate commitments can lead to additional capital, personnel, operations and maintenance expenditures and could significantly affect the economic position of existing facilities and proposed projects. Our failure or perceived failure to pursue or fulfill our sustainability-focused goals, targets and objectives within the timelines we announce, or at all, could adversely affect our business or reputation, as well as expose us to potential government enforcement actions and private litigation. We cannot predict the ultimate impact of achieving our emissions reduction goal, or the various implementation aspects, on our financial condition and results of operations.

Risks Related to Our Indebtedness

Our level of debt could adversely affect our ability to operate our business.

As of December 31, 2023, our indebtedness, including the current portion, totaled $4.9 billion, and our debt represented approximately 31.3% of our total capitalization. Our annual interest expense for 2023 was $165.0 million, net of interest capitalized of $8.0 million. Our level of debt and the limitations imposed on us by our existing or future debt agreements could have significant consequences on our business and future prospects, including the following:

- a portion of our cash flows from operations will be dedicated to the payment of interest and principal on our debt and will not be available for other purposes;
- we may not be able to obtain necessary financing in the future for working capital, capital expenditures, acquisitions, debt service requirements or other purposes;
- our less leveraged competitors could have a competitive advantage because they have greater flexibility to utilize their cash flows to improve their operations;
- we may be exposed to risks inherent in interest rate fluctuations because some of our borrowings are at variable rates of interest, which would result in higher interest expense in the event of increases in interest rates;
- we could be vulnerable in the event of a downturn in our business that would leave us less able to take advantage of significant business opportunities and to react to changes in our business and in market or industry conditions; and
- should we pursue additional expansions of existing assets or acquisition of third-party assets, we may not be able to obtain additional liquidity at cost effective interest rates.

These factors could be magnified or accelerated to the extent we were to finance future acquisitions with significant amounts of debt.

To service our indebtedness and fund our capital requirements, we will require a significant amount of cash. Our ability to generate cash depends on many factors beyond our control.

Our ability to make payments on and to refinance our indebtedness and to fund planned capital expenditures and pay cash dividends will depend on our ability to generate cash in the future, including any distributions that we may receive from Westlake Partners. This is subject to general economic, financial, currency, competitive, legislative, regulatory and other factors that are beyond our control.

Our business may not generate sufficient cash flows from operations, we may not receive sufficient distributions from Westlake Partners, and currently anticipated cost savings and operating improvements may not be realized on schedule. We also generate revenues denominated in currencies other than that of our indebtedness and may have difficulty converting those revenues into the currency of our indebtedness. We may need to refinance all or a portion of our indebtedness on or before maturity. In addition, we may not be able to refinance any of our indebtedness, including our credit facility and our senior notes, on commercially reasonable terms or at all. All of these factors could be magnified if we were to finance any future acquisitions with significant amounts of debt.

The Credit Agreement and the indenture governing certain of our senior notes impose significant operating and financial restrictions, which may prevent us from capitalizing on business opportunities and taking some actions.

The Credit Agreement and the indenture governing certain of our senior notes impose significant operating and financial restrictions on us. These restrictions limit our ability to:

- incur additional indebtedness;
- create liens;
- sell all or substantially all of our assets or consolidate or merge with or into other companies; and
- engage in sale-leaseback transactions.

These limitations are subject to a number of important qualifications and exceptions. The Credit Agreement also requires us to maintain a quarterly total leverage ratio.

These covenants may adversely affect our ability to finance future business opportunities or acquisitions. A breach of any of these covenants could result in a default in respect of the related debt. If a default occurred, the relevant lenders could elect to declare the debt, together with accrued interest and other fees, to be immediately due and payable. In addition, any acceleration of debt under the Credit Agreement will constitute a default under some of our other debt, including the indentures governing our senior notes.

Risks Related to Taxes

A change in tax laws, treaties or regulations, or their interpretation or application, could have a negative impact on our business and results of operations.

We operate in many different countries and in many states within the United States, and we are subject to changes in applicable tax laws, treaties or regulations in the jurisdictions in which we operate. A material change in these tax laws, treaties or regulations, or their interpretation or application, could have a negative impact on our business and results of operations.

On August 16, 2022, the Inflation Reduction Act of 2022 (the "IRA") was signed into law. The IRA contains several revisions to the Internal Revenue Code of 1986, as amended (the "Code"), including a 15% corporate minimum income tax for corporations with average annual adjusted financial statement income over a three-tax-year period in excess of $1 billion and is effective for the tax years beginning after December 31, 2022, a 1% excise tax on stock repurchases made by publicly traded U.S. corporations after December 31, 2022, and business tax credits and incentives for the development of clean energy projects and the production of clean energy. At this time, we do not expect the IRA will have a material impact on our consolidated financial statements. We continue to evaluate its impacts as new information and guidance becomes available.

We depend in part on distributions from Westlake Partners to generate cash for our operations, capital expenditures, debt service and other uses. Westlake Partners' tax treatment depends on its status as a partnership for federal income tax purposes, and it not being subject to a material amount of entity-level taxation. If the Internal Revenue Service ("IRS") were to treat Westlake Partners as a corporation for federal income tax purposes, or if Westlake Partners became subject to entity-level taxation for state tax purposes, its cash available for distribution would be substantially reduced, which would also likely cause a substantial reduction in the value of its common units that we hold.

The anticipated after-tax economic benefit of an investment in the common units of Westlake Partners depends largely on Westlake Partners being treated as a partnership for U.S. federal income tax purposes. Despite the fact that Westlake Partners is organized as a limited partnership under Delaware law, it would be treated as a corporation for U.S. federal income tax purposes unless it satisfies a "qualifying income" requirement. Based on Westlake Partners' current operations and current Treasury Regulations, Westlake Partners believes it satisfies the qualifying income requirement.

Prior to its initial public offering, Westlake Partners requested and obtained a favorable private letter ruling from the IRS to the effect that, based on facts presented in the private letter ruling request, income from the production, transportation, storage and marketing of ethylene and its co-products constitutes "qualifying income" within the meaning of Section 7704 of the Code. However, no ruling has been or will be requested regarding Westlake Partners' treatment as a partnership for U.S. federal income tax purposes. Failing to meet the qualifying income requirement or a change in current law could cause Westlake Partners to be treated as a corporation for U.S. federal income tax purposes or otherwise subject Westlake Partners to taxation as an entity.

If Westlake Partners were treated as a corporation for federal income tax purposes, it would pay U.S. federal income tax on its taxable income at the corporate tax rate. Because a tax would be imposed upon Westlake Partners as a corporation, its cash available for distribution to its unitholders would be substantially reduced. Therefore, treatment of Westlake Partners as a corporation would result in a material reduction in the anticipated cash flow and after-tax return to its unitholders, likely causing a substantial reduction in the value of its common units.

Westlake Partners' partnership agreement provides that if a law is enacted or an existing law is modified or interpreted in a manner that subjects Westlake Partners to taxation as a corporation or otherwise subjects Westlake Partners to entity-level taxation for U.S. federal, state, local or foreign income tax purposes, the minimum quarterly distribution amount and the target distribution amounts may be adjusted to reflect the impact of that law or interpretation on Westlake Partners. At the state level, several states have been evaluating ways to subject partnerships to entity-level taxation through the imposition of state income, franchise or other forms of taxation. Westlake Partners currently owns assets and conducts business in several states, most of which impose entity-level franchise or gross receipt taxes on partnerships. Imposition of similar entity-level taxes on Westlake Partners in other jurisdictions in which Westlake Partners conducts operations in the future could substantially reduce its cash available for distribution.

Risks Related to the Ownership of Our Securities

We will be controlled by our principal stockholder and its affiliates as long as they own a majority of our common stock, and our other stockholders will be unable to affect the outcome of stockholder voting during that time. Our interests may conflict with those of the principal stockholder and its affiliates, and we may not be able to resolve these conflicts on terms possible in arms-length transactions.

As long as TTWF LP (the "principal stockholder") and certain of its affiliates (such affiliates, together with the principal stockholder, the "principal stockholder affiliates"), which as of December 31, 2023, beneficially owned approximately 72% of our common stock, own a majority of our outstanding common stock, they will be able to exert significant control over us, and our other stockholders, by themselves, will not be able to affect the outcome of any stockholder vote. As a result, the principal stockholder, subject to any fiduciary duty owed to our minority stockholders under Delaware law, will be able to control all matters affecting us (some of which may present conflicts of interest), including:

- the composition of our Board of Directors and, through the Board, any determination with respect to our business direction and policies, including the appointment and removal of officers and the determination of compensation;
- any determinations with respect to mergers or other business combinations or the acquisition or disposition of assets;
- our financing decisions, capital raising activities and the payment of dividends; and
- amendments to our restated certificate of incorporation or amended and restated bylaws.

The principal stockholder will be permitted to transfer a controlling interest in us without being required to offer our other stockholders the ability to participate or realize a premium for their shares of common stock. A sale of a controlling interest to a third party may adversely affect the market price of our common stock and our business and results of operations because the change in control may result in a change of management decisions and business policy. Because we have elected not to be subject to Section 203 of the General Corporation Law of the State of Delaware, the principal stockholder may find it easier to sell its controlling interest to a third party than if we had not so elected.

In addition to any conflicts of interest that arise in the foregoing areas, our interests may conflict with those of the principal stockholder affiliates in a number of other areas, including:

- business opportunities that may be presented to the principal stockholder affiliates and to our officers and directors associated with the principal stockholder affiliates, and competition between the principal stockholder affiliates and us within the same lines of business;
- the solicitation and hiring of employees from each other; and
- agreements with the principal stockholder affiliates relating to corporate services that may be material to our business.

We may not be able to resolve any potential conflicts with the principal stockholder affiliates, and even if we do, the resolution may be less favorable than if we were dealing with an unaffiliated party, particularly if the conflicts are resolved while we are controlled by the principal stockholder affiliates. Our restated certificate of incorporation provides that the principal stockholder affiliates have no duty to refrain from engaging in activities or lines of business similar to ours and that the principal stockholder affiliates will not be liable to us or our stockholders for failing to present specified corporate opportunities to us.

Item 1B. Unresolved Staff Comments

None.

Item 1C. Cybersecurity

The Company maintains a comprehensive approach to cybersecurity and data protection, based on a risk-based, defense-in-depth strategy. We regularly assess industry best practices and standards and endeavor to implement them in our efforts to manage cybersecurity risk. We follow industry standard cybersecurity frameworks, including the National Institute of Standards and Technology's Cybersecurity Framework, to design, assess and update our cybersecurity strategy, controls and processes. Our focus is on protecting our highest-value information assets, which include manufacturing systems, financial systems, and confidential, personal, and private information.

To safeguard our networks and systems, we have a dedicated cybersecurity organization overseen by our Chief Information Security Officer, which operates within our information technology department overseen by our Chief Information Officer. Our cybersecurity organization employs multiple security controls, such as firewalls, spam protection, web filtering, endpoint detection and response software, controlled access, vulnerability management, redundancies, patching, and regular onsite and offsite backups. Our cybersecurity organization also uses a variety of processes to address cybersecurity threats related to the use of third-party technology and services, including pre-acquisition diligence, imposition of contractual obligations, and risk-based performance monitoring.

Both our Chief Information Officer and our Chief Information Security Officer have extensive experience in assessing and managing cybersecurity risks, including decades of collective experience in information technology and cybersecurity roles of increasing responsibility both at the Company and in prior positions. We prioritize cybersecurity awareness among our employees and contractors through various training exercises, including formal programs and simulated phishing events. We maintain incident response plans, playbooks, and engage third-party cybersecurity firms for simulated cyberattacks and penetration testing to identify potential risks. We also have a third-party cybersecurity firm on retainer for incident assistance and response. Periodic internal self-assessments are conducted by our cybersecurity organization using the National Institute of Standards and Technology Cybersecurity Framework.

From time to time, we experience cybersecurity threats and attempted breaches and other incidents. We classify and track these events based on significance and implement remediation actions that we consider appropriate to address the risks relating to such incidents. Although we have not experienced material impacts to our business strategy, results of operations or financial condition from any such incidents in the past three years, we cannot guarantee that a material incident will not occur in the future. Refer to "*Failure to adequately protect critical data and technology systems could materially affect our operations*" under "Item 1A. Risk Factors—Legal, Government and Regulatory Risks" of this Form 10-K.

Our Board has charged the Corporate Risk and Sustainability Committee with assisting the Board with its oversight of cybersecurity risks, which is a component of our overall enterprise risk management program. The Corporate Risk and Sustainability Committee includes directors with cybersecurity experience and expertise, primarily through supervision of information technology departments as executive officers. The Corporate Risk and Sustainability Committee receives regular updates from senior management and our Chief Information Officer on cybersecurity risks, incidents and trends, and ongoing and planned projects. Regular status reports are also provided by the cybersecurity organization to our Chief Information Officer and other members of our senior management and incident updates are reported to senior management as the Chief Information Officer and cybersecurity organization considers appropriate depending on the severity of the incident.

As part of our incident response planning, we also maintain cross-functional response teams involving personnel outside of our cybersecurity organization, both globally and regionally, in order to be prepared to respond to an incident.

Item 2. Properties

Information concerning the principal locations from which our products are manufactured are included in the tables set forth under the captions "Performance and Essential Materials Business—*Products*" and "Housing and Infrastructure Products Business—*Products*" contained in "Item 1. Business."

Headquarters

Our principal executive offices are located in Houston, Texas. Some of our office space is leased, at market rates, from an affiliate of our principal stockholder. See Note 20 to the consolidated financial statements appearing elsewhere in this Form 10-K and "Certain Relationships and Related Party Transactions" in our proxy statement to be filed with the SEC pursuant to Regulation 14A with respect to our 2024 annual meeting of stockholders (the "Proxy Statement").

Item 3. Legal Proceedings

In addition to the matters described under "Item 1. Business—Environmental" and Note 22 to our consolidated financial statements included in Item 8 of this Form 10-K, we are involved in various legal proceedings incidental to the conduct of our business. We do not believe that any of these legal proceedings will have a material adverse effect on our financial condition, results of operations or cash flows.

Item 4. Mine Safety Disclosure

Not Applicable.

Information about our Executive Officers

James Y. Chao (age 76). Mr. Chao has been our Chairman of the Board of Directors since July 2004 and became a director in June 2003. From May 1996 to July 2004, he served as our Vice Chairman. Mr. Chao has over 45 years of global experience in the chemical industry. In addition, Mr. Chao has been the Chairman of the Board of Westlake Partners' general partner since its formation in March 2014. From June 2003 until November 2010, Mr. Chao was the executive chairman of Titan Chemicals Corp. Bhd. He has served as a Special Assistant to the Chairman of China General Plastics Group and worked in various financial, managerial and technical positions at Mattel Incorporated, Developmental Bank of Singapore, Singapore Gulf Plastics Pte. Ltd. and Gulf Oil Corporation. Mr. Chao, along with his brother Albert Chao, assisted their father T.T. Chao in founding Westlake Corporation (formerly known as Westlake Chemical Corporation). He is the brother of Albert Y. Chao, father of Catherine T. Chao and David T. Chao and uncle of John T. Chao and Carolyn C. Sabat. Mr. Chao received his B.S. degree from Massachusetts Institute of Technology and an M.B.A. from Columbia University.

Albert Y. Chao (age 74). Mr. Chao has been our President since May 1996 and a director since June 2003. Mr. Chao became our Chief Executive Officer in July 2004. Mr. Chao has over 40 years of global experience in the chemical industry. In 1985, Mr. Chao assisted his father T.T. Chao and his brother James Chao in founding Westlake Corporation (formerly known as Westlake Chemical Corporation), where he served as Executive Vice President until he succeeded James Chao as President. In addition, Mr. Chao has been the President, Chief Executive Officer and a director of Westlake Partners' general partner since its formation in March 2014. He has held positions in the Controller's Group of Mobil Oil Corporation, in the Technical Department of Hercules Incorporated, in the Plastics Group of Gulf Oil Corporation and has served as Assistant to the Chairman of China General Plastics Group and Deputy Managing Director of a plastics fabrication business in Singapore. He is the brother of James Y. Chao, father of John T. Chao and Carolyn C. Sabat and uncle of Catherine T. Chao and David T. Chao. Mr. Chao received a bachelor's degree from Brandeis University and an M.B.A. from Columbia University.

M. Steven Bender (age 67). Mr. Bender has been our Executive Vice President and Chief Financial Officer since July 2017. From February 2008 to July 2017, Mr. Bender served as our Senior Vice President and Chief Financial Officer. In addition, Mr. Bender served as our Treasurer from July 2011 to April 2017, a position he also held from February 2008 until December 2010. From February 2007 to February 2008, Mr. Bender served as our Vice President, Chief Financial Officer and Treasurer and from June 2005 to February 2007, he served as our Vice President and Treasurer. In addition, Mr. Bender has been a director of Westlake Partners' general partner since its formation in March 2014, its Executive Vice President and Chief Financial Officer since February 2021, and its Senior Vice President and Chief Financial Officer from March 2014 to February 2021. Prior to joining Westlake, from June 2002 until June 2005, Mr. Bender served as Vice President and Treasurer of KBR, Inc., and from 1996 to 2002 he held the position of Assistant Treasurer for Halliburton Company. Prior to that, he held various financial positions within that company. Additionally, he was employed by Texas Eastern Corporation for over a decade in a variety of increasingly responsible audit, finance and treasury positions. Mr. Bender received a Bachelor of Business Administration from Texas A&M University and an M.B.A. from Southern Methodist University. Mr. Bender is also a Certified Public Accountant.

Robert F. Buesinger (age 67). Mr. Buesinger has been our Executive Vice President, Housing and Infrastructure Products, IT and Digital since February 2022. From July 2017 to February 2022, Mr. Buesinger served as our Executive Vice President, Vinyl Products and, from April 2010 to July 2017, he served as our Senior Vice President, Vinyls. Prior to joining us, Mr. Buesinger served as the General Manager and President of Chevron Phillips Chemical Company L.P.'s Performance Pipe Division from February 2010 to March 2010. From June 2008 to January 2010, Mr. Buesinger held the position of General Manager in the Alpha Olefins and Poly Alpha Olefins business of Chevron Phillips Chemical Company L.P. From April 2005 to May 2008, he served as the President and Managing Director of Chevron Phillips Singapore Chemicals Pte. Ltd. and Asia Region General Manager for Chevron Phillips Chemical Company L.P. Prior to that, he held various technical and sales management positions within that company. Mr. Buesinger holds a B.S. in Chemical Engineering from Tulane University.

L. Benjamin Ederington (age 53). Mr. Ederington has been our Executive Vice President, Performance and Essential Materials, General Counsel and Chief Administrative Officer since April 2023. From February 2022 to April 2023, Mr. Ederington served as our Executive Vice President, General Counsel, Chief Administrative Officer and Corporate Secretary; from July 2017 to February 2022; he served as our Senior Vice President, General Counsel, Chief Administrative Officer and Corporate Secretary; from December 2015 to July 2017, he served as our Vice President, General Counsel, Chief Administrative Officer and Corporate Secretary; and, from October 2013 to December 2015, he served as our Vice President, General Counsel and Corporate Secretary. In addition, Mr. Ederington has been a director of Westlake Partners' general partner since its formation in March 2014 and its Executive Vice President, Performance and Essential Materials, General Counsel and Chief Administrative Officer since May 2023. Mr. Ederington was Westlake Partners' general partner's Executive Vice President, General Counsel, Chief Administrative Officer and Corporate Secretary from March 2022 to May 2023; Senior Vice President, General Counsel, Chief Administrative Officer and Corporate Secretary from February 2021 to March 2022; and, its Vice President, General Counsel and Secretary from March 2014 to February 2021. Prior to joining Westlake, he held a variety of senior legal positions at LyondellBasell Industries, N.V. and its predecessor companies, LyondellBasell Industries AF SCA and Lyondell Chemical Company, including most recently as Associate General Counsel, Commercial & Strategic Transactions. He began his legal career more than 25 years ago at the law firm of Steptoe & Johnson, LLP. Mr. Ederington holds a B.A. from Yale University and received his J.D. from Harvard University.

Thomas J, Janssens (age 58). Mr. Janssens has been our Senior Vice President, Operations—Performance and Essential Material & Corporate Logistics since March 2022. From January 2021 to March 2022, Mr. Janssens served as our Vice President, Olefins, Feedstocks & Energy; from December 2019 to December 2020, he was our Vice President, Olefins & Logistics; from July 2017 to November 2019, he was our Vice President, Corporate Development, Logistics & IT; from January 2016 to June 2017, he was our Vice President, Logistics & IT; and, from October 2015 to December 2015, he was our Vice President, Logistics & Business Process Improvement. Prior to joining Westlake, Mr. Janssens was a consultant, from September 2002 to June 2009, and later, a Partner, from July 2009 to September 2015, with McKinsey & Company, where he advised energy and chemicals clients on strategic, commercial, operational and business process improvement projects. He began his career with Shell International in 1991, where he held a variety of commercial, engineering and planning roles. Mr. Janssens holds a MSc in Chemical Engineering from Eindhoven University of Technology and an MBA from the University of Chicago.

Johnathan S. Zoeller (age 48). Mr. Zoeller has been our Vice President and Chief Accounting Officer since March 2020. From August 2018 to March 2020, Mr. Zoeller served as our Vice President and Corporate Controller. In addition, Mr. Zoeller has been the Vice President and Chief Accounting Officer of Westlake Partners' general partner since March 2020. Mr. Zoeller joined us with over 19 years of public accounting experience, the majority of which was spent at KPMG LLP, where he was responsible for clients in the chemicals, oilfield services and oil/gas exploration and production industries. Mr. Zoeller held a variety of senior accounting positions at KPMG, including most recently as Partner, Audit from October 2011 to August 2018. He began his career with Arthur Andersen LLP in 1998. Mr. Zoeller holds a Bachelor of Accounting degree and a Master of Accounting degree from the University of Mississippi. He is a Certified Public Accountant.

PART II

Item 5. Market for Registrant's Common Equity, Related Stockholder Matters and Issuer Purchases of Equity Securities

Stockholder Matters

As of February 14, 2024, there were 30 holders of record of our common stock. Our common stock is listed on the New York Stock Exchange under the symbol "WLK."

Unregistered Sales of Equity Securities

We did not have any unregistered sales of equity securities during the quarter or fiscal year ended December 31, 2023 that we have not previously reported on a Quarterly Report on Form 10-Q or a Current Report on Form 8-K.

Issuer Purchases of Equity Securities

The following table provides information on our purchase of equity securities during the quarter ended December 31, 2023:

Period	Total Number of Shares Purchased [1]	Average Price Paid Per Share	Total Number of Shares Purchased as Part of Publicly Announced Plans or Programs [2]	Maximum Number (or Approximate Dollar Value) of Shares that May Yet Be Purchased Under the Plans or Programs [2]
October 2023	3,582	$ 122.54	—	$ 476,162,426
November 2023	—	—	—	476,162,426
December 2023	—	—	—	476,162,426
Total	3,582	$ 122.54	—	

(1) Represents 3,582 shares withheld in October 2023 in satisfaction of withholding taxes due upon the vesting of restricted stock units granted to our employees under the 2013 Omnibus Incentive Plan.

(2) In November 2014, our Board of Directors authorized a $250 million stock repurchase program (the "2014 Program") with no expiration date. In November 2015, our Board of Directors approved the expansion of the 2014 Program by an additional $150 million. In August 2018, our Board of Directors approved the further expansion of the existing 2014 Program by an additional $150 million. In August 2022, our Board of Directors approved the further expansion of the existing 2014 Program by an additional $500 million. As of December 31, 2023, 8,722,550 shares of our common stock had been acquired at an aggregate purchase price of approximately $574 million under the 2014 Program. Transaction fees and commissions are not reported in the average price paid per share in the table above. Decisions regarding the amount and the timing of purchases under the 2014 Program will be influenced by our cash on hand, our cash flows from operations, general market conditions and other factors. The 2014 Program may be discontinued by our Board of Directors at any time.

Item 6. [Reserved]

Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations

Overview

We are a vertically integrated global manufacturer and marketer of performance and essential materials and housing and infrastructure products. We operate in two principal operating segments, Performance and Essential Materials and Housing and Infrastructure Products. The Performance and Essential Materials segment includes Westlake North American Vinyls, Westlake North American Chlor-alkali & Derivatives, Westlake European & Asian Chlorovinyls, Westlake Olefins, Westlake Polyethylene and Westlake Epoxy. The Housing and Infrastructure Products segment includes Westlake Royal Building Products, Westlake Pipe & Fittings, Westlake Global Compounds and Westlake Dimex. We are highly integrated along our materials chain with significant downstream integration from ethylene and chlor-alkali (chlorine and caustic soda) into vinyls, polyethylene, epoxy and styrene monomer. We also have substantial downstream integration from polyvinyl chloride ("PVC") into our building products, PVC pipes and fittings and PVC compounds in our Housing and Infrastructure Products segment.

During 2022 and continuing through the end of 2023, our European businesses have been impacted by higher energy prices, inflation and reduced demand. Our North American businesses, where we derive a significant portion of our revenue, have also experienced the impacts of high energy costs and interest rates and slower demand for most of our products since the second half of 2022. However, in recent months we have experienced lower energy costs as prices for electricity and natural gas have declined following the elevated 2022 levels and inflation appears to be easing. In the near term, we expect that the volatility in energy prices, higher interest rates, inflation and other macroeconomic conditions will continue to impact margins and demand for most of our products.

On February 1, 2022, we completed the acquisition of the global epoxy business of Hexion Inc. ("Westlake Epoxy") for total consideration of $1,207 million. The assets acquired and liabilities assumed and the results of operations of Westlake Epoxy are included in the Performance and Essential Materials segment. During the fourth quarter of 2023, the Westlake Epoxy business's sales volumes and prices, specifically base epoxy resins in Europe, continued to deteriorate. These lower sales volumes and prices have been primarily driven by record exports at lower prices of bisphenol-A, epichlorohydrin and base epoxy resins (constituting the epoxy value chain) out of Asia into Europe and North America during the time when demand in the European market was contracting. In addition, Westlake Epoxy operations in Europe have experienced sustained high energy and power costs. These factors negatively impacted Westlake Epoxy financial results during 2023. Based on these developments, along with management's outlook for the Westlake Epoxy business we determined, in the fourth quarter of 2023, that the carrying amount of long-lived assets of our base epoxy resin business in the Netherlands and all of the goodwill of the Westlake Epoxy business will not be recoverable in the foreseeable future. As a result of this assessment, a goodwill impairment charge of $128 million and a non-cash long-lived asset impairment charge of $347 million, related to our base epoxy resin business in the Netherlands, were recognized in the fourth quarter of 2023.

Performance and Essential Materials

Our performance and essential materials such as ethylene, PVC, polyethylene, epoxy and caustic soda are some of the most widely used materials in the world and are upgraded into a wide variety of higher value-added products used in many end-markets. Westlake is the second-largest chlor-alkali producer and the second-largest PVC producer in the world, which makes Westlake a leading global chlorovinyls producer. Our performance and essential materials are used by customers in food and specialty packaging; industrial and consumer packaging; medical health applications; PVC pipe applications; consumer durables; mobility and transportation; renewable wind energy; coatings; and housing and construction products. Chlor-alkali and petrochemicals are typically manufactured globally in large volume by a number of different producers using widely available technologies. The chlor-alkali and petrochemical industries exhibit cyclical commodity characteristics, and margins are influenced by changes in the balance between supply and demand and the resulting operating rates, the level of general economic activity and the price of raw materials. Due to the significant size of new plants, capacity additions are built in large increments and typically require several years of demand growth to be absorbed. The cycle is generally characterized by periods of tight supply, leading to high operating rates and margins, followed by a decline in operating rates and margins primarily as a result of excess new capacity additions. Westlake is one of the leading producers of epoxy specialty resins, modifiers and curing agents in Europe, the United States and Asia, with a global reach to our end markets. Epoxy resins are the fundamental component of many types of materials and are often used in the automotive, construction, wind energy, aerospace and electronics industries due to their superior adhesion, strength and durability. We are also a leading supplier of Liquid and Solid Epoxy Resin. These base epoxies are used in a wide variety of industrial coatings applications.

Global demand for most of our products started to recover from the effects of the COVID-19 pandemic in the second half of 2020 and remained strong through the first half of 2022. However, since the second half of 2022 and continuing through the end of 2023, we saw significant volatility in natural gas and electricity prices, particularly in Europe, as well as in ethane and ethylene prices. We have also experienced lower prices and reduced demand for most of our products globally. The ongoing conflict between Russia and Ukraine since Russia's invasion of Ukraine in 2022, conflict in the Middle East, slow economic growth in China, increase in bisphenol-A, epichlorohydrin and base epoxy resin exports out of Asia into European and North American markets, timing of certain new ethylene and polyethylene capacity additions in North America, Asia, and the Middle East, volatility in natural gas and electricity prices, volatility in crude oil prices and inflationary pressures could have a continuing negative impact on the performance of Performance and Essential Materials businesses.

We purchase significant amounts of ethane feedstock, natural gas, ethylene and salt from external suppliers for use in production of performance and essential materials. We also purchase significant amounts of electricity to supply the energy required in our production processes. While we have agreements providing for the supply of ethane feedstock, natural gas, ethylene, salt and electricity, the contractual prices for these raw materials and energy vary with market conditions and may be highly volatile. Factors that have caused volatility in our raw material prices in the past, and which may do so in the future include:

- the availability of feedstock from shale gas and oil drilling;
- supply and demand for crude oil and natural gas;
- shortages of raw materials due to increasing demand;
- ethane and liquefied natural gas exports;
- capacity constraints due to higher construction costs for investments, construction delays, strike action or involuntary shutdowns;
- the general level of business and economic activity; and
- the direct or indirect effect of governmental regulation.

Significant volatility in raw material costs tends to put pressure on product margins as sales price increases could lag behind raw material cost increases. Conversely, when raw material costs decrease, customers may seek immediate relief in the form of lower sales prices. We currently use derivative instruments to reduce price volatility risk on feedstock commodities and lower overall costs. Normally, there is a pricing relationship between a commodity that we process and the feedstock from which it is derived. When this pricing relationship deviates from historical norms, we have from time to time entered into derivative instruments and physical positions in an attempt to take advantage of this relationship.

Our historical results have been significantly affected by our plant production capacity, our efficient use of that capacity and our ability to increase capacity. Since our inception, we have followed a disciplined growth strategy that focuses on plant acquisitions, new plant construction and internal expansion. We evaluate each expansion project on the basis of its ability to produce sustained returns in excess of our cost of capital and its ability to improve efficiency or reduce operating costs. We also regularly look at acquisition opportunities that would be consistent with, or complimentary to, our overall business strategies. Depending on the size of the acquisition, any such acquisitions could require external financing.

As noted above in Item 1A, "Risk Factors," we are subject to extensive environmental regulations, which may impose significant additional costs on our operations in the future. Further, concerns about greenhouse gas emissions and their possible effects on climate change has led to the enactment of regulations, and to proposed legislation and additional regulations, that could affect us in the form of increased cost of feedstocks and fuel, other increased costs of production and decreased demand for our products. While we do not expect any of these enactments or proposals to have a material adverse effect on us in the near term, we cannot predict the longer-term effect of any of these regulations or proposals on our future financial condition, results of operations or cash flows.

Housing and Infrastructure Products

Our Housing and Infrastructure Products segment is primarily comprised of building products, PVC pipes and fittings and PVC compound products. Our sales are affected by the individual decisions of distributors and dealers on the levels of inventory they carry, their views on product demand, their financial condition and the manner in which they choose to manage inventory risk. A significant portion of our performance in this segment is driven by the activities in the residential construction and repair and remodeling markets in North America, which began to decline at the end of the second quarter of 2022 primarily due to the negative effect that rising mortgage rates in the United States had on buyer sentiment. Performance of our housing and infrastructure products businesses over time are generally reflective of the trends of building permits and housing starts in the New Residential Construction Survey by the U.S. Census Bureau and the Repair and Remodeling Index (RRI) provided by the National Association of Home Builders (the "NAHB") among others. Although we ultimately expect that the Infrastructure Investment and Jobs Act of 2021 and historically low residential housing construction may have a favorable long-term impact on certain industries related to our Housing and Infrastructure Products segment, the current inflationary environment impacting consumer spending and priorities, the possibility of recession or financial market instability, rapidly rising interest rates, decade high mortgage interest rates impacting consumer affordability, and ongoing labor and supply chain constraints are expected to have an unfavorable impact on the demand for housing construction in the near term and, as a result, our products produced by this segment. The lower demand for new housing construction as a result of the factors discussed above, however, could encourage homeowners to remodel and repair existing homes, which could have a favorable impact on our product sales

The following table presents annual historical housing starts per the U.S. Census Bureau and the 2024 and 2025 outlook per the NAHB:

Period	Single and Multi-family Housing Starts (in thousands of units)	% Change
2021	1,601	16%
2022	1,553	(3)%
2023	1,420	(9)%
2024 Outlook	1,366	
2025 Outlook	1,417	

North American PVC facilities within the Performance and Essential Materials segment supply most of the PVC required for our building products and PVC pipes and fittings plants. Our raw materials for stone, roofing and accessories, windows, shutters and specialty tool products are externally purchased. PVC required for the PVC compounds plants is either internally sourced from our North American and Asian facilities within the Performance and Essential Materials segment or externally purchased based on the location of the plants. The remaining feedstocks required, including pigments, fillers, stabilizers and other ingredients, are purchased under short-term contracts based on prevailing market prices.

Factors that have caused volatility in our raw material prices, energy costs and production processes in the past, and which may do so in the future, include significant fluctuation in prices of these raw materials in response to, among other things, variable worldwide supply and demand across different industries, speculation in commodities futures, general economic, business or environmental conditions, labor costs, competition, import duties, tariffs, worldwide currency fluctuations, freight, inflationary pressures, regulatory costs, and product and process evolutions that impact demand for the same materials. Increasing raw material prices directly impact our cost of sales and our ability to maintain margins depends on implementing price increases in response to increasing raw material costs. The market for our products may or may not accept price increases, and as such, our future financial condition, results of operations or cash flows could be materially impacted.

Non-GAAP Financial Measures

The body of accounting principles generally accepted in the United States is commonly referred to as "GAAP." For this purpose, a non-GAAP financial measure is generally defined by the Securities and Exchange Commission ("SEC") as one that purports to measure historical or future financial performance, financial position or cash flows that (1) excludes amounts, or is subject to adjustments that have the effect of excluding amounts, that are included in the most directly comparable measure calculated and presented in accordance with GAAP in the statement of income, balance sheet or statement of cash flows (or equivalent statements) of the registrant; or (2) includes amounts, or is subject to adjustments that have the effect of including amounts, that are excluded from the most directly comparable measure so calculated and presented. In this report, we disclose non-GAAP financial measures, primarily earnings before interest, taxes, depreciation and amortization ("EBITDA") and Free Cash Flow. We define EBITDA as net income before interest expense, income taxes, depreciation and amortization. We define Free Cash Flow as net cash provided by operating activities less additions to property, plant and equipment. The non-GAAP financial measures described in this Form 10-K are not substitutes for the GAAP measures of earnings and cash flows.

EBITDA is included in this Form 10-K because our management considers it an important supplemental measure of our performance and believes that it is frequently used by securities analysts, investors and other interested parties in the evaluation of companies in our industry, some of which present EBITDA when reporting their results. We regularly evaluate our performance as compared to other companies in our industry that have different financing and capital structures and/or tax rates by using EBITDA. In addition, we utilize EBITDA in evaluating acquisition targets. Management also believes that EBITDA is a useful tool for measuring our ability to meet our future debt service and satisfy capital expenditure and working capital requirements, and EBITDA is commonly used by us and our investors to measure our ability to service indebtedness.

Free Cash Flow is included in this Form 10-K because our management considers it an important supplemental measure of our performance and believes that it is frequently used by securities analysts, investors and other interested parties in the evaluation of companies in our industry, some of which present Free Cash Flow when reporting their results. We regularly evaluate our performance as compared to other companies in our industry that have different financing and capital structures and/or tax rates by using Free Cash Flow. Management also believes that Free Cash Flow is useful to investors and securities analysts to evaluate our liquidity, evaluate strategic investment, evaluate our stock buyback plan and measure our ability to meet our future debt service.

EBITDA and Free Cash Flow are not substitutes for the GAAP measures of net income, income from operations and net cash provided by operating activities and are not necessarily measures of our ability to fund our cash needs. In addition, companies calculate EBITDA and Free Cash Flow differently and, therefore, EBITDA and Free Cash Flow as presented for us may not be comparable to EBITDA and Free Cash Flow reported by other companies. EBITDA has material limitations as a performance measure because it excludes interest expense, depreciation and amortization and income taxes. Free Cash Flow has material limitations as a performance measure because it only considers net cash provided by operating activities, and not net income or income from operations. For instance, it applies the entire cost of capital expenditure in the period in which the property or equipment is acquired, rather than spreading it over several periods as is the case with net income and income from operations.

Reconciliations of EBITDA to net income, income from operations and net cash provided by operating activities, and Free Cash Flow to net cash provided by operating activities, are included in the "Results of Operations" section below.

Results of Operations

Segment Data

The table below and descriptions that follow represent the consolidated results of operations of the Company for the years ended December 31, 2023, 2022 and 2021.

Net External Sales

The table below presents net external sales on a disaggregated basis for our two principal operating segments. Performance Materials net external sales primarily consist of sales of PVC, polyethylene and epoxy. Essential Materials net external sales primarily consist of sales of caustic soda, styrene, and related derivative materials. Housing Products net external sales primarily consist of sales of housing exterior and interior products, residential pipes and fittings and residential PVC compounds. Infrastructure Products net external sales primarily consist of sales of non-residential pipes and fittings and non-residential PVC compounds.

	Year Ended December 31,		
	2023	2022	2021
	(In millions of dollars, except per share data)		
Net external sales			
Performance and Essential Materials			
Performance Materials	$ 4,656	$ 6,964	$ 5,997
Essential Materials	3,680	4,044	2,673
Total Performance and Essential Materials	8,336	11,008	8,670
Housing and Infrastructure Products			
Housing Products	3,494	3,864	2,334
Infrastructure Products	718	922	774
Total Housing and Infrastructure Products	4,212	4,786	3,108
Total net external sales	$ 12,548	$ 15,794	$ 11,778
Income (loss) from operations			
Performance and Essential Materials	$ 59	$ 2,416	$ 2,549
Housing and Infrastructure Products	710	675	356
Corporate and other	(40)	(41)	(105)
Total income from operations	729	3,050	2,800
Interest expense	(165)	(177)	(176)
Other income, net	136	73	53
Provision for income taxes	178	649	607
Net income	522	2,297	2,070
Net income attributable to noncontrolling interests	43	50	55
Net income attributable to Westlake Corporation	$ 479	$ 2,247	$ 2,015
Diluted earnings per share	$ 3.70	$ 17.34	$ 15.58
EBITDA [1]	$ 1,962	$ 4,179	$ 3,693
Free Cash Flow [2]	$ 1,302	$ 2,287	$ 1,736

(1) See above for discussions on non-GAAP financial measures. See "Reconciliation of EBITDA to Net Income, Income from Operations and Net Cash Provided by Operating Activities" below.

(2) See above for discussions on non-GAAP financial measures. See "Reconciliation of Free Cash Flow to Net Cash Provided by Operating Activities" below.

	Year Ended December 31,			
	2023		2022	
	Average Sales Price	Volume	Average Sales Price	Volume
Product sales price and volume percentage change from prior year				
Performance and Essential Materials	-21 %	-3 %	+15 %	+13 %
Housing and Infrastructure Products	-3 %	-9 %	+35 %	+19 %
Company average	-16 %	-5 %	+20 %	+14 %

	Year Ended December 31,	
	2023	2022
Domestic US prices percentage change from prior-year period for fuel cost and feedstock		
Fuel cost (Natural Gas)	-59 %	+67 %
Feedstock (Ethane)	-49 %	+56 %

Reconciliation of EBITDA to Net Income, Income from Operations and Net Cash Provided by Operating Activities

The following table presents the reconciliation of EBITDA to net income, income from operations and net cash provided by operating activities, the most directly comparable GAAP financial measures, for each of the periods indicated.

	Year Ended December 31,		
	2023	2022	2021
	(In millions of dollars)		
Net cash provided by operating activities	$ 2,336	$ 3,395	$ 2,394
Changes in operating assets and liabilities and other	(1,989)	(1,119)	(301)
Deferred income taxes	175	21	(23)
Net income	522	2,297	2,070
Less:			
Other income, net	136	73	53
Interest expense	(165)	(177)	(176)
Provision for income taxes	(178)	(649)	(607)
Income from operations	729	3,050	2,800
Add:			
Depreciation and amortization	1,097	1,056	840
Other income, net	136	73	53
EBITDA	$ 1,962	$ 4,179	$ 3,693

Reconciliation of Free Cash Flow to Net Cash Provided by Operating Activities

The following table presents the reconciliation of Free Cash Flow to net cash provided by operating activities, the most directly comparable GAAP financial measure, for each of the periods indicated.

	Year Ended December 31,		
	2023	2022	2021
	(In millions of dollars)		
Net cash provided by operating activities	$ 2,336	$ 3,395	$ 2,394
Less:			
Additions to property, plant and equipment	1,034	1,108	658
Free Cash Flow	$ 1,302	$ 2,287	$ 1,736

2023 Compared with 2022

Summary

For the year ended December 31, 2023, net income attributable to Westlake Corporation was $479 million, or $3.70 per diluted share, on net sales of $12,548 million. This represents a decrease in net income attributable to Westlake Corporation of $1,768 million, or $13.64 per diluted share, compared to 2022 net income attributable to Westlake Corporation of $2,247 million, or $17.34 per diluted share, on net sales of $15,794 million. Income from operations was $729 million for the year ended December 31, 2023, as compared to $3,050 million for the year ended December 31, 2022, a decrease of $2,321 million. The decrease in net income and income from operations was primarily due to lower sales prices and volumes for most of our products across both segments as a result of weaker global demand, a non-cash impairment charge of $475 million that comprised of Westlake Epoxy goodwill and long-lived assets of our epoxy base resin business in the Netherlands in the fourth quarter of 2023, and a pre-tax litigation charge of approximately $150 million related to a final settlement to fully resolve lawsuits involving certain liability claims. The decreases in net income and income from operations in 2023 were partially offset by lower natural gas and feedstock costs and lower amortization of intangibles. Net sales decreased by $3,246 million to $12,548 million in 2023 from $15,794 million in 2022, primarily due to lower sales prices and volumes for most of our products across both segments.

Net Sales. Net sales decreased by $3,246 million, or 21%, to $12,548 million in 2023 from $15,794 million in 2022, primarily due to lower sales prices and volumes for most of our products. Average sales prices for 2023 decreased by 16% as compared to 2022, due to continuing weaker demand for most of our products across both segments. Sales volumes decreased by 5% in 2023 as compared to 2022, substantially due to weaker global demand for many of our products across both segments.

Gross Profit. Gross profit margin percentage was 18% in 2023 as compared to 26% in 2022. The decline in gross margin percentage for the year ended December 31, 2023 as compared to the year ended December 31, 2022 was primarily due to lower prices for most of our products and the litigation charge, as discussed above, partially offset lower natural gas and feedstock costs.

Selling, General and Administrative Expenses. Selling, general and administrative expenses increased by $30 million to $865 million in 2023 from $835 million in 2022. This increase was mainly due to higher compensation and professional consulting expenses and a full year of selling, general and administrative expenses in the current year for the epoxy business acquired in February 2022.

Amortization of Intangibles. Amortization expense decreased by $33 million to $122 million in 2023 from $155 million in 2022, primarily due to the end of useful lives of certain intangible assets within the Housing and Infrastructure Products segment during 2022.

Impairment of Goodwill and Long-Lived Assets. The impairment of $475 million comprised of Westlake Epoxy goodwill impairment of $128 million and long-lived asset impairment of $347 million related to our base epoxy resin business in the Netherlands, recognized within the Performance and Infrastructure Materials segment.

Restructuring, Transaction and Integration-related Costs. Restructuring, transaction and integration-related costs decreased by $5 million to $28 million in 2023 from $33 million in 2022. The costs in 2023 are restructuring and integration costs primarily related to plant closures resulting from the Company's manufacturing footprint optimization efforts. The 2022 expenses primarily related to the integration-related consulting fees and restructuring expenses related to the businesses acquired in previous years.

Interest Expense. Interest expense decreased by $12 million to $165 million in 2023 from $177 million in 2022, primarily due to the repayment of $269 million of debt during 2022.

Other Income, Net. Other income, net of $136 million in 2023 was higher than other income, net of $73 million in 2022, substantially due to higher interest income attributable to a higher average cash and cash equivalents balance and higher interest rates in the current period and insurance recoveries in 2023.

Income Taxes. The effective income tax rate was 25% in 2023, which was higher than the 22% effective income tax rate in 2022. The effective income tax rate in 2023 was higher as compared to 2022, primarily because Westlake Epoxy's goodwill impairment was not deductible for tax purposes, and because of a valuation allowance that was recorded against Westlake Epoxy Netherlands' deferred tax assets including net operating loss carryforwards, mostly generated from the impairment of long-lived assets. Such increase in the effective income tax rate was partially offset by an increase in U.S. federal research and development credits available to the Company and a decrease in state and foreign taxes.

Performance and Essential Materials Segment

Net Sales. Net sales for the Performance and Essential Materials segment decreased by $2,672 million, or 24%, to $8,336 million in 2023 from $11,008 million in 2022, primarily resulting from lower global demand for most of our products across our businesses under this segment. Average sales prices for the Performance and Essential Materials segment decreased by 21% in 2023 as compared to 2022. Lower Performance Materials sales prices were primarily due to lower PVC resin, polyethylene and epoxy sales prices. Lower Essential Materials sales prices were due to lower caustic soda sales prices. Sales volumes for the Performance and Essential Materials segment decreased by 3% in 2023 as compared to 2022, primarily due to lower epoxy and caustic soda sales volumes.

Income from Operations. Income from operations for the Performance and Essential Materials segment decreased by $2,357 million to $59 million in 2023 from $2,416 million in 2022. This decrease in income from operations was primarily due to lower sales prices and volumes for most of our products, a non-cash impairment charge of $475 million related to Westlake Epoxy goodwill and long-lived assets of our epoxy base resin business in the Netherlands, and a litigation charge of approximately $150 million related to a final settlement to fully resolve lawsuits involving certain liability claims, both of which were recorded in the fourth quarter of 2023, and planned and unplanned outages. These decreases were partially offset by lower natural gas and feedstock costs.

Housing and Infrastructure Products Segment

Net Sales. Net sales for the Housing and Infrastructure Products segment decreased by $574 million, or 12%, to $4,212 million in 2023 from $4,786 million in 2022. Housing Products net sales of $3,494 million decreased by $370 million as compared to 2022 primarily due to lower demand for most of our products across our housing businesses due to the downturn in U.S. residential construction throughout 2023, partially offset by higher sales prices for some of our building products. Infrastructure Products net sales of $718 million decreased by $204 million as compared to 2022, due to a continuing lower demand for our products attributable to customers destocking from elevated inventory levels in the prior year. Average sales prices for the Housing and Infrastructure Products segment decreased slightly by 3% in 2023 as compared to 2022. Sales volumes for the Housing and Infrastructure Products segment decreased by 9% in 2023, as compared to 2022.

Income from Operations. Income from operations for the Housing and Infrastructure Products segment increased by $35 million to $710 million in 2023 from $675 million in 2022. The increase in income from operations was primarily due to lower raw material costs and higher sales prices for some of our building products, partially offset by lower sales prices for PVC compounds and pipe and fittings and volumes across most of our housing and infrastructure businesses.

2022 Compared with 2021

Summary

For the year ended December 31, 2022, net income attributable to Westlake Corporation was $2,247 million, or $17.34 per diluted share, on net sales of $15,794 million. This represents an increase in net income attributable to Westlake Corporation of $232 million, or $1.76 per diluted share, compared to 2021 net income attributable to Westlake Corporation of $2,015 million, or $15.58 per diluted share, on net sales of $11,778 million. Income from operations was $3,050 million for the year ended December 31, 2022, as compared to $2,800 million for the year ended December 31, 2021, an increase of $250 million. The increase in net income and income from operations was primarily due to significantly higher global sales prices for caustic soda and chlorine caused by higher demand through the year, and higher prices and margins on many of our products in the Housing and Infrastructure Products segment, due to strong demand for our products primarily during the first half of 2022. In addition, net income and income from operations included the contributions from the businesses acquired in the second half of 2021 and in the first quarter of 2022. Net income and income from operations for the year ended December 31, 2022 was negatively impacted by softening demand for PVC resin during the second half of 2022 due to slowing global economic conditions. In addition, higher natural gas, power and feedstock costs and a $70 million pre-tax litigation charge negatively impacted net income and income from operations during the year ended December 31, 2022. Selling, general and administrative expenses were also higher for the year ended December 31, 2022 primarily due to the businesses acquired in the second half of 2021 and in the first quarter of 2022. Net sales increased by $4,016 million to $15,794 million in 2022 from $11,778 million in 2021, due to higher sales prices for caustic soda and chlorine and for many of our products in the Housing and Infrastructure Products segment, partially offset by lower polyethylene sales prices. Sales volumes in the year ended December 31, 2022 were also higher in both segments as compared to the year ended December 31, 2021, primarily due to the contributions from the acquired businesses in the second half of 2021 and in the first quarter of 2022, partially offset by lower PVC resin, caustic soda and PVC compounds sales volumes.

Net Sales. Net sales increased by $4,016 million, or 34%, to $15,794 million in 2022 from $11,778 million in 2021, primarily due to higher sales prices for caustic soda and chlorine and for many of our products in the Housing and Infrastructure Products segment as well as higher sales volumes due to the businesses acquired in the second half of 2021 and in the first quarter of 2022, partially offset by lower polyethylene sales prices. Average sales prices for 2022 increased by 20% as compared to 2021 due to strong demand for caustic soda and strong residential construction, repair and remodeling markets in North America during the first half of 2022. Sales volumes increased by 14% in 2022 as compared to 2021, primarily due to the businesses acquired in the second half of 2021 and in the first quarter of 2022, partially offset by lower PVC resin, caustic soda and PVC compounds sales volumes.

Gross Profit. Gross profit margin percentage was 26% in 2022 as compared to 30% in 2021. The decline in gross margin percentage for the year ended December 31, 2022 as compared to the year ended December 31, 2021 was primarily due to higher natural gas, power, feedstock and raw material costs and lower polyethylene sales prices. These decreases versus the prior year were partially offset by higher sales prices for caustic soda, chlorine, derivative products, pipes and fittings and PVC compounds.

Selling, General and Administrative Expenses. Selling, general and administrative expenses increased by $284 million to $835 million in 2022 from $551 million in 2021. This increase was mainly due to the inclusion of expenses related to the businesses acquired in the second half of 2021 and in the first quarter of 2022 as well as higher employee-related costs.

Amortization of Intangibles. Amortization expense increased by $32 million to $155 million in 2022 from $123 million in 2021, primarily due to the amortization of intangibles associated with the businesses acquired in the second half of 2021 and in the first quarter of 2022.

Restructuring, Transaction and Integration-related Costs. The restructuring, transaction and integration-related costs of $33 million and $21 million for the years ended December 31, 2022 and December 31, 2021, respectively, primarily consisted of restructuring, transaction and integration-related costs associated with the businesses acquired in the second half of 2021 and in the first quarter of 2022.

Interest Expense. Interest expense of $177 million in 2022 was comparable to $176 million in 2021. The full year interest expense in 2022 associated with the issuance of $1,700 million aggregate principal amount of senior notes in August 2021 was substantially offset by the realization of losses in 2021 on the settlement of interest rate lock arrangements associated with the senior notes issued in August 2021.

Other Income, Net. Other income, net of $73 million in 2022 was higher than other income, net of $53 million in 2021, primarily due to higher interest income.

Income Taxes. The effective income tax rate was 22% in 2022, which was slightly lower than the 23% effective tax rate in 2021. The effective tax rate in 2022 was lower as compared to 2021, primarily due to an increase in U.S. federal research and development credits available to the Company in 2022.

Performance and Essential Materials Segment

Net Sales. Net sales for the Performance and Essential Materials segment increased by $2,338 million, or 27%, to $11,008 million in 2022 from $8,670 million in 2021. Average sales prices for the Performance and Essential Materials segment increased by 15% in 2022 as compared to 2021. The increase in average sales prices were primarily driven by higher prices for caustic soda, chlorine and derivative products, partially offset by lower polyethylene sales prices. Sales volumes for the Performance and Essential Materials segment increased by 13% in 2022 as compared to 2021, primarily due to contributions from the recently acquired Westlake Epoxy business, which were partially offset by lower PVC resin, caustic soda and styrene sales volumes.

Income from Operations. Income from operations for the Performance and Essential Materials segment decreased by $133 million to $2,416 million in 2022 from $2,549 million in 2021. This decrease in income from operations was primarily due to lower polyethylene sales prices and integrated margins and lower sales volumes for PVC resin and caustic soda, higher natural gas, power and feedstock costs and a $70 million pre-tax litigation charge in the third quarter of 2022. The decrease in income from operations versus the prior-year was partially offset by higher sales prices for caustic soda, chlorine and derivative products and income from Westlake Epoxy, which was acquired in the first quarter of 2022. Trading activity in 2022 resulted in a loss of approximately $28 million as compared to a gain of $12 million in 2021.

Housing and Infrastructure Products Segment

Net Sales. Net sales for the Housing and Infrastructure Products segment increased by $1,678 million, or 54%, to $4,786 million in 2022 from $3,108 million in 2021. Housing Products net sales of $3,864 million increased by $1,530 million as compared to 2021 primarily due to higher sales prices for all products as well as a full year of sales contribution from the businesses acquired in the second half of 2021. Infrastructure Products net sales of $922 million increased by $148 million as compared to 2021 primarily due to significantly higher sales prices. Average sales prices for the Housing and Infrastructure Products segment increased by 35% in 2022 as compared to 2021, primarily due to strong residential construction and repair and remodeling activity during the first half of 2022. Sales volumes for the Housing and Infrastructure Products segment increased by 19% in 2022, as compared to 2021, primarily due to contributions from the businesses acquired in the second half of 2021, partially offset by lower sales volumes for PVC compounds.

Income from Operations. Income from operations for the Housing and Infrastructure Products segment increased by $319 million to $675 million in 2022 from $356 million in 2021. The increase in income from operations was primarily due to higher sales prices driven by higher housing construction and remodeling activity during the first half of 2022. Income from operations was also higher due to contributions from the businesses acquired in the second half of 2021. The higher income from operations in 2022 as compared to 2021 was partially offset by higher raw material, power and production costs.

Cash Flows

Operating Activities

Operating activities provided cash of $2,336 million in 2023 as compared to cash provided by operating activities of $3,395 million in 2022. The $1,059 million decrease in cash flow from operating activities was mainly due to lower prices and demand for most of our products, partially offset by favorable changes in working capital. Changes in components of working capital, which we define for purposes of this cash flow discussion as accounts receivable, inventories, prepaid expenses and other current assets, less accounts payable and accrued and other liabilities, provided cash of $600 million in 2023, as compared to $174 million of cash provided in 2022, a favorable change of $426 million. The favorable changes in 2023 were substantially due to lower inventory costs driven by lower feedstock and fuel costs, as compared to 2022.

Operating activities provided cash of $3,395 million in 2022 as compared to cash provided by operating activities of $2,394 million in 2021. The $1,001 million increase in cash flows from operating activities was mainly due to the increase in income from operations, which was partially offset by working capital changes and an unfavorable change related to the turnaround at OpCo's Petro 2 facility and other turnaround activities. Changes in components of working capital, which we define for purposes of this cash flow discussion as accounts receivable, inventories, prepaid expenses and other current assets, less accounts payable and accrued and other liabilities, provided cash of $174 million in 2022, as compared to $383 million of cash used in 2021, a favorable change of $557 million. The majority of the unfavorable changes in 2021 were driven by higher accounts receivable and higher inventories, partially offset by higher accounts payable and accrued and other liabilities. The unfavorable change in accounts receivable was primarily driven by higher sales prices resulting in higher trade customer balances. The higher inventories, accounts payable and accrued and other liabilities in 2022 were primarily driven by higher inventory cost and an increase in operating activities, as compared to 2021.

Investing Activities

Net cash used for investing activities during 2023 was $1,037 million as compared to net cash used of $2,479 million in 2022. The decrease in investing activities during 2023 was primarily because of the epoxy acquisition in February 2022 for $1,163 million. Capital expenditures were $1,034 million during 2023 as compared to $1,108 million during 2022. Capital expenditures in 2023 and 2022 were primarily related to projects to increase production capacity or reduce costs, maintenance and safety projects and environmental projects at our various facilities. The investing activities during 2022 were comprised primarily of the acquisition of Westlake Epoxy in February 2022 for $1,163 million, the purchase of an additional 3.2% interest in LACC for $89 million, aggregate contributions of $87 million to LACC, capital expenditures, and other asset acquisitions.

Net cash used for investing activities during 2022 was $2,479 million as compared to net cash used of $3,213 million in 2021. Investing activities during 2022 were comprised primarily of the acquisition of Westlake Epoxy in February 2022 for $1,163 million, the purchase of an additional 3.2% interest in LACC for $89 million, aggregate contributions of $87 million to LACC, capital expenditures, and other asset acquisitions. Our 2021 acquisition related activities were higher, as compared to 2022, due to the acquisitions of the Boral Target Companies, LASCO and Dimex for $2,554 million in the aggregate. Capital expenditures were $1,108 million in 2022 compared to $658 million in 2021. Capital expenditures in 2022 and 2021 were primarily related to expansion projects at various chlor-alkali plants as well as projects to improve production capacity or reduce costs, maintenance and safety projects and environmental projects at our various facilities.

Financing Activities

Net cash used by financing activities during 2023 was $245 million as compared to net cash used by financing activities of $587 million in 2022. The decrease in cash used for financing activities during 2023 as compared to 2022 was primarily due to the redemption of $250 million aggregate principal amount of the 3.60% Senior Notes due 2022 and higher repurchases of our common stock in 2022 as compared to 2023. The financing activities during 2023 were primarily related to the $221 million payment of cash dividends, the $54 million payment of cash distributions to noncontrolling interests, repurchases of shares of our common stock for an aggregate purchase price of $23 million and inflows of $44 million from the exercise of stock options. The financing activities during 2022 were primarily related to the redemption of $250 million aggregate principal amount of the 3.60% 2022 Senior Notes, $169 million payment of cash dividends, the $60 million payment of cash distributions to noncontrolling interests and repurchases of shares of our common stock for an aggregate purchase price of $101 million.

Net cash used by financing activities during 2022 was $587 million as compared to net cash provided of $1,437 million in 2021. Financing activities during 2022 were primarily related to the redemption of $250 million aggregate principal amount of the 3.60% 2022 Senior Notes, the $169 million payment of cash dividends, the $60 million payment of cash distributions to noncontrolling interests and repurchases of our common stock for an aggregate purchase price of $101 million. The activities during 2021 were primarily related to the registered public offering of $300 million aggregate principal amount of the 0.875% 2024 Senior Notes, $350 million aggregate principal amount of the 2.875% 2041 Senior Notes, $600 million aggregate principal amount of the 3.125% 2051 Senior Notes and $450 million aggregate principal amount of the 3.375% 2061 Senior Notes and the payment of debt issuance costs of $18 million related to such notes. The remaining activities in 2021 were primarily related to the $145 million payment of cash dividends, the $48 million payment of cash distributions to noncontrolling interests and repurchases of our common stock for an aggregate purchase price of $30 million.

Liquidity and Capital Resources

Liquidity and Financing Arrangements

Our principal sources of liquidity are from cash and cash equivalents, cash from operations, short-term borrowings under the Credit Agreement and our long-term financing.

In November 2014, our Board of Directors authorized a $250 million stock repurchase program (the "2014 Program"). In November 2015, our Board of Directors approved the expansion of the 2014 Program by an additional $150 million. In August 2018, our Board of Directors approved the further expansion of the existing 2014 Program by an additional $150 million. In August 2022, our Board of Directors approved the further expansion of the existing 2014 Program by an additional $500 million. As of December 31, 2023, we had repurchased 8,722,550 shares of our common stock for an aggregate purchase price of approximately $574 million under the 2014 Program. During the year ended December 31, 2023, 211,294 shares of our common stock were repurchased for an aggregate purchase price of $23 million under the 2014 Program. Purchases under the 2014 Program may be made either through the open market or in privately negotiated transactions. Decisions regarding the amount and the timing of purchases under the 2014 Program will be influenced by our cash on hand, our cash flows from operations, general market conditions and other factors. The 2014 Program may be discontinued by our Board of Directors at any time.

On October 4, 2018, Westlake Chemical Partners LP ("Westlake Partners") and Westlake Chemical Partners GP LLC, the general partner of Westlake Partners, entered into an Equity Distribution Agreement with UBS Securities LLC, Barclays Capital Inc., Citigroup Global Markets Inc., Deutsche Bank Securities Inc., RBC Capital Markets, LLC, Merrill Lynch, Pierce, Fenner & Smith Incorporated and Wells Fargo Securities, LLC to offer and sell Westlake Partners common units, from time to time, up to an aggregate offering amount of $50 million. This Equity Distribution Agreement was amended on February 28, 2020 to reference a new shelf registration and subsequent renewals thereof for utilization under this agreement. No common units were issued under this program in 2023, 2022 or 2021.

We believe that our sources of liquidity as described above are adequate to fund our normal operations and ongoing capital expenditures and turnaround activities. Funding of any potential large expansions such as our recent acquisitions or potential future acquisitions or the redemption of debt may likely necessitate, and therefore depend on our ability to obtain, additional financing in the future. We may not be able to access additional liquidity at favorable interest rates due to volatility of the commercial credit markets.

Cash and Cash Equivalents

As of December 31, 2023, our cash and cash equivalents totaled $3,304 million.

In addition to our cash and cash equivalents, our credit agreement is available to provide liquidity as needed, as described under "Debt" below.

Debt

As of December 31, 2023, the carrying value of our indebtedness totaled $4,906 million. See Note 11 to the consolidated financial statements appearing elsewhere in this Form 10-K for a discussion of our long-term indebtedness. Defined terms used in this section have the definitions assigned to such terms in Note 11 to the consolidated financial statements included in Item 8 of this Form 10-K.

Our ability to make payments on our indebtedness and to fund planned capital expenditures will depend on our ability to generate cash in the future, which is subject to general economic, financial, competitive, legislative, regulatory and other factors that are beyond our control. Based on our current level of operations and unless we were to undertake a new expansion or large acquisition, we believe our cash flows from operations, available cash and available borrowings under our credit agreement will be adequate to meet our normal operating needs for the foreseeable future.

Our long-term debt consisted of the following as of December 31, 2023:

	Principal Amount (in millions)	Debt Issuance Date	Maturity Date	Par Call Date	Optional Redemption Terms and Other Matters
0.875% senior notes due 2024 (the "0.875% 2024 Senior Notes")	$ 300	August 2021	August 2024	August 15, 2022	(1)
3.60% senior notes due 2026 (the "3.60% 2026 Senior Notes")	750	August 2016	August 2026	May 15, 2026	(1) (4)
Loan related to tax-exempt waste disposal revenue bonds due 2027	11	December 1997	December 2027		(6)
1.625% €700 million senior notes due 2029 (the "1.625% 2029 Senior Notes")	773	July 2019	July 2029	April 17, 2029	(1) (2)
3.375% senior notes due 2030 (the "3.375% 2030 Senior Notes")	300	June 2020	June 2030	March 15, 2030	(1) (3)
3.50% senior notes due 2032 (the "3.50% 2032 tax-exempt GO Zone Refunding Senior Notes")	250	November 2017	November 2032	November 1, 2027	(5)
2.875% senior notes due 2041 (the "2.875% 2041 Senior Notes")	350	August 2021	August 2041	February 15, 2041	(1) (3)
5.00% senior notes due 2046 (the "5.00% 2046 Senior Notes")	700	August 2016	August 2046	February 15, 2046	(1) (4)
4.375% senior notes due 2047 (the "4.375% 2047 Senior Notes")	500	November 2017	November 2047	May 15, 2047	(1) (3)
3.125% senior notes due 2051 (the "3.125% 2051 Senior Notes")	600	August 2021	August 2051	February 15, 2051	(1) (3)
3.375% senior notes due 2061 (the "3.375% 2061 Senior Notes")	450	August 2021	August 2061	February 15, 2061	(1) (3)
Term loan 2026 (the "2026 Term Loan")	13	March 2021	March 2026		(7)
Total long-term debt	$ 4,997				

(1) At our option, we may redeem the notes at any time on or after the specified par call date at a redemption price equal to 100% of the principal amount of the notes being redeemed plus accrued and unpaid interest.

(2) At our option, we may redeem the notes at any time prior to the specified par call date at a redemption price equal to the greater of (i) 100% of the principal amount of the notes being redeemed plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the notes being redeemed that would be due if the notes matured on the specified par call date (not including any portion of such payments of interest accrued as of the redemption date), discounted to the redemption date on an annual basis at the applicable comparable government bond rate plus 30 basis points plus accrued and unpaid interest.

(3) At our option, we may redeem the notes at any time prior to the specified par call date at a redemption price equal to the greater of (i) 100% of the principal amount of the notes being redeemed plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the notes being redeemed that would be due if the notes matured on the specified par call date (excluding accrued and unpaid interest to the redemption date), discounted to the redemption date on a semi-annual basis at the treasury rate plus 20 to 40 basis points plus accrued and unpaid interest.

(4) At our option, we may redeem the notes at any time prior to the specified par call date at a redemption price equal to the greater of (i) 100% of the principal amount of the notes being redeemed plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the notes being redeemed (excluding accrued and unpaid interest to the redemption date), discounted to the redemption date on a semi-annual basis at the treasury rate plus 35 to 45 basis points, plus accrued and unpaid interest.

(5) In the event of a redemption of certain bonds (the "GO Zone Bonds") issued by the Louisiana Local Government Environmental Facility and Development Authority (the "Authority") in 2017, we will redeem notes equal in principal amount to the GO Zone Bonds to be redeemed at a redemption price equal to the redemption price of the GO Zone Bonds to be redeemed, plus accrued interest to the redemption date. The GO Zone Bonds are subject to optional redemption by the Authority upon the direction of the Company at any time on or after November 1, 2027, for 100% of the principal amount plus accrued interest to the redemption date.

(6) The waste disposal revenue bonds expire in December 2027 and are subject to redemption and mandatory tender for purchase prior to maturity under certain conditions. Interest on the waste disposal revenue bonds accrues at a rate determined by a remarketing agent and is payable quarterly. The interest rate on the waste disposal revenue bonds at December 31, 2023 was 4.30%.

(7) The 2026 Term Loan has a 5-year maturity and includes a government rate subsidy. The interest rate on the 2026 Term Loan as of December 31, 2023 was 0.95%.

The holders of the 0.875% 2024 Senior Notes, the 3.60% 2026 Senior Notes, the 1.625% 2029 Senior Notes, the 3.375% 2030 Senior Notes, the 3.50% 2032 tax-exempt GO Zone Refunding Senior Notes, the 2.875% 2041 Senior Notes, the 5.00% 2046 Senior Notes, the 4.375% 2047 Senior Notes, the 3.125% 2051 Senior Notes and the 3.375% 2061 Senior Notes may require us to repurchase the notes at a price equal to 101% of their principal amount, plus accrued and unpaid interest to, but not including, the date of repurchase, upon the occurrence of both a "change of control" and, within 60 days of such change of control, a "below investment grade rating event" (as such terms are defined in the respective indentures governing these notes).

The indenture governing the 0.875% 2024 Senior Notes, the 3.60% 2026 Senior Notes, the 1.625% 2029 Senior Notes, the 3.375% 2030 Senior Notes, the 3.50% 2032 tax-exempt GO Zone Refunding Senior Notes, the 2.875% 2041 Senior Notes, the 5.00% 2046 Senior Notes, the 4.375% 2047 Senior Notes, the 3.125% 2051 Senior Notes, and the 3.375% 2061 Senior Notes contains customary events of default and covenants that, among other things and subject to certain exceptions, restrict us and certain of our subsidiaries' ability to (1) incur certain secured indebtedness, (2) engage in certain sale and leaseback transactions and (3) consolidate, merge or transfer all or substantially all of our assets.

As of December 31, 2023, we were in compliance with all of our long-term debt covenants.

Credit Agreement

On June 9, 2022, we entered into a new $1.5 billion revolving credit facility that is scheduled to mature on June 9, 2027 (the "Credit Agreement") and, in connection therewith, terminated our then existing revolving credit agreement. The Credit Agreement bears interest at either (a) Adjusted Term SOFR (as defined in the Credit Agreement) plus a margin ranging from 1.00% to 1.625% per annum or (b) Alternate Base Rate (as defined in the Credit Agreement) plus a margin ranging from 0.00% to 0.625% per annum, in each case depending on the credit rating of the Company. The Credit Agreement contains certain affirmative and negative covenants, including a quarterly total leverage ratio financial maintenance covenant. As of December 31, 2023, we were in compliance with the total leverage ratio financial maintenance covenant.

The Credit Agreement also contains certain events of default and, if and for so long as certain events of default have occurred and are continuing, any overdue amounts outstanding under the Credit Agreement will accrue interest at an increased rate, the lenders can terminate their commitments to lend thereunder and payments of any outstanding amounts thereunder could be accelerated by the lenders. None of our subsidiaries are required to guarantee our obligations under the Credit Agreement.

The Credit Agreement includes a $150 million sub-limit for letters of credit, and any outstanding letters of credit will be deducted from availability under the facility. The Credit Agreement also provides for a discretionary $50 million commitment for swingline loans to be provided on a same-day basis. We may also increase the size of the facility, in increments of at least $25 million, up to a maximum of $500 million, subject to certain conditions and if certain lenders agree to commit to such an increase.

Westlake Chemical Partners LP Credit Arrangements

Our subsidiary, Westlake Chemical Finance Corporation, is the lender party to a $600 million revolving credit facility with Westlake Chemical Partners LP ("Westlake Partners") (the "MLP Revolver") that is scheduled to mature on July 12, 2027. As of December 31, 2023, outstanding borrowings under the credit facility totaled $377 million and bore interest at Secured Overnight Financing Rate, as administered by the Federal Reserve Bank of New York ("SOFR") plus the Applicable Margin plus a 0.10% credit spread adjustment. On July 12, 2022, Westlake Partners entered into the Fourth Amendment (the "MLP Revolver Amendment") to the MLP Revolver. The MLP Revolver Amendment, among other things, extended the maturity date to July 12, 2027 and provided for the replacement of LIBOR with SOFR. Borrowings under the MLP Revolver now bear interest at a variable rate of either (a) SOFR plus the Applicable Margin plus a 0.10% credit spread adjustment or, if SOFR is no longer available, (b) the Alternate Base Rate plus the Applicable Margin minus 1.0%. The Applicable Margin under the MLP Revolver varies between 1.75% and 2.75%, depending on the Partnership's Consolidated Leverage Ratio.

Our subsidiary, Westlake Polymers LLC, is the administrative agent to a $600 million revolving credit facility with Westlake Chemical OpCo LP ("OpCo") (the "OpCo Revolver") that is scheduled to mature on July 12, 2027. As of December 31, 2023, outstanding borrowings under the credit facility totaled $23 million and bore interest at SOFR plus the Applicable Margin of 1.75% plus a 0.10% credit spread adjustment. On July 12, 2022, OpCo entered into the Second Amendment (the "OpCo Revolver Amendment") to the OpCo Revolver. The OpCo Revolver Amendment, among other things, extended the maturity date to July 12, 2027 and provided for the replacement of LIBOR with SOFR. Borrowings under the OpCo Revolver now bear interest at a variable rate of either (a) SOFR plus the Applicable Margin plus a 0.10% credit spread adjustment or, if SOFR is no longer available, (b) the Alternate Base Rate plus the Applicable Margin minus 1.0%. The Applicable Margin under the OpCo Revolver is 1.75%

We consolidate Westlake Partners and OpCo for financial reporting purposes as we have a controlling financial interest. As such, the revolving credit facilities described above between our subsidiaries and Westlake Partners and OpCo are eliminated from the financial statements upon consolidation.

Contractual and Other Obligations

The Company's material cash requirements for contractual and other obligations in the near term (next 12 months) and the long term period (2024 and thereafter) include long-term debt, interest payments, operating leases, pension benefits funding, post-retirement healthcare benefits, purchase obligations, asset retirement obligations and letters of credit.

Debt Obligations and Interest Payments. As of December 31, 2023, we had $300 million debt obligations due within the near term, and debt obligations of $4,697 million due over the long-term period. At December 31, 2023, long-term debt related interest expense of $158 million was due within the near term, and related interest expense of $2,692 million was due over the long-term period. Maturities of our debt consist of $300 million in 2024, $763 million in 2026 and $11 million in 2027. There are no other scheduled maturities of debt in 2024 through 2028. See Note 11, "Long-Term Debt," in the Notes to Consolidated Financial Statements in Item 8 of this Form 10-K for further information on our debt obligations and the expected timing of future principal and interest payments.

Operating leases. As of December 31, 2023, there was $142 million in operating lease obligations due within the near term, and $768 million due over the long-term period. See Note 7, "Leases," in the Notes to Consolidated Financial Statements in Item 8 of this Form 10-K for further detail of our obligations and the timing of expected future payments.

Pension Benefits Funding and Post-retirement Healthcare Benefits. Pension benefits funding obligations due within the near term were $8 million while post-retirement healthcare benefit payment obligations due within the near term were $8 million as of December 31, 2023. As of December 31, 2023, we had $135 million and $64 million of pension benefit funding and post-retirement healthcare benefit obligations due over the long-term period, respectively. The estimate of the timing of future payments under our defined benefit pension plans which cover certain eligible employees in the United States and non-U.S. countries and our post-retirement healthcare benefits to the employees of certain subsidiaries who meet certain minimum age and service requirements involves the use of certain assumptions, including retirement ages and payout periods. See Note 14, "Employee Benefits," in the Notes to Consolidated Financial Statements in Item 8 of this Form 10-K for further information on our obligations and the timing of expected future payments.

Purchase Obligations. Purchase obligations include agreements to purchase goods and services that are enforceable and legally binding and that specify all significant terms, including a minimum quantity and price. We are party to various obligations to purchase goods and services, including commitments to purchase various feedstock, utilities, nitrogen, oxygen, product storage, pipeline usage and logistic support, in each case in the ordinary course of our business, as well as various purchase commitments for our capital projects. As of December 31, 2023, we had $2,152 million of enforceable and legally binding purchase commitments due within the near term, and $5,976 million due over the long-term period.

Asset Retirement Obligations. As of December 31, 2023, we had $3 million asset retirement obligations due within the near term, and $55 million due over the long-term period. Asset retirement obligations includes the estimated costs and timing of payments to satisfy our recognized asset retirement obligations. We recognize asset retirement obligations in the period in which the liability becomes probable and reasonably estimable. Initially, the asset retirement obligation is recorded at fair value and capitalized as a component of the carrying value of the long-lived asset to which the obligation relates. See Note 1, "Description of Business and Significant Accounting Policies," in the Notes to Consolidated Financial Statements in Item 8 of this Form 10-K for further detail of our asset retirement obligations.

Letters of Credit. As of December 31, 2023, we had $39 million standby letters of credit, made in the ordinary course of business, maturing within the near term, and no standby letters of credit maturing over the long-term period. We had no letters of credit outstanding under our Credit Agreement.

Critical Accounting Policies and Estimates

Critical accounting policies are those that are important to our financial condition and require management's most difficult, subjective or complex judgments. Different amounts would be reported under different operating conditions or under alternative assumptions. We have evaluated the accounting policies used in the preparation of the accompanying consolidated financial statements and related notes and believe those policies are reasonable and appropriate. Our significant accounting policies are summarized in Note 1 to the consolidated financial statements appearing elsewhere in this Form 10-K.

Critical accounting estimates are those estimates made in accordance with the accounting principles generally accepted in the United States ("GAAP") that involve a significant level of estimation uncertainty and have had or are reasonably likely to have a material impact on our financial condition or results of operation. Our more critical accounting estimates include those related to business combinations, fair values, long-lived assets, goodwill, accruals for long-term employee benefits, accounts receivable, income taxes and environmental and legal obligations. Inherent in such estimates are certain key assumptions. We periodically update the estimates used in the preparation of the financial statements based on our latest assessment of the current and projected business and general economic environment. We believe the following to be our most critical accounting estimates required for the preparation of our financial statements.

Business Combinations and Intangible Assets Including Goodwill. We account for business combinations using the acquisition method of accounting, and accordingly, the assets and liabilities of the acquired business are recorded at their fair values at the date of acquisition. The excess of the purchase price over the estimated fair value is recorded as goodwill. Any changes in the estimated fair values of the net assets recorded for acquisitions prior to the finalization of more detailed analysis, but not to exceed one year from the date of acquisition, will change the amount of the purchase price allocable to goodwill. Any subsequent changes to any purchase price allocations that are material to our consolidated financial results will be adjusted in the same period's financial statements, including the effect on earnings of changes in depreciation, amortization or other income effects, if any, as a result of the change to the provisional amounts, calculated as if the accounting had been completed at the acquisition date. All acquisition costs are expensed as incurred and in-process research and development costs are recorded at fair value as an indefinite-lived intangible asset and assessed for impairment thereafter until completion, at which point the asset is amortized over its expected useful life. Separately recognized transactions associated with business combinations are generally expensed subsequent to the acquisition date. The application of business combination accounting requires the use of significant estimates and assumptions. Our estimates of fair value are based upon assumptions we believe to be reasonable, but which are inherently uncertain and unpredictable. The fair value of the customer relationships acquired are estimated by management through a discounted cash flow model using the multi-period excess earnings methodology, which involves the use of significant estimates and assumptions related to revenue growth rates, operating margins, discount rates, and customer attrition rates, among other items. The fair value of the technology and trade names acquired is estimated by management through a discounted cash flow model using the relief from royalty methodology, which involves the use of significant estimates and assumptions related to revenue growth rates, and discount rates. The results of operations of acquired businesses are included in our consolidated financial statements from the acquisition date.

Fair Value Estimates. We develop estimates of fair value to allocate the purchase price paid to acquire a business to the assets acquired and liabilities assumed in an acquisition, to assess impairment of long-lived assets and goodwill and to record marketable securities and pension plan assets. We use all available information to make these fair value determinations, including the engagement of third-party consultants. In addition, we record all pension plan assets and certain marketable securities at fair value. The fair value of these items is determined by quoted market prices or from observable market-based inputs. See Note 16 to the consolidated financial statements appearing elsewhere in this Form 10-K for more information.

Long-Lived Assets. Key estimates related to long-lived assets include useful lives, recoverability of carrying values and existence of any retirement obligations. Such estimates could be significantly modified. The carrying values of long-lived assets could be impaired by significant changes or projected changes in supply and demand fundamentals (which could have a negative impact on operating rates or margins), new technological developments, new competitors with significant raw material or other cost advantages, adverse changes associated with the United States and global economies, the cyclical nature of the chemical and refining industries and uncertainties associated with governmental actions.

We evaluate long-lived assets for potential impairment indicators whenever events or changes in circumstances indicate that the carrying amount of an asset may not be recoverable, including when negative conditions such as significant current or projected operating losses exist. Our judgments regarding the existence of impairment indicators are based on legal factors, market conditions and the operational performance of our businesses. Actual impairment losses incurred could vary significantly from amounts estimated. Long-lived assets are assessed for impairment by asset group, the lowest level for which identifiable cash flows are largely independent of the cash flows of other groups of assets and liabilities. Additionally, future events could cause us to conclude that impairment indicators exist and that associated long-lived assets of our businesses are impaired. Any resulting impairment loss could have a material adverse impact on our financial condition and results of operations.

During the fourth quarter of 2023, the Westlake Epoxy business's sales volumes and prices, specifically base epoxy resins in Europe, continued to deteriorate. These lower sales volumes and prices were primarily driven by record exports out of Asia into Europe and North America. In addition, Westlake Epoxy operations in Europe have experienced sustained high energy and power costs. These factors negatively impacted Westlake Epoxy financial results during 2023. The Company identified these developments, along with management's outlook for the Westlake Epoxy business over the foreseeable future, as impairment indicators during the fourth quarter of 2023. Recoverability tests were performed for each of Westlake Epoxy's asset groups to compare the carrying amounts of assets to the net undiscounted cash flow projections of the asset group generated from its use and eventual disposition. The undiscounted cash flow projections were based on historical results, estimates made by management of future market conditions, current and future strategic and operational plans and future financial performance projected through the remaining useful life of the primary asset in the asset group. Based on the recoverability tests, we determined that the carrying amount of the primary assets of Westlake Epoxy's Netherlands asset group is not recoverable, and as such, an impairment loss was recorded to reduce the carrying amount of the asset group to its fair value. See Note 6 in the notes to the consolidated financial statements for further details.

The estimated useful lives of long-lived assets range from one to 40 years. Depreciation and amortization of these assets, including amortization of deferred turnaround costs, under the straight-line method over their estimated useful lives totaled $1,097 million, $1,056 million and $840 million in 2023, 2022 and 2021, respectively. If the useful lives of the assets were found to be shorter than originally estimated, depreciation or amortization charges would be accelerated.

We defer the costs of planned major maintenance activities, or turnarounds, and amortize the costs over the period until the next planned turnaround of the affected unit. Total costs deferred on turnarounds were $179 million, $178 million and $215 million in 2023, 2022 and 2021, respectively. As of December 31, 2023, deferred turnaround costs, net of accumulated amortization, totaled $391 million. Amortization in 2023, 2022 and 2021 of deferred turnaround costs was $137 million, $80 million and $56 million, respectively. Expensing turnaround costs as incurred would likely result in greater variability of our quarterly operating results and would adversely affect our financial position and results of operations.

Additional information concerning long-lived assets and related depreciation and amortization appears in Notes 6 and 8 to the consolidated financial statements appearing elsewhere in this Form 10-K.

Goodwill. At December 31, 2023, our recorded goodwill was $2,041 million. Goodwill is evaluated for impairment when events or changes in circumstances indicate the fair value of a reporting unit with goodwill has been reduced below its carrying value, and otherwise at least annually. We perform our annual impairment assessment for both the Performance and Essential Materials and Housing and Infrastructure Products reporting units in October each year. We may elect to perform an optional qualitative assessment to determine whether a quantitative impairment analysis is required. The qualitative assessment considers factors such as macroeconomic conditions, industry and market considerations, cost factors related to raw materials and labor, current and projected financial performance, changes in management or strategy, and market capitalization. Alternatively, we may unconditionally elect to bypass the qualitative assessment and perform a quantitative goodwill impairment assessment in any period.

We evaluated the Westlake Epoxy reporting unit's goodwill for impairment during the fourth quarter of 2023 in line with the timing of our annual goodwill impairment assessment for all reporting units. We performed the quantitative assessment for each of our reporting units within both of our segments during the fourth quarter of 2023. The quantitative analysis compares a reporting unit's fair value to its carrying amount to determine whether goodwill is impaired. The fair values of the reporting units are calculated using both a discounted cash flow methodology and a market value methodology. The discounted cash flow projections are based on a forecast to reflect the cyclicality of the business. The forecast is based on historical results, estimates by management of future market conditions, current and future strategic and operational plans and future financial performance. Significant assumptions used in the discounted cash flow projection included projected sales volumes based on production capacities, future sales prices, feedstock, energy and power costs and capital expenditures to maintain safe and reliable plant operations. The future cash flows are discounted to present value using an applicable discount rate. The significant assumptions used in determining the fair value of the reporting unit using the market value methodology include the determination of appropriate market comparables and the estimated multiples of EBITDA a willing buyer is likely to pay. Based on the quantitative tests performed during the fourth quarter of 2023, an impairment of the Westlake Epoxy reporting unit was identified as the carrying amount of the reporting unit exceeded its fair value. See Note 8 in the notes to the consolidated financial statements for further details. For all reporting units other than Westlake Epoxy, even if the fair values of the reporting units decreased by 10% from the fair values determined for the quantitative tests, the carrying amounts of the reporting units would not have exceeded their fair values. See Item 1A, "Risk Factors—If our goodwill or other long-lived assets become impaired in the future, we may be required to record non-cash charges to earnings, which could be significant."

Long-Term Employee Benefit Costs. Our costs for long-term employee benefits, particularly pension and postretirement medical and life benefits, are incurred over long periods of time and involve many uncertainties over those periods. The net periodic benefit cost attributable to current periods is based on several assumptions about such future uncertainties and is sensitive to changes in those assumptions. It is our responsibility, often with the assistance of independent experts, to select assumptions that represent the best estimates of those uncertainties. It is also our responsibility to review those assumptions periodically and, if necessary, adjust the assumptions to reflect changes in economic or other factors.

Accounting for employee retirement plans involves estimating the cost of benefits that are to be provided in the future and attempting to match, for each employee, that estimated cost to the period worked. To accomplish this, we rely extensively on advice from actuaries, and we make assumptions about inflation, investment returns, mortality, employee turnover and discount rates that ultimately impact amounts recorded. Changes in these assumptions may result in different expense and liability amounts. One of the more significant assumptions relates to the discount rate for measuring benefit obligations. At December 31, 2023, the projected pension benefit obligations for U.S. and non-U.S. plans were calculated using assumed weighted average discount rates of 5.0% and 3.2%, respectively. The discount rates were determined using a benchmark pension discount curve and applying spot rates from the curve to each year of expected benefit payments to determine the appropriate discount rate. As a result of the funding relief provided by the enactment of the Bipartisan Budget Act of 2015, no minimum funding requirements are expected during 2024 for the U.S. pension plans. Additional information on the 2024 funding requirements and key assumptions underlying these benefit costs appear in Note 14 to the consolidated financial statements appearing elsewhere in this Form 10-K.

The following table reflects the sensitivity of the benefit obligation of our pension plans to changes in the actuarial assumptions:

	2023	
	U.S. Plans	Non-U.S. Plans
	(In millions of dollars)	
Projected benefit obligation, end of year	$ 494	$ 629
Discount rate increases by 100 basis points	(41)	(82)
Discount rate decreases by 100 basis points	48	104

A one-percentage point increase or decrease in assumed healthcare trend rates would not have a significant effect on the amounts reported for the healthcare plans.

While we believe that the amounts recorded in the consolidated financial statements appearing elsewhere in this Form 10-K related to these retirement plans are based on the best estimates and judgments available, the actual outcomes could differ from these estimates.

Income Taxes. We utilize the balance sheet method of accounting for deferred income taxes. Under this method, deferred tax assets or liabilities are recorded based upon temporary differences between the tax basis of assets and liabilities and their carrying values for financial reporting purposes. Deferred tax expense or benefit is the result of changes in the deferred tax assets and liabilities during the period. Valuation allowances are recorded against deferred tax assets when it is considered more likely than not that the deferred tax assets will not be realized. Additional information on income taxes appears in Note 17 to the consolidated financial statements appearing elsewhere in this Form 10-K.

Environmental and Legal Obligations. We consult with various professionals to assist us in making estimates relating to environmental costs and legal proceedings. We accrue an expense when we determine that it is probable that a liability has been incurred and the amount is reasonably estimable. While we believe that the amounts recorded in the accompanying consolidated financial statements related to these contingencies are based on the best estimates and judgments available, the actual outcomes could differ from our estimates. Additional information about certain legal proceedings and environmental matters appears in Note 22 to the consolidated financial statements appearing elsewhere in this Form 10-K.

Asset Retirement Obligations. We recognize asset retirement obligations in the period in which the liability becomes probable and reasonably estimable. Initially, the asset retirement obligation is recorded at fair value and capitalized as a component of the carrying value of the long-lived asset to which the obligation relates. The liability is recorded at its future value each period, and the capitalized cost is depreciated over the estimated useful life of the related asset. Upon settlement of the liability, a gain or loss is recorded. We have conditional asset retirement obligations for the removal and disposal of hazardous materials from certain of our manufacturing facilities. Additional information on asset retirement obligations appears in Note 1, under Asset Retirement Obligations, to the consolidated financial statements appearing elsewhere in this Form 10-K.

We also have conditional asset retirement obligations that have not been recognized because the fair values of the conditional legal obligations cannot be measured due to the indeterminate settlement date of the obligations. Settlements of the unrecognized conditional asset retirement obligations are not expected to have a material adverse effect on our financial condition, results of operations or cash flows in any individual reporting period.

Recent Accounting Pronouncements

See Note 1 to the consolidated financial statements included in Item 8 of this Form 10-K for a full description of recent accounting pronouncements, including expected date of adoption and estimated effect on results of operations and financial condition.

Item 7A. Quantitative and Qualitative Disclosures about Market Risk

Commodity Price Risk

A substantial portion of our products and raw materials are commodities whose prices fluctuate as market supply and demand fundamentals change. Accordingly, product margins and the level of our profitability tend to fluctuate with changes in the business cycle. We try to protect against such instability through various business strategies. Our strategies include ethylene product feedstock flexibility and moving downstream into our other products where pricing is more stable. We use derivative instruments (including commodity swaps and options) in certain instances to reduce price volatility risk on feedstocks and products. Based on our open derivative positions at December 31, 2023, a hypothetical $0.10 increase in the price of a gallon of ethane and a hypothetical $0.10 increase in the price of a million British thermal units of natural gas would not have a material impact on our income before income taxes.

Interest Rate Risk

We are exposed to interest rate risk with respect to fixed and variable rate debt. At December 31, 2023, we had $4,973 million aggregate principal amount of fixed rate debt. We are subject to the risk of higher interest cost if and when this debt is refinanced. If interest rates were 1.0% higher at the time of refinancing, our annual interest expense would increase by approximately $50 million. Also, at December 31, 2023, we had $24 million principal amount of variable rate debt outstanding, which represents the 2026 term loans due 2026 and the tax-exempt waste disposal revenue bonds due 2027. We do not currently hedge our variable interest rate debt, but we may do so in the future. The weighted average variable interest rate for our variable rate debt of $24 million as of December 31, 2023 was 2.48%. A hypothetical 100 basis point increase in the average interest rate on our variable rate debt would not result in a material change in the interest expense.

Secured Overnight Financing Rate ("SOFR") is used as a reference rate for borrowings under our revolving line of credit. We did not have any SOFR-based borrowings outstanding at December 31, 2023.

Foreign Currency Exchange Rate Risk

We are exposed to foreign currency exchange rate risk associated with our international operations. However, the effect of fluctuations in foreign currency exchange rates caused by our international operations has not had a material impact on our overall operating results. We may engage in activities to mitigate our exposure to foreign currency exchange risk in certain instances through the use of currency exchange derivative instruments, including forward exchange contracts, cross-currency swaps or spot purchases. A forward exchange contract obligates us to exchange predetermined amounts of specified currencies at a stated exchange rate on a stated date. A cross-currency swap obligates us to make periodic payments in the local currency and receive periodic payments in our functional currency based on the notional amount of the instrument. In January 2018, we entered into foreign exchange hedging contracts designated as net investment hedges with an aggregate notional value of €220 million designed to reduce the volatility in stockholders' equity from changes in currency exchange rates associated with our net investments in foreign operations. In July 2019, we terminated a portion of the foreign exchange hedging contract with a notional value of €70 million. The notional value of the remaining net investment hedges was €150 million at December 31, 2023. The arrangement is scheduled to settle in 2026.

In July 2019, we completed the registered public offering of €700 million aggregate principal amount of the 1.625% 2029 Senior Notes. We designated this euro-denominated debt as a non-derivative net investment hedge of a portion of our net investments in euro functional-currency denominated subsidiaries to offset foreign currency fluctuations.

Item 8. Financial Statements and Supplementary Data

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Financial statement schedules not included in this Form 10-K have been omitted because they are not applicable or because the required information is shown in the financial statements or notes thereto.

MANAGEMENT'S REPORT ON INTERNAL CONTROL OVER FINANCIAL REPORTING

The management of Westlake Corporation is responsible for establishing and maintaining adequate internal control over financial reporting. Westlake's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles.

Westlake management assessed the effectiveness of the Company's internal control over financial reporting as of December 31, 2023. In making this assessment, management used the criteria set forth by the Committee of Sponsoring Organizations of the Treadway Commission in *Internal Control—Integrated Framework* (2013). Based on its assessment, Westlake's management has concluded that the Company's internal control over financial reporting was effective as of December 31, 2023 based on those criteria.

PricewaterhouseCoopers LLP, the independent registered public accounting firm that audited the financial statements included in this Annual Report on Form 10-K, has also audited the effectiveness of internal control over financial reporting as of December 31, 2023 as stated in their report that appears on the following page.

Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of Westlake Corporation

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Westlake Corporation and its subsidiaries (the "Company") as of December 31, 2023 and 2022, and the related consolidated statements of operations, of comprehensive income, of changes in stockholders' equity and of cash flows for each of the three years in the period ended December 31, 2023, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control—Integrated Framework* (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in *Internal Control—Integrated Framework* (2013) issued by the COSO.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management's Report on Internal Control over Financial Reporting appearing under Item 8. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Goodwill Impairment Assessment—Westlake Epoxy Reporting Unit

As described in Notes 1 and 8 to the consolidated financial statements, management tests goodwill for impairment at least annually in the fourth quarter of each year, or when events or changes in circumstances indicate the fair value of a reporting unit with goodwill has been reduced below its carrying amount. The fair value of the reporting unit was determined using both a discounted cash flow methodology and a market value methodology. Significant assumptions used in the discounted cash flow projection included projected sales volumes based on production capacities, future sales prices, feedstock, energy and power costs and capital expenditures to maintain safe and reliable plant operations. The future cash flows were discounted to present value using a discount rate. The significant assumptions used in determining the fair value of the reporting unit using the market value methodology include the determination of appropriate market comparables and the estimated multiples of net income before interest expense, income taxes, depreciation and amortization (EBITDA) a willing buyer is likely to pay. Based on the quantitative tests performed during the fourth quarter of 2023, management determined that the fair value of the Westlake Epoxy reporting unit did not exceed its carrying amount, and as such, a goodwill impairment charge of $128 million was recognized.

The principal considerations for our determination that performing procedures relating to the goodwill impairment assessment of the Westlake Epoxy reporting unit is a critical audit matter are (i) the significant judgment by management when developing the fair value estimate of the Westlake Epoxy reporting unit; (ii) a high degree of auditor judgment, subjectivity, and effort in performing procedures and evaluating management's significant assumptions related to future sales prices, feedstock cost, the discount rate, and estimated EBITDA; and (iii) the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to management's goodwill impairment assessment, including controls over the valuation of the Westlake Epoxy reporting unit. These procedures also included, among others (i) testing management's process for developing the fair value estimate of the reporting unit; (ii) evaluating the appropriateness of the discounted cash flow and market value methodologies used by management; (iii) testing the completeness and accuracy of underlying data used in the discounted cash flow and market value methodologies; and (iv) evaluating the reasonableness of the significant assumptions used by management related to future sales prices, feedstock cost, the discount rate, and estimated EBITDA. Evaluating management's assumptions related to future sales prices, feedstock cost, and estimated EBITDA involved evaluating whether the assumptions used by management were reasonable considering (i) the current and past performance of the Westlake Epoxy reporting unit; (ii) the consistency with external industry data; and (iii) whether the assumptions were consistent with evidence obtained in other areas of the audit. Professionals with specialized skill and knowledge were used to assist in evaluating (i) the appropriateness of the discounted cash flow and market value methodologies and (ii) the reasonableness of the discount rate assumption.

/s/PricewaterhouseCoopers LLP
Houston, Texas
February 21, 2024

We have served as the Company's auditor since 1986, which includes periods before the Company became subject to SEC reporting requirements.

WESTLAKE CORPORATION
CONSOLIDATED BALANCE SHEETS

	December 31,	
	2023	2022
	(in millions of dollars, except par values and share amounts)	
ASSETS		
Current assets		
Cash and cash equivalents	$ 3,304	$ 2,228
Accounts receivable, net	1,601	1,801
Inventories	1,622	1,866
Prepaid expenses and other current assets	82	78
Total current assets	6,609	5,973
Property, plant and equipment, net	8,519	8,477
Operating lease right-of-use assets	697	615
Goodwill	2,041	2,161
Customer relationships, net	910	993
Other intangible assets, net	493	572
Equity method investments	1,115	1,142
Other assets, net	651	617
Total assets	$ 21,035	$ 20,550
LIABILITIES AND EQUITY		
Current liabilities		
Accounts payable	$ 877	$ 889
Accrued and other liabilities	1,614	1,409
Current portion of long-term debt, net	299	—
Total current liabilities	2,790	2,298
Long-term debt, net	4,607	4,879
Deferred income taxes	1,560	1,735
Pension and other post-retirement benefits	363	355
Operating lease liabilities	611	504
Other liabilities	340	314
Total liabilities	10,271	10,085
Commitments and contingencies (Note 22)		
Stockholders' equity		
Preferred stock, $0.01 par value, 50,000,000 shares authorized; no shares issued and outstanding	—	—
Common stock, $0.01 par value, 300,000,000 shares authorized; 134,651,380 and 134,651,380 shares issued at December 31, 2023 and 2022, respectively	1	1
Common stock, held in treasury, at cost; 6,439,289 and 7,278,651 shares at December 31, 2023 and 2022, respectively	(435)	(467)
Additional paid-in capital	630	601
Retained earnings	10,143	9,885
Accumulated other comprehensive loss	(98)	(89)
Total Westlake Corporation stockholders' equity	10,241	9,931
Noncontrolling interests	523	534
Total equity	10,764	10,465
Total liabilities and equity	$ 21,035	$ 20,550

The accompanying notes are an integral part of these consolidated financial statements.

WESTLAKE CORPORATION

CONSOLIDATED STATEMENTS OF OPERATIONS

	Year Ended December 31,		
	2023	2022	2021
	(in millions of dollars, except per share data and share amounts)		
Net sales	$ 12,548	$ 15,794	$ 11,778
Cost of sales	10,329	11,721	8,283
Gross profit	2,219	4,073	3,495
Selling, general and administrative expenses	865	835	551
Impairment of goodwill and long-lived assets	475	—	—
Amortization of intangibles	122	155	123
Restructuring, transaction and integration-related costs	28	33	21
Income from operations	729	3,050	2,800
Other income (expense)			
Interest expense	(165)	(177)	(176)
Other income, net	136	73	53
Income before income taxes	700	2,946	2,677
Provision for income taxes	178	649	607
Net income	522	2,297	2,070
Net income attributable to noncontrolling interests	43	50	55
Net income attributable to Westlake Corporation	**$ 479**	**$ 2,247**	**$ 2,015**
Earnings per common share attributable to Westlake Corporation:			
Basic	$ 3.73	$ 17.46	$ 15.66
Diluted	$ 3.70	$ 17.34	$ 15.58
Weighted average common shares outstanding:			
Basic	127,806,317	127,970,445	128,002,911
Diluted	128,598,441	128,845,562	128,697,982

The accompanying notes are an integral part of these consolidated financial statements.

WESTLAKE CORPORATION
CONSOLIDATED STATEMENTS OF COMPREHENSIVE INCOME

	Year Ended December 31,		
	2023	2022	2021
	(in millions of dollars)		
Net income	$ 522	$ 2,297	$ 2,070
Other comprehensive income (loss), net of income taxes			
Pension and other post-retirement benefits			
Pension and other post-retirement benefits reserves adjustment	(40)	29	60
Income tax benefit (provision) on pension and other post-retirement benefits	(8)	3	(16)
Foreign currency translation adjustments			
Foreign currency translation	32	(78)	2
Income tax benefit (provision) on foreign currency translation	7	(12)	(17)
Other comprehensive income (loss), net of income taxes	(9)	(58)	29
Comprehensive income	513	2,239	2,099
Comprehensive income attributable to noncontrolling interests, net of tax of $3, $3 and $2 for 2023, 2022 and 2021, respectively	43	45	56
Comprehensive income attributable to Westlake Corporation	$ 470	$ 2,194	$ 2,043

The accompanying notes are an integral part of these consolidated financial statements.

WESTLAKE CORPORATION

CONSOLIDATED STATEMENTS OF CHANGES IN STOCKHOLDERS' EQUITY

	Common Stock		Common Stock, Held in Treasury						
	Number of Shares	Amount	Number of Shares	At Cost	Additional Paid-in Capital	Retained Earnings	Accumulated Other Comprehensive Income (Loss)	Noncontrolling Interests	Total
	(in millions of dollars, except share amounts)								
Balances at December 31, 2020	134,651,380	$ 1	6,821,174	$ (401)	$ 569	$ 5,938	$ (64)	$ 535	$ 6,578
Net income	—	—	—	—	—	2,015	—	55	2,070
Other comprehensive income	—	—	—	—	—	—	28	1	29
Common stock repurchased	—	—	355,800	(30)	—	—	—	—	(30)
Shares issued—stock-based compensation	—	—	(441,335)	32	(19)	—	—	—	13
Stock-based compensation	—	—	—	—	31	—	—	—	31
Dividends declared	—	—	—	—	—	(145)	—	—	(145)
Distributions to noncontrolling interests	—	—	—	—	—	—	—	(48)	(48)
Noncontrolling interests	—	—	—	—	—	—	—	30	30
Balances at December 31, 2021	134,651,380	$ 1	6,735,639	$ (399)	$ 581	$ 7,808	$ (36)	$ 573	$ 8,528
Net income	—	—	—	—	—	2,247	—	50	2,297
Other comprehensive income	—	—	—	—	—	—	(53)	(5)	(58)
Common stock repurchased	—	—	1,079,736	(102)	—	—	—	—	(102)
Shares issued—stock-based compensation	—	—	(536,724)	34	(16)	—	—	—	18
Stock-based compensation	—	—	—	—	35	—	—	—	35
Dividends declared	—	—	—	—	—	(169)	—	—	(169)
Distributions to noncontrolling interests	—	—	—	—	—	—	—	(60)	(60)
Noncontrolling interests	—	—	—	—	1	(1)	—	(24)	(24)
Balances at December 31, 2022	134,651,380	$ 1	7,278,651	$ (467)	$ 601	$ 9,885	$ (89)	$ 534	$ 10,465
Net income	—	—	—	—	—	479	—	43	522
Other comprehensive income	—	—	—	—	—	—	(9)	—	(9)
Common stock repurchased	—	—	211,294	(23)	—	—	—	—	(23)
Shares issued—stock-based compensation	—	—	(1,050,656)	55	(11)	—	—	—	44
Stock-based compensation	—	—	—	—	40	—	—	—	40
Dividends declared	—	—	—	—	—	(221)	—	—	(221)
Distributions to noncontrolling interests	—	—	—	—	—	—	—	(54)	(54)
Balances at December 31, 2023	134,651,380	$ 1	6,439,289	$ (435)	$ 630	$ 10,143	$ (98)	$ 523	$ 10,764

The accompanying notes are an integral part of these consolidated financial statements.

WESTLAKE CORPORATION

CONSOLIDATED STATEMENTS OF CASH FLOWS

	Year Ended December 31,		
	2023	2022	2021
		(in millions of dollars)	
Cash flows from operating activities			
Net income	$ 522	$ 2,297	$ 2,070
Adjustments to reconcile net income to net cash provided by operating activities			
Depreciation and amortization	1,097	1,056	840
Stock-based compensation expense	43	36	31
Loss from disposition and write-off of property, plant and equipment	45	46	28
Impairment of goodwill and long-lived assets	475	—	—
Deferred income taxes	(175)	(21)	23
Other (gains) losses, net	(3)	5	16
Changes in operating assets and liabilities, net of effect of business acquisitions			
Accounts receivable	225	325	(528)
Inventories	250	(140)	(309)
Prepaid expenses and other current assets	(19)	5	(27)
Accounts payable	(26)	(153)	242
Accrued and other liabilities	170	137	239
Other, net	(268)	(198)	(231)
Net cash provided by operating activities	2,336	3,395	2,394
Cash flows from investing activities			
Acquisition of businesses, net of cash acquired	—	(1,203)	(2,554)
Additions to investments in unconsolidated subsidiaries	(25)	(180)	(24)
Additions to property, plant and equipment	(1,034)	(1,108)	(658)
Other, net	22	12	23
Net cash used for investing activities	(1,037)	(2,479)	(3,213)
Cash flows from financing activities			
Debt issuance costs	—	—	(18)
Distributions to noncontrolling interests	(54)	(60)	(48)
Dividends paid	(221)	(169)	(145)
Proceeds from debt issuance and drawdown of revolver, net	—	—	1,671
Proceeds from exercise of stock options	44	18	13
Repayment of revolver and senior notes	—	(250)	—
Repurchase of common stock for treasury	(23)	(101)	(30)
Other, net	9	(25)	(6)
Net cash provided by (used for) financing activities	(245)	(587)	1,437
Effect of exchange rate changes on cash, cash equivalents and restricted cash	19	(24)	(14)
Net increase in cash, cash equivalents and restricted cash	1,073	305	604
Cash, cash equivalents and restricted cash at beginning of the year	2,246	1,941	1,337
Cash, cash equivalents and restricted cash at end of the year	$ 3,319	$ 2,246	$ 1,941

The accompanying notes are an integral part of these consolidated financial statements.

WESTLAKE CORPORATION

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS
(in millions of dollars, except share amounts and per share data)

1. Description of Business and Significant Accounting Policies

Description of Business

Westlake Corporation, formerly known as Westlake Chemical Corporation (the "Company") operates as an integrated global manufacturer and marketer of performance and essential materials and housing and infrastructure products. These products include some of the most widely used materials in the world, which are fundamental to many diverse consumer and industrial markets, including residential construction, flexible and rigid packaging, automotive products, healthcare products, water treatment, coatings as well as other durable and non-durable goods. The Company's customers range from large chemical processors and plastics fabricators to small construction contractors, municipalities and supply warehouses throughout North America, Europe and Asia. The industries in which the Company operates are subject to price fluctuations and volatile feedstock pricing typical of a commodity-based industry, the effects of which may not be immediately passed along to customers.

Westlake Chemical Partners LP

In 2014, the Company formed Westlake Chemical Partners LP ("Westlake Partners") to operate, acquire and develop ethylene production facilities and related assets. Westlake Partners' assets consist of a limited partner interest in Westlake Chemical OpCo LP ("OpCo"), as well as the general partner interest in OpCo. OpCo's assets include two ethylene production facilities at the Company's Lake Charles, Louisiana site,one ethylene production facility at the Company's Calvert City, Kentucky site and a 200-mile common carrier ethylene pipeline that runs from Mont Belvieu, Texas to the Company's Longview, Texas site. As of December 31, 2023, the Company held a 77.2% limited partner interest in OpCo and a controlling interest in Westlake Partners. The operations of Westlake Partners are consolidated in the Company's financial statements.

Principles of Consolidation

The consolidated financial statements include the accounts of the Company and subsidiaries in which the Company directly or indirectly owns more than a 50% voting interest and exercises control and, when applicable, entities for which the Company has a controlling financial interest or is the primary beneficiary. Investments in majority-owned companies where the Company does not exercise control and investments in nonconsolidated affiliates (20%-50% owned companies, joint ventures and partnerships) are accounted for using the equity method of accounting. Undistributed earnings from joint ventures included in retained earnings were immaterial as of December 31, 2023. All intercompany transactions and balances are eliminated in consolidation.

Certain reclassifications have been made to the prior-year financial statements to conform to the current year presentation.

Noncontrolling interests represent the direct equity interests held by investors in the Company's consolidated subsidiaries, Westlake Partners, Taiwan Chlorine Industries, Ltd. and Suzhou Huasu Plastics Co., Ltd.

Effective January 2021, the Company consolidated RS Cogen, LLC ("RS Cogen") into its consolidated financial statements. Effective October 31, 2022, the Company acquired the remaining 50% interest in RS Cogen.

Basis of Presentation

The accompanying consolidated financial statements have been prepared in conformity with the accounting principles generally accepted in the United States.

Cash and Cash Equivalents

Cash equivalents consist of highly liquid investments that are readily convertible into cash and have a maturity of three months or less at the date of acquisition.

Concentration of Credit Risk

Financial instruments which potentially subject the Company to concentration of risk consist principally of trade receivables from customers engaged in manufacturing polyethylene products, polyvinyl chloride ("PVC") products, epoxy products, chlor-alkali and derivative products as well as housing and infrastructure products such as PVC compounds, PVC pipe and fittings, trim and mouldings, stone, windows, roofing and exterior products. The Company's large number of customers, the diversity of these customers' businesses and the markets they serve and customers geographic dispersion limits the concentrations of credit risk with respect to receivables. The Company performs periodic credit evaluations of the customers' financial condition and generally does not require collateral. The Company maintains allowances for potential losses.

Allowance for Credit Losses

The determination of the allowance for credit losses is based on estimation of the amount of accounts receivable that the Company believes are unlikely to be collected. Estimating this amount requires analysis of the financial strength of the Company's customers, the use of historical experience, the Company's accounts receivable aged trial balance, customer specific collectability analysis and an evaluation of economic conditions. The allowance for credit losses is reviewed quarterly. Past due balances over 90 days and high-risk accounts as determined by the analysis of financial strength of customers are reviewed individually for collectability.

Inventories

Inventories primarily include product, material and supplies. Inventories are stated at lower of cost or net realizable value. Cost is determined using the first-in, first-out ("FIFO") or average method.

Property, Plant and Equipment

Property, plant and equipment are carried at cost, net of accumulated depreciation. Cost includes expenditures for improvements and betterments that extend the useful lives of the assets and interest capitalized on significant capital projects. Capitalized interest was $8, $4 and $3 for the years ended December 31, 2023, 2022 and 2021, respectively. Repair and maintenance costs are charged to operations as incurred. Gains and losses on the disposition or retirement of fixed assets are reflected in the consolidated statement of operations when the assets are sold or retired.

Depreciation is provided by utilizing the straight-line method over the estimated useful lives of the assets as follows:

Classification	Years
Buildings and improvements	40
Plant and equipment	10-25
Other	3-15

Impairment of Long-Lived Assets

The Company reviews long-lived assets, including tangible assets and intangible assets with finite lives, for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset may not be recoverable. Long-lived assets assessed for impairment are grouped by asset group, the lowest level for which identifiable cash flows are largely independent of the cash flows of other groups of assets and liabilities. Recoverability of assets to be held and used is measured by a comparison of the carrying amount of an asset to future net undiscounted cash flows expected to be generated by the asset group from its use and eventual disposition of that asset group. Assets are considered to be impaired if the carrying amount of an asset exceeds the future undiscounted cash flows. The impairment recognized is measured by the amount by which the carrying amount of the assets exceeds the fair value of the assets. Assets to be disposed of are reported at the lower of the carrying amount or estimated fair value less costs to sell. Intangible assets with finite lives are amortized over their estimated useful lives and evaluated with the associated long-lived asset group for impairment whenever impairment indicators exist. See Note 6 for more information on the Company's long-lived assets impairment tests.

Impairment of Goodwill

Goodwill is tested for impairment at least annually, or when events or changes in circumstances indicate the fair value of a reporting unit with goodwill has been reduced below its carrying amount. The Company performed its annual goodwill impairment tests for each of the reporting units within the Performance and Essential Materials and Housing and Infrastructure Products segments in the fourth quarter of 2023. The tests indicated impairment of the Westlake Epoxy reporting unit under the Performance and Essential Materials segment as the carrying amount of the reporting unit exceeded its fair value. See Note 8 for more information on the Company's annual goodwill impairment tests.

Equity Method Investments

The Company accounts for investments using the equity method of accounting if the Company has the ability to exercise significant influence over, but not control of, an investee. Significant influence generally exists if the Company has an ownership interest representing between 20% and 50% of the voting rights. Under the equity method of accounting, investments are stated initially at cost and are adjusted for subsequent additional investments and the proportionate share of profit or losses and distributions. The Company records its share of the profits or losses of the equity investments, net of income taxes, in the consolidated statements of income. The equity method investments are evaluated for impairment when events or changes in circumstances indicate, in management's judgment, that the carrying value of such investments may have experienced an other-than-temporary decline in value. When evidence of loss in value has occurred, management compares the estimated fair value of the investment to the carrying value of the investment to determine whether an impairment has occurred. If the estimated fair value is less than the carrying value and management considers the decline in value to be other-than-temporary, the excess of the carrying value over the estimated fair value is recognized in the consolidated financial statements as an impairment.

Other Assets, net

Other assets, net include turnaround costs, investments in unconsolidated subsidiaries, restricted cash, deferred charges and other long-term assets.

The Company accounts for turnaround costs under the deferral method. Turnarounds are the scheduled and required shutdowns of specific operating units in order to perform planned major maintenance activities. The costs related to the significant overhaul and refurbishment activities include maintenance materials, parts and direct labor costs. The costs of the turnaround are deferred when incurred at the time of the turnaround and amortized (within depreciation and amortization) on a straight-line basis until the next planned turnaround, which typically range from three to six years. Deferred turnaround costs are presented as a component of other assets, net. The cash outflows related to these costs are included in operating activities in the consolidated statement of cash flows.

Business Combinations

The Company records business combinations using the acquisition method of accounting. Under the acquisition method of accounting, identifiable assets acquired and liabilities assumed are recorded at their acquisition date fair values. The excess of the purchase price over the estimated fair value is recorded as goodwill. Changes in the estimated fair values of net assets recorded for acquisitions prior to the finalization of more detailed analysis, but not to exceed one year from the date of acquisition, will adjust the amount of the purchase price allocable to goodwill. Measurement period adjustments are reflected in the period in which they occur.

Income Taxes

The Company utilizes the balance sheet method of accounting for deferred income taxes. Under this method, deferred tax assets or liabilities are recorded based upon temporary differences between the tax basis of assets and liabilities and their carrying values for financial reporting purposes. Deferred tax expense or benefit is the result of changes in the deferred tax assets and liabilities during the period. Valuation allowances are recorded against deferred tax assets when it is considered more likely than not that the deferred tax assets will not be realized.

Foreign Currency Translation

Assets and liabilities of foreign subsidiaries are translated to U.S. dollars at the exchange rate as of the end of the year. Statement of operations items are translated at the average exchange rate for the year. The resulting translation adjustment is recorded as a separate component of stockholders' equity.

Revenue Recognition

Revenue is recognized when the Company transfers control of inventories to its customers. Amounts recognized as revenues reflect the consideration to which the Company expects to be entitled in exchange for those inventories. Provisions for discounts, rebates and returns are incorporated in the estimate of variable consideration and reflected as a reduction to revenue in the same period as the related sales.

Control of inventories generally transfers upon shipment for domestic sales. The Company excludes taxes collected on behalf of customers from the estimated contract price. For export contracts, the point at which control passes to the customer varies depending on the terms specified in the customer contract.

The Company generally invoices customers and recognizes revenue and accounts receivable upon transferring control of inventories. In limited circumstances, the Company transfers control of inventories shortly before it has an unconditional right to receive consideration, resulting in recognition of contract assets. The Company also receives advance payments from certain customers, resulting in recognition of contract liabilities. Contract assets and liabilities are generally settled within the period and are not material to the consolidated balance sheets. The Company expenses sales commissions when incurred. These costs are recorded within selling, general and administrative expenses. Aside from the amounts disclosed within Note 9, the Company does not disclose the value of unsatisfied performance obligations because its contracts with customers (1) have an original expected duration of one year or less or (2) have only variable consideration that is calculated based on market prices at specified dates and is allocated to wholly unsatisfied performance obligations.

Revenue from Contracts with Customers ("ASC 606") requires disclosure of disaggregated revenue into categories that depict the nature of how the Company's revenue and cash flows are affected by economic factors. The Company discloses revenues by business and segment in Note 23.

Leases

The Company is obligated under various long-term and short-term operating leases for rail cars, buildings, land and other transportation and storage assets. The Company determines whether an arrangement is, or contains, a lease at contract inception. Some of the Company's arrangements contain both lease and non-lease components. For certain transportation equipment leases, the Company accounts for the lease and non-lease components as a single lease component. The Company records right-of-use assets and corresponding lease liabilities for operating leases with terms greater than one year. Operating lease right-of-use assets and liabilities are recorded at the present value of the fixed lease payments over the life of the lease. The majority of the Company's leases do not provide an implicit rate. Therefore, the Company uses its incremental borrowing rate at lease commencement to measure operating lease right-of-use assets and lease liabilities. Certain of the Company's leases provide for renewal and purchase options. Renewal and purchase options are evaluated at lease commencement and included in the lease term if they are reasonably certain to be exercised. Short-term leases are recognized in rental expense on a straight-line basis over the lease term and are not recorded in the consolidated balance sheets. The Company's finance leases are not material to the consolidated financial statements.

Transportation and Freight

Amounts billed to customers for freight and handling costs on outbound shipments are included in net sales in the consolidated statements of operations. Transportation and freight costs incurred by the Company on outbound shipments are included in cost of sales in the consolidated statements of operations.

Price Risk Management

The Company recognizes derivative instruments on the balance sheet at fair value, and changes in a derivative's fair value are currently recognized in earnings or comprehensive income, depending on the designation of the derivative. If the derivative is designated as a fair value hedge, the changes in the fair value of the derivative and of the hedged item attributable to the hedged risk are recognized in earnings. If the derivative is designated as a cash flow hedge, the effective portion of the change in the fair value of the derivative is recorded in comprehensive income and is recognized in the statement of operations when the hedged item affects earnings. Ineffective portions of changes in the fair value of cash flow hedges are recognized in earnings currently. The derivative instruments did not have a material impact on the Company's consolidated financial statements.

Asset Retirement Obligations

The Company has conditional asset retirement obligations for the removal and disposal of hazardous materials from certain of the Company's manufacturing facilities.

The Company recognizes asset retirement obligations in the period in which the liability becomes probable and reasonably estimable. Recognized asset retirement obligations are initially recorded at fair value and capitalized as a component of the carrying value of the long-lived asset to which the obligation relates. The liability is accreted to its future value each period, and the capitalized cost is depreciated over the estimated useful life of the related asset. Upon settlement of the liability, a gain or loss is recorded. As of December 31, 2023, the Company had $3 and $37 of asset retirement obligations recorded as accrued and other liabilities and other liabilities, respectively. As of December 31, 2022, the Company had no asset retirement obligations recorded as accrued and other liabilities and had $31 of asset retirement obligations recorded as other liabilities.

The Company also has conditional asset retirement obligations that have not been recognized because the fair values of the conditional legal obligations cannot be measured due to the indeterminate settlement date of the obligations. Settlements of the unrecognized conditional asset retirement obligations are not expected to have a material adverse effect on the Company's financial condition, results of operations or cash flows in any individual reporting period.

Environmental Costs

Environmental costs relating to current operations are expensed or capitalized, as appropriate, depending on whether such costs provide future economic benefits. Remediation liabilities are recognized when the costs are considered probable and can be reasonably estimated. Measurement of liabilities is based on currently enacted laws and regulations, existing technology and undiscounted site-specific costs. Environmental liabilities in connection with properties that are sold or closed are realized upon such sale or closure, to the extent they are probable and estimable and not previously reserved. Recognition of any joint and several liabilities is based upon the Company's best estimate of its final pro rata share of the liability.

Use of Estimates

The preparation of financial statements in conformity with accounting principles generally accepted in the United States of America requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues and expenses, and the disclosure of contingent assets and liabilities. Actual results could differ from those estimates.

Warranty Costs

The Company provides warranties for certain products, primarily under the housing and infrastructure segment, against defects in material and performance. The accrual for warranty claims is recorded at the time of sale based on historical warranty claims experience. Warranty liabilities are included in accrued liabilities and other liabilities in the consolidated balance sheets.

The warranty liabilities activity for the years ended December 31, 2023 and 2022 is as follows:

	Year Ended December 31,	
	2023	2022
Beginning balance, January 1,	$ 58	$ 55
Estimated fair value of warranty liability assumed in acquisition	—	1
Warranty provisions	24	15
Warranty claims paid	(13)	(13)
Ending balance, December 31,	$ 69	$ 58

Recently Issued Accounting Pronouncements

Income Taxes (Topic 740): Improvements to Income Tax Disclosures (ASU No.2023-09)

In December 2023, the Financial Accounting Standards Board ("FASB") issued accounting standards update No. 2023-09 to enhance the transparency and decision-usefulness of income tax disclosures and to provide information to better assess how an entity's operations and related tax risks and tax planning and operational opportunities affect its tax rate and prospects for future cash flows. For public business entities, the amendments in this update are effective for annual periods beginning after December 15, 2024. The Company is in the process of evaluating the impact of this standard on the disclosures in the Company's financial statements.

Segment Reporting (Topic 280): Improvements to Reportable Segment Disclosures (ASU 2023-07)

In November 2023, the FASB issued accounting standards update No. 2023-07 to improve reportable segment disclosure requirements, primarily through enhanced disclosures about significant segment expenses. The update requires public entities to disclose, on an annual and interim basis, significant segment expenses that are regularly provided to the chief operating decision maker (CODM), the title and position of the CODM and an explanation of how the CODM uses the reported measure(s) of segment profit or loss in assessing segment performance and deciding how to allocate resources. The amendments in this update are effective for public entities in fiscal years beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024 and are to be applied retrospectively to all prior periods presented in the financial statements. Early adoption is permitted. The Company is in the process of evaluating the impact of this standard on the disclosures in the Company's financial statements.

Business Combinations—Joint Venture Formations (Subtopic 805-60): Recognition and Initial Measurement (ASU 2023-05)

In August 2023, the FASB issued accounting standards update No. 2023-05 to address the accounting for contributions made to a joint venture, upon formation, in a joint venture's financial statements, and to (1) provide decision-useful information to investors and other allocators of capital (collectively, investors) in a joint venture's financial statements and (2) reduce diversity in practice. Under the ASU, upon formation, a joint venture should recognize and initially measure its assets and liabilities at fair value (with exceptions to fair value measurement that are consistent with guidance for business combinations). The amendments in this update become effective for all joint venture formations with a formation date on or after January 1, 2025. Additionally, a joint venture that was formed before January 1, 2025 may elect to apply the amendments retrospectively if it has sufficient information. Early adoption is permitted in any interim or annual period in which financial statements have not yet been issued (or made available for issuance), either prospectively or retrospectively. The Company does not expect that this accounting standard will have a material impact, upon adoption, on the Company's consolidated financial position, results of operations and cash flows.

Leases (Topic 842): Common Control Arrangements (ASU 2023-01)

In March 2023, the FASB issued accounting standards update No. 2023-01 to amend certain provisions of ASC 842 that apply to arrangements between related parties under common control. The update requires all companies to amortize leasehold improvements associated with common control leases over the asset's useful life to the common control group regardless of the lease term. The amendment in this update is effective for all entities in fiscal years beginning after December 15, 2023, including interim periods within those fiscal years. Early application is permitted. The Company does not expect that this accounting standard will have a material impact, upon adoption, on the Company's consolidated financial position, results of operations and cash flows.

Recently Adopted Accounting Standards

Liabilities—Supplier Finance Programs (ASU No. 2022-04)

In September 2022, the FASB issued accounting standards update No. 2022-04 to enhance transparency of supplier finance programs. Under the ASU, the buyer in a supplier finance program is required to disclose information about the key terms of the program, outstanding confirmed amounts as of the end of the period, a rollforward of the amount of obligations confirmed and the amount of obligations subsequently paid, and a description of where in the financial statements outstanding amounts are presented. The amendments in this update became effective for all entities for fiscal years beginning after December 15, 2022, including interim periods within those fiscal years, except for the disclosure of rollforward information, which is effective for fiscal years beginning after December 15, 2023. The Company adopted this accounting standard effective January 1, 2023, and the adoption did not have a material impact on the Company's consolidated financial position, results of operations and cash flows.

Business Combinations—Accounting for Contract Assets and Contract Liabilities from Contracts with Customers Update (ASU No.2021-08)

In October 2021, the FASB issued an accounting standards update that requires acquiring entities to recognize and measure contract assets and contract liabilities in a business combination in accordance with the accounting guidance on Revenue from Contracts with Customers (ASC 606). The guidance in this update improves comparability for both the recognition and measurement of acquired revenue contracts with customers at the date of and after a business combination. The accounting standard became effective for reporting periods beginning after December 15, 2022. The Company adopted this accounting standard effective January 1, 2023, and the adoption did not have a material impact on the Company's consolidated financial position, results of operations and cash flows. The standard will be applicable to future business combinations.

2. Acquisitions

Hexion Epoxy Business.

On February 1, 2022, the Company completed its acquisition of Hexion's global epoxy business ("Westlake Epoxy") for total consideration of $1,207. The assets acquired and liabilities assumed and the results of operations of the Westlake Epoxy business are included in the Performance and Essential Materials segment. The purchase accounting adjustments for the year ended December 31, 2023 resulted in a $4 increase in goodwill. The intangible assets that have been acquired are being amortized over periods of 11 to 19 years, except for certain intangible assets that were subject to impairment in the fourth quarter of 2023 as discussed in Note 6.

The information below represents the purchase price allocation:

Cash	$	42
Accounts receivable		299
Inventories		346
Prepaid expenses and other current assets		22
Property, plant and equipment		664
Operating lease right-of-use assets		59
Intangible assets		145
Other assets		104
Total assets acquired		1,681
Accounts payable		189
Accrued and other liabilities		88
Deferred income taxes		96
Pensions and other post-retirement benefits		163
Operating lease liabilities		48
Other liabilities		19
Total liabilities assumed		603
Total identifiable net assets acquired		1,078
Noncontrolling interest		(2)
Goodwill		131
Total Westlake Corporation purchase consideration	$	1,207

Boral Target Companies in North America

On October 1, 2021, the Company completed its acquisition of Boral Limited's North American building products businesses in roofing, siding, trim and shutters, decorative stone and windows (the "Boral Target Companies") for total consideration of $2,140. The assets acquired and liabilities assumed and the results of operations of the Boral Target Companies are included in the Housing and Infrastructure Products segment.

LASCO Fittings, Inc.

On August 19, 2021, the Company completed its acquisition of LASCO Fittings, Inc., a Delaware corporation ("LASCO"), a manufacturer of injected-molded polyvinyl chloride ("PVC") fittings that serve the plumbing, pool and spa, industrial, irrigation and retail markets in the United States, for total consideration of $277. The assets acquired and liabilities assumed and the results of operations of LASCO are included in the Housing and Infrastructure Products segment.

Dimex LLC.

On September 10, 2021, the Company completed its acquisition of DX Acquisition Corp., a Delaware corporation ("Dimex"), a producer of various consumer products made from post-industrial-recycled PVC, polyethylene and thermoplastic elastomer materials, including, landscape edging; home, office and industrial matting; marine dock edging; and masonry joint controls. The total consideration was $172. The assets acquired and liabilities assumed and the results of operations of Dimex are included in the Housing and Infrastructure Products segment.

3. Financial Instruments

Cash Equivalents

The Company had $360 of held-to-maturity securities with original maturities of three months or less, primarily consisting of corporate debt securities, classified as cash equivalents at December 31, 2023. There were no held-to-maturity securities at December 31, 2022. The Company's investments in held-to-maturity securities were held at amortized cost, which approximates fair value.

Restricted Cash and Cash Equivalents

The Company had restricted cash and cash equivalents of $15 and $18 at December 31, 2023 and 2022, respectively. The Company's restricted cash and cash equivalents are primarily related to balances that are restricted for payment of distributions to certain of the Company's current and former employees and are reflected primarily in other assets, net in the consolidated balance sheets.

4. Accounts Receivable

Accounts receivable consist of the following at December 31:

	2023	2022
Trade customers	$ 1,413	$ 1,676
Related parties	7	3
Allowance for credit losses	(27)	(28)
	1,393	1,651
Federal and state taxes	65	69
Other	143	81
Accounts receivable, net	$ 1,601	$ 1,801

5. Inventories

Inventories consist of the following at December 31:

	2023	2022
Finished products	$ 989	$ 1,157
Feedstock, additives, chemicals and other raw materials	401	496
Materials and supplies	232	213
Inventories	$ 1,622	$ 1,866

6. Property, Plant and Equipment

Property, plant and equipment consist of the following at December 31:

	2023	2022
Land	$ 304	$ 299
Buildings and improvements	951	895
Plant and equipment	10,771	10,408
Other	803	711
	12,829	12,313
Less: Accumulated depreciation	5,240	4,655
	7,589	7,658
Construction in progress	930	819
Property, plant and equipment, net	$ 8,519	$ 8,477

Depreciation expense on property, plant and equipment of $763, $762 and $604 is included primarily in cost of sales in the consolidated statements of operations for the years ended December 31, 2023, 2022 and 2021, respectively.

Westlake Epoxy

During the fourth quarter of 2023, the Westlake Epoxy business's sales volumes and prices, specifically base epoxy resins in Europe, continued to deteriorate. These lower sales volumes and prices were primarily driven by record exports out of Asia into Europe and North America. In addition, Westlake Epoxy operations in Europe have experienced sustained high energy and power costs. These factors negatively impacted Westlake Epoxy financial results during 2023. The Company identified these developments, along with management's outlook for the Westlake Epoxy business over the foreseeable future, as impairment indicators during the fourth quarter of 2023. Based on this triggering event, the Company performed a quantitative impairment analysis of Westlake Epoxy's long-lived assets along with the annual goodwill impairment assessment during the fourth quarter of 2023.

Long-Lived Assets Impairment

The Company evaluated Westlake Epoxy's long-lived asset groups for impairment during the fourth quarter of 2023 in line with the identification of the triggering event discussed above. Recoverability tests were performed for each of Westlake Epoxy's asset groups to compare the carrying amounts to the net undiscounted cash flow projections of the respective asset groups. The undiscounted cash flow projections were based on historical results, estimates made by management of future market conditions, current and future strategic and operational plans and future financial performance projected through the remaining useful life of the primary asset in the asset group. Based on the recoverability tests, the Company determined that the carrying amount of the primary assets of the Westlake Epoxy Netherlands asset group is not recoverable. We determined all other Epoxy asset groups to be recoverable. The fair value of the asset group was calculated using a discounted cash flow methodology and a non-cash impairment charge of $347, related to the Company's base epoxy resin business in the Netherlands, was recognized within the Performance and Essential Materials segment to reduce the carrying amount of the asset group to its fair value. The long-lived assets impairment within the Westlake Epoxy Netherlands asset group consists of non-cash charges of $256 in property, plant and equipment, $32 in operating lease right-of-use assets, $6 in customer relationships, $43 in other intangible assets, and $10 in other assets. The long-lived assets impairment charge is reported in impairment of goodwill and long-lived assets on the consolidated statements of operations.

7. Leases

Lease-related asset and liability balances consist of the following at December 31:

	2023	2022
Operating Leases		
Right-of-use assets	$ 697	$ 615
Accrued and other liabilities	$ 124	$ 116
Operating lease liabilities	611	504
Total operating lease liabilities	$ 735	$ 620
Weighted Average Remaining Term (in years)	9	9
Weighted Average Lease Discount Rate	3.8 %	2.9 %

The Company's operating lease cost is comprised of payments related to operating leases recorded in the consolidated balance sheet and short-term rental payments for leases that are not recorded in the consolidated balance sheet. Variable operating lease cost was not material to the consolidated statements of operations for the years ended December 31, 2023, 2022, and 2021. The components of operating lease expense were as follows:

	Year Ended December 31,		
	2023	2022	2021
Operating lease cost [1]	$ 142	$ 139	$ 117
Short-term lease cost	121	101	85
Total operating lease cost	$ 263	$ 240	$ 202

(1) Includes fixed lease payments for operating leases recorded in the consolidated balance sheet.

Maturities of lease liabilities were as follows at December 31, 2023:

	Operating Leases
2024	$ 142
2025	122
2026	108
2027	95
2028	81
Thereafter	362
Total lease payments	910
Less: imputed interest	(175)
Present value of lease liabilities	$ 735

Related Party Leases

The Company leases certain assets under operating leases with related parties. Right-of-use assets and the associated operating lease liabilities for related party operating leases were approximately $26 and $29 as of December 31, 2023 and December 31, 2022, respectively. The Company recognized operating lease cost for fixed lease payments to related parties of $5 and $6 for the years ended December 31, 2023 and 2022, respectively.

8. Goodwill and Other Intangible Assets

Goodwill

The following table summarizes gross carrying amounts and changes in the carrying amount of goodwill for the years ended December 31, 2023 and 2022.

	Performance and Essential Materials Segment	Housing and Infrastructure Products Segment	Total
Balance at December 31, 2021	$ 902	$ 1,122	$ 2,024
Goodwill acquired during the year	119	5	124
Measurement period adjustment	8	16	24
Effects of changes in foreign exchange rates	(9)	(2)	(11)
Balance at December 31, 2022	1,020	1,141	2,161
Measurement period adjustment	4	1	5
Impairment of goodwill	(128)	—	(128)
Effects of changes in foreign exchange rates	1	2	3
Balance at December 31, 2023	$ 897	$ 1,144	$ 2,041

Goodwill Impairment

The Company performed a quantitative assessment for the purposes of its annual goodwill impairment analysis for each of the reporting units within the Performance and Essential Materials and the Housing and Infrastructure Products segments during the fourth quarter of 2023. The fair values of the reporting units were determined using both a discounted cash flow methodology and a market value methodology. The discounted cash flow projections were based on a long-term forecast to reflect the cyclicality of the business. The forecast is based on historical results, estimates by management of future market conditions, current and future strategic and operational plans and future financial performance. Significant assumptions used in the discounted cash flow projection included projected sales volumes based on production capacities, future sales prices, feedstock, energy and power costs and capital expenditures to maintain safe and reliable plant operations. The future cash flows were discounted to present value using a discount rate ranging from 10.0% to 12.0%. The significant assumptions used in determining the fair value of the reporting unit using the market value methodology include the determination of appropriate market comparables and the estimated multiples of net income before interest expense, income taxes, depreciation and amortization (EBITDA) a willing buyer is likely to pay. Based on the quantitative tests performed during the fourth quarter of 2023, the Company determined that the fair value of the Westlake Epoxy reporting unit did not exceed its carrying amount, and as such, a goodwill impairment charge of $128 was recognized within the Performance and Essential Materials segment. The goodwill impairment charge is reported in impairment of goodwill and long-lived assets on the consolidated statements of operations. For all other reporting units, the Company determined that the fair value of each of the reporting units were substantially in excess of carrying value.

Intangible Assets

Intangible assets consist of the following at December 31:

	2023			2022			Weighted Average Life
	Cost	Accumulated Amortization	Net	Cost	Accumulated Amortization	Net	
Customer relationships	$ 1,586	$ (676)	$ 910	$ 1,589	$ (596)	$ 993	15
Other intangible assets:							
Licenses and intellectual property	264	(136)	128	306	(124)	182	15
Trade name	385	(104)	281	411	(86)	325	18
Other	112	(28)	84	84	(19)	65	17
Total other intangible assets	$ 761	$ (268)	$ 493	$ 801	$ (229)	$ 572	

Scheduled amortization of intangible assets for the next five years is as follows: $142, $113, $105, $103 and $103 in 2024, 2025, 2026, 2027 and 2028, respectively.

9. Equity Method Investments

LACC, LLC Joint Venture

In 2015, Eagle US 2 LLC ("Eagle"), a wholly-owned subsidiary of the Company, and Lotte Chemical USA Corporation, a subsidiary of Lotte Chemical Corporation ("Lotte"), formed a joint venture, LACC, LLC ("LACC"), to design, build and operate an ethylene facility with 2.2 billion pounds per year of ethylene production capacity. Pursuant to a contribution and subscription agreement between Eagle and LACC, Eagle contributed $225 to LACC to fund construction costs of the ethylene plant, representing approximately 12% of the membership interests in LACC. On November 12, 2019, the Company, through Eagle, completed the acquisition of an additional 34.8% of the membership interests in LACC from Lotte for approximately $817. On March 15, 2022, the Company acquired an additional 3.2% of the membership interests in LACC from Lotte for $89. As of December 31, 2023, the Company owned an aggregate 50% membership interest in LACC. As of December 31, 2023, the Company's investment exceeded the underlying equity in net assets by approximately $200 which was assigned to goodwill and not amortized.

The ethylene plant was built adjacent to the Company's chlor-alkali facility in Lake Charles. During the third quarter of 2019, the ethylene plant began commercial operations.

The Company accounts for its investment in LACC under the equity method of accounting. The LACC joint venture is a cost-sharing arrangement between the members of LACC. The members of LACC receive their proportionate shares of ethylene offtake each month and fund cash operating costs, excluding depreciation and amortization. As a result, LACC recognizes net losses equal to depreciation and amortization each period. The Company's equity in losses from LACC, which is equal to its share of depreciation and amortization expenses, is recognized in cost of sales in the consolidated statements of operations. The Company's investment in LACC is classified as an equity method investment in the consolidated balance sheets. The Company's capital contributions to fund its share of capital expenditures are classified within investing activities in the consolidated statements of cash flows.

The Company's ethylene offtake from LACC was approximately935 million and 766 million pounds during the years ended December 31, 2023 and 2022, respectively.

Changes in the Company's investment in LACC for the years ended December 31, 2023 and 2022 were as follows:

	Investment in LACC
Balance at December 31, 2021	$ 943
Cash contributions	87
Additional interest purchased	89
Depreciation and amortization	(44)
Balance at December 31, 2022	1,075
Cash contributions	21
Depreciation and amortization	(49)
Balance at December 31, 2023	$ 1,047

Services Provided to LACC and Lotte

The Company provides certain utilities and other services to LACC and Lotte. Pursuant to a construction and reimbursement agreement, LACC and Lotte agreed to reimburse the Company for construction costs over a 6.5-year period beginning in 2020. In addition to the reimbursements for construction costs, the Company charges LACC and Lotte certain fixed fees under an operating, maintenance and logistics agreement. The Company accounts for the reimbursement of construction costs and the fixed fees as components of the total transaction price and recognizes it ratably in net sales over approximately 25 years. The remaining performance obligations at December 31, 2023, representing these fixed components of the transaction price, totaled $52 and $72 from LACC and Lotte, respectively. The associated contract liabilities recorded from LACC and Lotte totaled $23 and $29 as of December 31, 2023, respectively, and $19 and $24 as of December 31, 2022, respectively. In addition to the reimbursements for construction costs and other fixed fees, the Company charges LACC and Lotte certain variable fees.

Other Equity Method Investments

In addition to LACC, the Company has other equity method investments amounting to $68 and $67 as of December 31, 2023 and 2022, respectively. See Note 20 for more detailed information.

10. Accounts Payable

Accounts payable consist of the following at December 31:

	2023	2022
Accounts payable—third parties	$ 849	$ 870
Accounts payable to related parties	15	16
Notes payable	13	3
Accounts payable	$ 877	$ 889

11. Long-Term Debt

Long-term debt consist of the following at December 31:

	December 31, 2023	December 31, 2022
0.875% senior notes due 2024 (the "0.875% 2024 Senior Notes")	$ 300	$ 300
3.60% senior notes due 2026 (the "3.60% 2026 Senior Notes")	750	750
Loan related to tax-exempt waste disposal revenue bonds due 2027	11	11
1.625% €700 million senior notes due 2029 (the "1.625% 2029 Senior Notes")	773	750
3.375% senior notes due 2030 (the "3.375% 2030 Senior Notes")	300	300
3.50% senior notes due 2032 (the "3.50% 2032 tax-exempt GO Zone Refunding Senior Notes")	250	250
2.875% senior notes due 2041 (the "2.875% 2041 Senior Notes")	350	350
5.00% senior notes due 2046 (the "5.00% 2046 Senior Notes")	700	700
4.375% senior notes due 2047 (the "4.375% 2047 Senior Notes")	500	500
3.125% senior notes due 2051 (the "3.125% 2051 Senior Notes")	600	600
3.375% senior notes due 2061 (the "3.375% 2061 Senior Notes")	450	450
Term loans due 2026	13	15
Long-term debt, principal amount	4,997	4,976
Less:		
Unamortized discount and debt issuance costs	91	97
Long-term debt, carrying value	4,906	4,879
Less current portion:		
0.875% senior notes due 2024	299	—
Long-term debt, carrying value, net of current portion	$ 4,607	$ 4,879

Credit Agreement

On June 9, 2022, the Company entered into a new $1,500 revolving credit facility that is scheduled to mature on June 9, 2027 (the "Credit Agreement") and, in connection therewith, terminated the Company's then existing revolving credit agreement. The Credit Agreement bears interest at either (a) Adjusted Term SOFR (as defined in the Credit Agreement) plus a margin ranging from 1.00% to 1.625% per annum or (b) Alternate Base Rate (as defined in the Credit Agreement) plus a margin ranging from 0.00% to 0.625% per annum, in each case depending on the credit rating of the Company. The Credit Agreement contains certain affirmative and negative covenants, including a quarterly total leverage ratio financial maintenance covenant. As of December 31, 2023, the Company was in compliance with the total leverage ratio financial maintenance covenant. The Credit Agreement also contains certain events of default and, if and for so long as certain events of default have occurred and are continuing, any overdue amounts outstanding under the Credit Agreement will accrue interest at an increased rate, the lenders can terminate their commitments to lend thereunder and payments of any outstanding amounts thereunder could be accelerated by the lenders. None of the Company's subsidiaries are required to guarantee the obligations of the Company under the Credit Agreement.

The Credit Agreement includes a $150 sub-limit for letters of credit, and any outstanding letters of credit will be deducted from availability under the facility. The Credit Agreement also provides for a discretionary $50 commitment for swingline loans to be provided on a same-day basis. The Company may also increase the size of the facility, in increments of at least $25, up to a maximum of $500, subject to certain conditions and if certain lenders agree to commit to such an increase.

As of December 31, 2023, the Company had no borrowings and no letters of credit outstanding, and had borrowing availability of $1,500, under the Credit Agreement.

0.875% Senior Notes due 2024

In August 2021, the Company completed the registered public offering of $300 aggregate principal amount of the 0.875% 2024 Senior Notes. The Company may optionally redeem the 0.875% 2024 Senior Notes at any time and from time to time on or after August 15, 2022 for 100% of the principal amount plus accrued and unpaid interest.

3.60% Senior Notes due 2026 and 5.00% Senior Notes due 2046

In August 2016, the Company issued $750 aggregate principal amount of the 3.60% 2026 Senior Notes and $700 aggregate principal amount of the 5.00% 2046 Senior Notes. In March 2017, the Company commenced registered exchange offers to exchange the 3.60% 2026 Senior Notes and the 5.00% 2046 Senior Notes for new notes that are identical in all material respects to the 3.60% 2026 Senior Notes and the 5.00% 2046 Senior Notes, except that the offer and issuance of the new Securities and Exchange Commission-registered notes have been registered under the Securities Act of 1933, as amended (the "Securities Act"). The exchange offers expired on April 24, 2017, and approximately 99.97% of the 3.60% 2026 Senior Notes and 100% of the 5.00% 2046 Senior Notes were exchanged. The 3.60% 2026 Senior Notes that were not exchanged in the 3.60% 2026 Senior Notes exchange offer have not been registered under the Securities Act or any state securities laws and may not be offered or sold in the U.S. absent registration or an applicable exemption from registration requirements or a transaction not subject to the registration requirements of the Securities Act or any state securities law.

Revenue Bonds

In December 1997, the Company entered into a loan agreement with a public trust established for public purposes for the benefit of the Parish of Calcasieu, Louisiana. The public trust issued $11 principal amount of tax-exempt waste disposal revenue bonds in order to finance the Company's construction of waste disposal facilities for an ethylene plant. The waste disposal revenue bonds expire in December 2027 and are subject to redemption and mandatory tender for purchase prior to maturity under certain conditions. The interest rate on the waste disposal revenue bonds at December 31, 2023 and 2022 was 4.30% and 3.88%, respectively.

1.625% Senior Notes due 2029

On July 17, 2019, the Company completed the registered public offering of €700 million aggregate principal amount of the 1.625% 2029 Senior Notes. The Company received approximately $779 of net proceeds from the offering. The 1.625% 2029 Senior Notes will accrue interest from July 17, 2019 at a rate of 1.625% per annum, payable annually in arrears on July 17 of each year, beginning July 17, 2020. The Company may optionally redeem the 1.625% 2029 Senior Notes in accordance with the terms of the 1.625% 2029 Senior Notes. The Company designated this euro-denominated debt as a non-derivative net investment hedge of a portion of the Company's net investments in euro functional-currency denominated subsidiaries to offset foreign currency fluctuations.

3.375% Senior Notes due 2030

On June 12, 2020, the Company completed the registered public offering of $300 aggregate principal amount of the 3.375% 2030 Senior Notes. There is no sinking fund and no scheduled amortization of the 3.375% 2030 Senior Notes prior to maturity. The 3.375% 2030 Senior Notes accrue interest from June 12, 2020 at a rate of 3.375% per annum, payable semi-annually in arrears on June 15 and December 15 of each year, beginning December 15, 2020. The Company may optionally redeem the 3.375% 2030 Senior Notes in accordance with the terms of the 3.375% 2030 Senior Notes.

3.50% 2032 GO Zone Refunding Bonds

In November 2017, the Louisiana Local Government Environmental Facility and Development Authority (the "Authority") completed the remarketing of $250 aggregate principal amount of 3.50% tax-exempt revenue refunding bonds due November 1, 2032 (the "3.50% 2032 GO Zone Bonds"), the net proceeds of which were used to redeem $250 aggregate principal amount of the Authority's 6 ¾% tax-exempt revenue bonds due November 1, 2032 issued by the Authority under the GO Zone Act in December 2007. In connection with the remarketing of the 3.50% 2032 GO Zone Bonds, the Company issued $250 aggregate principal amount of the 3.50% 2032 GO Zone Refunding Senior Notes. The 3.50% 2032 GO Zone Bonds are subject to optional redemption by the Authority upon the direction of the Company at any time on or after November 1, 2027, for 100% of the principal amount plus accrued interest.

2.875% Senior Notes due 2041

In August 2021, the Company completed the registered public offering of $350 aggregate principal amount of the 2.875% 2041 Senior Notes. The Company may optionally redeem the 2.875% 2041 Senior Notes at any time and from time to time prior to February 15, 2041 (six months prior to the maturity date) for a redemption price equal to the greater of (i) 100% of the principal amount plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the 2.875% 2041 Senior Notes being redeemed that would be due if the 2.875% 2041 Senior Notes matured on February 15, 2041, discounted to the redemption date on a semi-annual basis, plus 20 basis points, and plus accrued and unpaid interest. In addition, the Company may optionally redeem the 2.875% 2041 Senior Notes at any time on or after February 15, 2041 for 100% of the principal amount plus accrued and unpaid interest.

4.375% Senior Notes due 2047

In November 2017, the Company completed the registered public offering of $500 aggregate principal amount of the 4.375% 2047 Senior Notes. The 4.375% 2047 Senior Notes are unsecured and mature on November 15, 2047. There is no sinking fund and no scheduled amortization of the 4.375% 2047 Senior Notes prior to maturity. The Company may optionally redeem the 4.375% 2047 Senior Notes in accordance with the terms of the 4.375% 2047 Senior Notes.

3.125% Senior Notes due 2051

In August 2021, the Company completed the registered public offering of $600 aggregate principal amount of the 3.125% 2051 Senior Notes. The Company may optionally redeem the 3.125% 2051 Senior Notes at any time and from time to time prior to February 15, 2051 (six months prior to the maturity date) for a redemption price equal to the greater of (i) 100% of the principal amount plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the 3.125% 2051 Senior Notes being redeemed that would be due if the 3.125% 2051 Senior Notes matured on February 15, 2051, discounted to the redemption date on a semi-annual basis, plus 25 basis points, and plus accrued and unpaid interest. In addition, the Company may optionally redeem the 3.125% 2051 Senior Notes at any time on or after February 15, 2051 for 100% of the principal amount plus accrued and unpaid interest.

3.375% Senior Notes due 2061

In August 2021, the Company completed the registered public offering of $450 aggregate principal amount of the 3.375% 2061 Senior Notes. The Company may optionally redeem the 3.375% 2061 Senior Notes at any time and from time to time prior to February 15, 2061 (six months prior to the maturity date) for a redemption price equal to the greater of (i) 100% of the principal amount plus accrued and unpaid interest and (ii) the sum of the present values of the remaining scheduled payments on the 3.375% 2061 Senior Notes being redeemed that would be due if the 3.375% 2061 Senior Notes matured on February 15, 2061, discounted to the redemption date on a semi-annual basis, plus 25 basis points, and plus accrued and unpaid interest. In addition, the Company may optionally redeem the 3.375% 2061 Senior Notes at any time on or after February 15, 2061 for 100% of the principal amount plus accrued and unpaid interest.

The holders of the 0.875% 2024 Senior Notes, the 3.60% 2026 Senior Notes, the 1.625% 2029 Senior Notes, the 3.375% 2030 Senior Notes, the 3.50% 2032 GO Zone Refunding Senior Notes, the 2.875% 2041 Senior Notes, the 5.00% 2046 Senior Notes, the 4.375% 2047 Senior Notes, the 3.125% 2051 Senior Notes and the 3.375% 2061 Senior Notes may require the Company to repurchase the notes at a price equal to 101% of their principal amount, plus accrued and unpaid interest to, but not including, the date of repurchase, upon the occurrence of both a "change of control" and, within 60 days of such change of control, a "below investment grade rating event" (as such terms are defined in the respective indentures governing these notes).

The indenture governing the 0.875% 2024 Senior Notes, the 3.60% 2026 Senior Notes, the 1.625% 2029 Senior Notes, the 3.375% 2030 Senior Notes, the 3.50% 2032 GO Zone Refunding Senior Notes, the 2.875% 2041 Senior Notes, the 5.00% 2046 Senior Notes, the 4.375% 2047 Senior Notes, the 3.125% 2051 Senior Notes, and the 3.375% 2061 Senior Notes (together, the "Notes") contains customary events of default and covenants that, among other things and subject to certain exceptions, restrict the Company and certain of the Company's subsidiaries' ability to (1) incur certain secured indebtedness, (2) engage in certain sale and leaseback transactions and (3) consolidate, merge or transfer all or substantially all of its assets. The Notes are unsecured and none of the Company's subsidiaries have guaranteed any series of the Notes.

2026 Term Loans

In March 2021, Taiwan Chlorine Industries, Ltd., the Company's 60%-owned joint venture, entered into five-year loan agreements for a maximum total limit of approximately $22. The interest rate on these loans at December 31, 2023 was 0.95%. The unsecured loans include a government rate subsidy and have a 5-year maturity. The balance outstanding under these loans was approximately $13 at December 31, 2023.

As of December 31, 2023, the Company was in compliance with all of its long-term debt covenants.

The weighted average interest rate on all long-term debt was 3.2% at December 31, 2023 and 2022. Unamortized debt issuance costs on long-term debt were $36 and $40 at December 31, 2023 and 2022, respectively.

Aggregate scheduled maturities of long-term debt during the next five years consist of $300 in 2024, $763 in 2026 and $11 in 2027. There are no other scheduled maturities of debt in 2024 through 2028.

12. Stockholders' Equity

The Company's Board of Directors has declared regular quarterly dividends to holders of its common stock aggregating $221, $169 and $145 for the years ended December 31, 2023, 2022 and 2021, respectively.

Common Stock

Each share of common stock entitles the holder to one vote on all matters on which holders are permitted to vote, including the election of directors. There are no cumulative voting rights. Accordingly, holders of a majority of the total votes entitled to vote in an election of directors will be able to elect all of the directors standing for election. Subject to preferences that may be applicable to any outstanding preferred stock, the holders of the common stock will share equally on a per share basis any dividends when, as and if declared by the Board of Directors out of funds legally available for that purpose. If the Company is liquidated, dissolved or wound up, the holders of the Company's common stock will be entitled to a ratable share of any distribution to stockholders, after satisfaction of all the Company's liabilities and of the prior rights of any outstanding class of the Company's preferred stock. The Company's common stock has no preemptive or conversion rights or other subscription rights. There are no redemption or sinking fund provisions applicable to the Company's common stock.

Preferred Stock

The Company's charter authorizes the issuance of shares of preferred stock. The Company's Board of Directors has the authority, without shareholder approval, to issue preferred shares from time to time in one or more series, and to fix the number of shares and terms of each such series. The Board may determine the designations and other terms of each series including dividend rates, whether dividends will be cumulative or non-cumulative, redemption rights, liquidation rights, sinking fund provisions, conversion or exchange rights and voting rights.

Stock Repurchase Program

In November 2014, the Company's Board of Directors approved a $250 share repurchase program (the "2014 Program"). In November 2015, the Company's Board of Directors approved the expansion of the 2014 Program by an additional $150. In August 2018, the Company's Board of Directors approved the expansion of the 2014 Program by an additional $150. In August 2022, the Company's Board of Directors approved the further expansion of the existing 2014 Program by an additional $500. The number of shares repurchased by the Company under the 2014 Program was 211,294, 1,079,736 and 355,800 for the years ended December 31, 2023, 2022 and 2021, respectively. As of December 31, 2023, the Company had repurchased a total of 8,722,550 shares of its common stock for an aggregate purchase price of approximately $574.

Any shares repurchased under the 2014 Program are held by the Company as treasury stock and may be used for general corporate purposes, including for the 2013 Omnibus Incentive Plan. In 2014, the Company began delivering treasury shares to employees and non-employee directors for options exercised, for the settlement of restricted stock units and for the settlement of performance stock units. The cost of treasury shares delivered is determined using the specific identification method.

13. Accumulated Other Comprehensive Income (Loss)

Changes in accumulated other comprehensive income (loss) by component were as follows:

	Pension and Other Post-Retirement Benefits Liability, Net of Tax	Cumulative Foreign Currency Exchange, Net of Tax	Total
Balances at December 31, 2021	$ 20	$ (56)	$ (36)
Net other comprehensive income (loss) attributable to Westlake Corporation	32	(85)	(53)
Balances at December 31, 2022	52	(141)	(89)
Net other comprehensive income (loss) attributable to Westlake Corporation	(48)	39	(9)
Balances at December 31, 2023	$ 4	$ (102)	$ (98)

14. Employee Benefits

Defined Contribution Plans

U.S. Plans

The Company has a defined contribution savings plan covering the eligible U.S. regular full-time and part-time employees, whereby eligible employees may elect to contribute up to 100% of their annual eligible compensation, subject to an annual plan limit in line with the annual elective contribution limit as determined by the Internal Revenue Service. The Company matches its employee's contribution up to a certain percentage of such employee's compensation, per the terms of the plan. The Company may, at its discretion and per the terms of the plan, make an additional non-matching contribution in an amount as the Board of Directors may determine. For the years ended December 31, 2023, 2022 and 2021, the Company recorded approximately $36, $33 and $24, respectively, to expense for these contributions.

Further, within the plan, the Company also makes an annual retirement contribution to substantially all employees of certain subsidiaries. The Company's contributions to the plan are determined as a percentage of employees' pay. For the years ended December 31, 2023, 2022 and 2021, the Company charged approximately $48, $46 and $35, respectively, to expense for these contributions.

Non-U.S. Plans

The Company has defined contribution plans in several countries covering eligible employees of the Company. The Company's contributions to the plans are based on applicable laws in each country and eligibility of employees of certain subsidiaries for the annual retirement contribution. Contributions to the Company's non-U.S. defined contribution plans are made by both the employee and the Company. For the years ended December 31, 2023, 2022 and 2021, the Company charged approximately $13, $3 and $4, respectively, to expense for its contributions to these plans. For the years ended December 31, 2023, 2022 and 2021, the Company charged an additional **$**, $2 and $1, respectively, to expense related to the annual retirement contributions to these plans.

Defined Benefit Plans

U.S. Plans

The Company has noncontributory defined benefit pension plans that cover certain eligible salaried and wage employees of certain subsidiaries. However, eligibility for the Company's plans has been frozen. Benefits for salaried employees under these plans are based primarily on years of service and employees' pay near retirement. Benefits for wage employees are based upon years of service and a fixed amount as periodically adjusted. The Company recognizes the years of service prior to the Company's acquisition of the subsidiary's facilities for purposes of determining vesting, eligibility and benefit levels for certain employees of the subsidiary and for determining vesting and eligibility for certain other employees of the subsidiary. The measurement date for these plans is December 31.

Non-U.S. Plans

The Company has defined benefit pension plans covering current and former employees associated with the Company's operations. In conjunction with the Westlake Epoxy acquisition on February 1, 2022, as disclosed in Note 2, the Company assumed certain defined benefit pension plans that are incorporated in the information disclosed. Several non-U.S. pension plans are unfunded and have no plan assets. These pension plans are closed to new participants. Benefits for employees for these plans are based primarily on employees' pay near retirement. The measurement date for the non-U.S. plans is December 31.

Details of the changes in benefit obligations, plan assets and funded status of the Company's pension plans are as follows:

	2023		2022	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Change in benefit obligation				
Benefit obligation, beginning of year	$ 518	$ 530	$ 673	$ 152
Benefit obligation assumed with acquisition	—	—	—	612
Service cost	1	4	1	3
Interest cost	24	19	14	7
Actuarial loss (gain)	(7)	82	(121)	(188)
Benefits paid	(42)	(21)	(49)	(17)
Settlements	—	(3)	—	(1)
Foreign exchange effects	—	18	—	(38)
Benefit obligation, end of year	$ 494	$ 629	$ 518	$ 530
Change in plan assets				
Fair value of plan assets, beginning of year	$ 415	$ 370	$ 562	$ 21
Plan assets assumed with acquisition	—	—	—	538
Actual return	55	19	(97)	(145)
Employer contribution	2	15	2	1
Benefits paid	(42)	(21)	(49)	(17)
Administrative expenses paid	(3)	—	(3)	—
Settlements	—	(3)	—	(1)
Foreign exchange effects	—	11	—	(27)
Fair value of plan assets, end of year	$ 427	$ 391	$ 415	$ 370
Funded status, end of year	$ (67)	$ (238)	$ (103)	$ (160)

The actuarial loss (gain) in the benefit obligation for the periods presented is primarily driven by discount rate assumption changes.

	2023		2022	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Amounts recognized in the consolidated balance sheet at December 31				
Noncurrent assets	$ —	$ 36	$ —	$ 63
Current liabilities	(2)	(11)	(2)	(9)
Noncurrent liabilities	(65)	(263)	(101)	(214)
Net amount recognized	$ (67)	$ (238)	$ (103)	$ (160)

	2023		2022	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Amounts recognized in accumulated other comprehensive income (loss)				
Net loss (gain)	$ (66)	$ 67	$ (30)	$ (10)
Prior service credit	(1)	(4)	(2)	(4)
Total before tax [1]	$ (67)	$ 63	$ (32)	$ (14)

(1) After-tax totals for pension benefits were a loss of $4 and a gain of $45 for 2023 and 2022, respectively, and are reflected in stockholders' equity as accumulated other comprehensive income (loss).

In the U.S., the Pension Protection Act of 2006 (the "Pension Protection Act") established a relationship between a qualified pension plan's funded status and the actual benefits that can be provided. Restrictions on plan benefits and additional funding and notice requirements are imposed when a plan's funded status is less than certain threshold levels. For the 2023 plan year, the funded status for the Company's U.S. pension plans are above 80% and, as such, are exempt from the Pension Protection Act's benefit restrictions.

Pension plans with an accumulated benefit obligation in excess of plan assets at December 31 are as follows:

	2023		2022	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Information for pension plans with an accumulated benefit obligation in excess of plan assets				
Projected benefit obligation	$ (494)	$ (273)	$ (518)	$ (223)
Accumulated benefit obligation	(494)	(271)	(518)	(221)
Fair value of plan assets	427	—	415	—

The following table provides the components of net periodic benefit costs, other changes in plan assets and benefit obligation recognized in other comprehensive income.

	Year Ended December 31,					
	2023		2022		2021	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Components of net periodic benefit cost						
Service cost	$ 1	$ 4	$ 1	$ 3	$ 3	$ 2
Administrative expenses	3	—	3	—	3	—
Interest cost	24	19	14	7	11	1
Expected return on plan assets	(26)	(12)	(38)	(12)	(38)	(1)
Net amortization	(1)	(2)	—	1	—	3
Net periodic benefit cost (gain)	$ 1	$ 9	$ (20)	$ (1)	$ (21)	$ 5
Other changes in plan assets and benefit obligation recognized in other comprehensive income (OCI)						
Net loss (gain) emerging	$ (36)	$ 74	$ 13	$ (31)	$ (49)	$ (9)
Prior service credit	—	—	—	—	(2)	—
Amortization of prior service credit	1	—	—	—	—	—
Amortization of net gain (loss)	—	2	—	(1)	—	(2)
Foreign exchange effects	—	1	—	(1)	—	—
Total recognized in OCI	$ (35)	$ 77	$ 13	$ (33)	$ (51)	$ (11)
Total net periodic benefit cost and OCI	$ (34)	$ 86	$ (7)	$ (34)	$ (72)	$ (6)

The estimated prior service credit and net gain (loss) for the defined benefit plans to be amortized from accumulated other comprehensive income (loss) into net periodic benefit cost during 2024 are expected to be $1 and $0, respectively.

The weighted-average assumptions used to determine pension plan obligations and net periodic benefit costs for the plans are as follows:

	2023		2022		2021	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Weighted average assumptions used to determine benefit obligations at December 31						
Discount rate	5.0 %	3.2 %	4.9 %	3.7 %	2.6 %	1.4 %
Rate of compensation increase	— %	3.1 %	— %	2.9 %	— %	2.6 %
Weighted average assumptions used to determine net periodic benefit costs for years ended December 31						
Discount rate for benefit obligations	4.9 %	3.7 %	2.5 %	1.2 %	2.1 %	0.8 %
Discount rate for service cost	5.0 %	3.8 %	2.8 %	1.4 %	2.4 %	0.8 %
Discount rate for interest cost	4.8 %	3.7 %	2.1 %	1.2 %	1.5 %	0.8 %
Expected return on plan assets	6.5 %	3.3 %	7.0 %	2.6 %	7.0 %	4.0 %
Rate of compensation increase	— %	2.9 %	— %	2.8 %	— %	2.6 %

The discount rates for the Company's U.S. and non-U.S. plans are determined using a benchmark pension discount curve and applying spot rates from the curve to each year of expected benefit payments to determine the appropriate discount rate for the Company. The assumed long-term return on plan assets is estimated by considering factors such as the plan's overall investment strategy, current economic conditions and historical averages.

The Company's U.S. pension plan investments are held in the Westlake Defined Benefit Plan. The Company's overall investment strategy for these pension plan assets is to achieve a balance between moderate income generation and capital appreciation. The investment strategy includes a mix of approximately 60% of investments for long-term growth, and 40% for near-term benefit payments and liability hedging, with a diversification of asset types. These pension funds' investment policies target asset allocations of approximately 60% equity securities and 40% fixed income securities in order to pursue a balance between moderate income generation, capital appreciation and a reduction in funded status volatility.

The Company's non-U.S. pension plan investments are primarily held in three pension plans in the Netherlands (collectively, the "Netherlands Plan"). Per the terms of the Netherlands Plan asset management agreement between the Company and the asset manager, the Netherlands Plan's current portfolio is strategically weighted towards fixed income securities. The Netherlands Plan's investment policy targets asset allocations of approximately 20% equity securities and 80% fixed income securities. The Netherlands Plan's investment strategy allows for re-allocations of either of the equity or the fixed income securities from 0% up to 100%.

Equity securities primarily include investments in large-cap and small-cap companies located in the U.S. and international developed and emerging markets stocks. Fixed income securities are comprised of investment and non-investment grade bonds, including U.S. Treasuries and U.S. and non-U.S. corporate bonds of companies from diversified industries. These bonds also include longer duration securities to reduce funding volatility and reduce the asset/liability mismatch in terms of interest rate sensitivity. Fixed income funds also include international government bonds and mortgage funds. The U.S. pension fund investment policy allows a discretionary range in various asset classes within the asset allocation model of up to 10%. The Company does not believe that there are significant concentrations of risk in the pension plan assets due to its strategy of asset diversification. At December 31, 2023, plan assets did not include direct ownership of the Company's common stock.

Under the accounting guidance for fair value measurements, inputs used to measure fair value are classified in one of three levels:

Level 1: Quoted market prices in active markets for identical assets or liabilities.

Level 2: Observable market-based inputs or unobservable inputs that are corroborated by market data.

Level 3: Unobservable inputs that are not corroborated by market data.

WESTLAKE CORPORATION

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS—(Continued)
(in millions of dollars, except share amounts and per share data)

The investments in the collective trust and mutual funds are valued using a market approach based on the net asset value of units held. The fair values of the Company's plan assets at December 31, by asset category, are as follows:

	2023							
	U.S. Plans				Non-U.S. Plans			
	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Cash and common stock:								
Cash and cash equivalents	$ —	$ —	$ —	$ —	$ 3	$ —	$ —	$ 3
Collective investment trust and mutual funds —Equity securities:								
Large-cap funds [1]	47	92	—	139	—	1	—	1
Small-cap funds [2]	—	8	—	8	—	—	—	—
International funds [3]	70	23	—	93	—	74	—	74
Collective investment trust and mutual funds —Fixed income:								
Bond funds [4]	39	138	—	177	—	303	—	303
Short-term investment funds	—	10	—	10	—	—	—	—
Group insurance contract	—	—	—	—	—	—	10	10
	$ 156	$ 271	$ —	$ 427	$ 3	$ 378	$ 10	$ 391

	2022							
	U.S. Plans				Non-U.S. Plans			
	Level 1	Level 2	Level 3	Total	Level 1	Level 2	Level 3	Total
Cash and common stock:								
Cash and cash equivalents	$ —	$ —	$ —	$ —	$ 7	$ —	$ —	$ 7
Collective investment trust and mutual funds —Equity securities:								
Large-cap funds [1]	42	90	—	132	—	1	—	1
Small-cap funds [2]	—	10	—	10	—	—	—	—
International funds [3]	72	24	—	96	—	73	—	73
Collective investment trust and mutual funds —Fixed income:								
Bond funds [4]	37	128	—	165	—	280	—	280
Short-term investment funds	—	12	—	12	—	—	—	—
Group insurance contract	—	—	—	—	—	—	9	9
	$ 151	$ 264	$ —	$ 415	$ 7	$ 354	$ 9	$ 370

(1) Substantially all of the assets of these funds are invested in large-cap U.S. companies. The remainder of the assets of these funds is invested in cash reserves.

(2) Substantially all of the assets of these funds are invested in small-cap U.S. companies. The remainder of the assets of these funds is invested in cash reserves.

(3) Substantially all of the assets of the U.S. Plans' funds are invested in international companies in developed markets (excluding the U.S.), and the remainder of the assets of these funds is invested in cash reserves. The assets of the non-U.S. Plans' funds are primarily invested in diversified global equities, real estate and private equities.

(4) The assets of the U.S. Plans' funds represent investment grade bonds of U.S. issuers, including U.S. Treasury notes. The assets of the non-U.S. Plans represent fixed income funds that are primarily invested in international government bonds and mortgage funds.

The Company's funding policy for its U.S. plans is consistent with the minimum funding requirements of federal law and regulations. Based on preliminary estimates, the Company expects to make contributions of approximately $0 and $5 for the U.S. and Non-U.S. pension plans, respectively, in 2024.

Multi-employer Plans

Non-U.S. Plans

The Company participates in two multi-employer plans, Pensionskasse der Mitarbeiter der Hoechst-Gruppe VVaG and Pensionskasse der Wacker-Chemie GmbH VVaG, which provide benefits to certain of the Company's employees in Germany. These multi-employer plans are closed to new participants. The plans provide fixed, monthly retirement payments on the basis of the credits earned by the participating employees. To the extent that the plans are underfunded, future contributions to the plans may increase and may be used to fund retirement benefits for employees related to other employers. The benefit obligations are covered up to a certain salary threshold by contributions made by the Company and employees to the plans. Contributions to the Company's multi-employer plans are expensed as incurred.

Other Post-retirement Benefits

In the U.S., the Company provides post-retirement healthcare and life insurance benefits for certain employees and their dependents who meet minimum age and service requirements. The Company has the right to modify or terminate some of these benefits. The Company has a post-retirement plan in Canada which is unfunded and provides medical and life insurance benefits for certain employees and their dependents. The Company also has an unfunded post-retirement benefit plan in the Netherlands. The amounts recognized in the consolidated balance sheets related to other post-retirement benefits plans were as follows:

	2023		2022	
	U.S. Plans	Non-U.S. Plans	U.S. Plans	Non-U.S. Plans
Amounts recognized in the consolidated balance sheet at December 31				
Current liabilities	$ (8)	$ (1)	$ (8)	$ (1)
Noncurrent liabilities	(33)	(2)	(38)	(2)
Net amount recognized	$ (41)	$ (3)	$ (46)	$ (3)

Estimated Future Benefit Payments

The following benefit payments are expected to be paid:

	Pension Benefits	Other Post-retirement Benefits
Estimated future benefit payments:		
Year 1	$ 68	$ 8
Year 2	67	6
Year 3	67	4
Year 4	66	3
Year 5	65	3
Years 6 to 10	329	15

15. Stock-Based Compensation

Under the Westlake Corporation 2013 Omnibus Incentive Plan (as amended and restated in 2023, the "2013 Plan"), all employees and non-employee directors of the Company, as well as certain individuals who have agreed to become the Company's employees, are eligible for awards. Shares of common stock may be issued as authorized in the 2013 Plan. At the discretion of the administrator of the 2013 Plan, employees and non-employee directors may be granted awards in the form of stock options, stock appreciation rights, stock awards, restricted stock units or cash awards (any of which may be a performance award). Outstanding stock option awards have a 10-year term and vest (1) ratably on an annual basis over a three-year period or (2) at the end of a five-year period. Outstanding restricted stock units and performance stock units vest either (1) ratably on an annual basis over a one to five-year period or (2) at the end of a three or six-year period. In accordance with accounting guidance related to share-based payments, stock-based compensation expense for all stock-based compensation awards is based on estimated grant-date fair value. The Company recognizes these stock-based compensation costs net of a forfeiture rate and on a straight-line basis over the requisite service period of the award for only those shares expected to vest. For the years ended December 31, 2023, 2022 and 2021, the total recognized stock-based compensation expense related to equity awards issued under the 2013 Plan was $40, $35 and $31, respectively.

Option activity and changes during the year ended December 31, 2023 were as follows:

	Options	Weighted Average Exercise Price	Weighted Average Remaining Term (Years)	Aggregate Intrinsic Value
Outstanding at December 31, 2022	1,531,070	$ 77.76		
Granted	197,251	122.65		
Exercised	(615,296)	73.31		
Cancelled	(39,136)	110.25		
Outstanding at December 31, 2023	1,073,889	$ 87.37	6.1	$ 56
Exercisable at December 31, 2023	683,794	$ 74.23	4.8	$ 45

For options outstanding at December 31, 2023, the options had the following range of exercise prices:

Range of Prices	Options Outstanding	Weighted Average Remaining Contractual Life (Years)
$44.42 - $61.87	169,789	2.6
$65.81 - $68.09	218,529	5.1
$79.83 - $86.54	262,357	6.3
$107.75 - $108.12	243,933	7.0
$122.65 - $122.65	179,281	9.1

The aggregate intrinsic value in the table above represents the total pretax intrinsic value (the difference between the Company's closing stock price on the last trading day of the year and the exercise price, multiplied by the number of in-the-money options) that would have been received by the option holders had all option holders exercised their options on December 31, 2023. This amount changes based on the fair market value of the Company's common stock. For the years ended December 31, 2023, 2022 and 2021, the total intrinsic value of options exercised was $34, $13 and $9, respectively.

As of December 31, 2023, $8 of total unrecognized compensation cost related to stock options is expected to be recognized over a weighted-average period of 1.8 years. Income tax benefits of $4, $2 and $2 were realized from the exercise of stock options during the years ended December 31, 2023, 2022 and 2021, respectively.

The Company used the Black-Scholes option pricing model to value its options. The table below presents the weighted average value and assumptions used in determining each option's fair value. Volatility was calculated using historical trends of the Company's common stock price.

	Stock Option Grants		
	Year Ended December 31,		
	2023	2022	2021
Weighted average fair value	$ 44.91	$ 34.20	$ 25.18
Risk-free interest rate	4.0 %	1.8 %	0.6 %
Expected life in years	5	5	5
Expected volatility	39.1 %	36.9 %	36.9 %
Expected dividend yield	1.1 %	1.1 %	1.2 %

Non-vested restricted stock units as of December 31, 2023 and changes during the year ended December 31, 2023 were as follows:

	Number of Units	Weighted Average Grant Date Fair Value
Non-vested at December 31, 2022	690,414	$ 85.77
Granted	247,131	120.77
Vested	(228,113)	69.45
Forfeited	(67,804)	97.87
Non-vested at December 31, 2023	641,628	$ 103.77

As of December 31, 2023, there was $31 of unrecognized stock-based compensation expense related to non-vested restricted stock units. This cost is expected to be recognized over a weighted-average period of 2.0 years. The total fair value of restricted stock units that vested during the years ended December 31, 2023, 2022 and 2021 was $28, $18 and $18, respectively.

Performance stock unit payout is based on the greater of the average annual economic-value added results for the Company (equal to net operating profit after tax less a capital charge based upon the weighted average cost of capital) and relative total shareholder return as compared to a peer group of companies. The units have payouts that range from zero to 200 percent of the target award.

Non-vested performance stock units as of December 31, 2023 and changes during the year ended December 31, 2023 were as follows:

	Number of Units	Weighted Average Grant Date Fair Value
Non-vested at December 31, 2022	250,369	$ 106.13
Granted	72,627	167.76
Vested	(106,705)	74.99
Forfeited	(19,370)	142.29
Non-vested at December 31, 2023	196,921	$ 142.18

As of December 31, 2023, there was $13 of unrecognized stock-based compensation expense related to non-vested performance stock units. This cost is expected to be recognized over a weighted-average period of 1.8 years. The total fair value of performance stock units that vested during the years ended December 31, 2023, 2022 and 2021 was $13, $8 and $0, respectively.

The Company used a Monte Carlo simulation model to value the performance stock units on the grant date.The table below presents the assumptions used in determining grant date fair value. Volatility was calculated using historical trends of the Company's common stock price.

	Performance Stock Units		
	Year Ended December 31,		
	2023	2022	2021
Risk-free interest rate	4.4 %	1.7 %	0.2 %
Expected life in years	2.87	2.87	2.87
Expected volatility of Westlake Corporation common stock	40.8 %	49.3 %	49.4 %
Expected volatility of peer companies	26.6% - 53.2%	30.8% - 67.2%	30.7% - 65.6%
Average correlation coefficient of peer companies	0.61	0.66	0.65
Grant date fair value	$ 167.76	$ 147.98	$ 109.94

Westlake Chemical Partners LP Awards

The Company's wholly-owned subsidiary and the general partner of Westlake Partners, Westlake Chemical Partners GP LLC ("Westlake Partners GP"), maintains a unit-based compensation plan for directors and employees of Westlake Partners GP and Westlake Partners.

The Westlake Partners 2014 Long-term Incentive Plan ("Westlake Partners 2014 Plan") permits various types of equity awards including but not limited to grants of phantom units and restricted units. Awards granted under the Westlake Partners 2014 Plan may be settled with Westlake Partners units or in cash or a combination thereof. Compensation expense for these awards was not material to the Company's consolidated financial statements for the years ended December 31, 2023, 2022 and 2021.

16. Fair Value Measurements

The Company has financial assets and liabilities subject to fair value measures. These financial assets and liabilities include cash and cash equivalents, accounts receivable, net, accounts payable and long-term debt, all of which are recorded at carrying value. The amounts reported in the consolidated balance sheets for cash and cash equivalents, accounts receivable, net and accounts payable approximate their fair value due to the short maturities of these instruments. The carrying and fair values of the Company's long-term debt at December 31, 2023 and 2022 are summarized in the table below. The Company's long-term debt instruments are publicly-traded. A market approach, based upon quotes from financial reporting services, is used to measure the fair value of the Company's long-term debt. Because the Company's long-term debt instruments may not be actively traded, the inputs used to measure the fair value of the Company's long-term debt are classified as Level 2 inputs within the fair value hierarchy.

	2023		2022	
	Carrying Value	Fair Value	Carrying Value	Fair Value
Long-term Debt	$ 4,906	$ 4,236	$ 4,879	$ 3,940

17. Income Taxes

The components of income before income taxes are as follows:

	Year Ended December 31,		
	2023	2022	2021
Domestic	$ 923	$ 2,523	$ 2,298
Foreign	(223)	423	379
	$ 700	$ 2,946	$ 2,677

The Company's provision for (benefit from) income taxes consist of the following:

	Year Ended December 31,		
	2023	2022	2021
Current			
Federal	$ 229	$ 473	$ 434
State	26	74	57
Foreign	98	123	93
Total current	353	670	584
Deferred			
Federal	(82)	(1)	19
State	(31)	(14)	13
Foreign	(62)	(6)	(9)
Total deferred	(175)	(21)	23
Total provision for (benefit from) income taxes	$ 178	$ 649	$ 607

A reconciliation of taxes computed at the statutory rate to the Company's income tax expense is as follows:

	Year Ended December 31,		
	2023	2022	2021
Provision for federal income tax, at statutory rate	$ 148	$ 619	$ 563
State income tax provision, net of federal income tax effect	7	59	56
Foreign income tax rate differential	(15)	30	22
Noncontrolling interests	(7)	(8)	(11)
Change in valuation allowance	76	—	(29)
U.S. federal research and development credits	(54)	(27)	(1)
Uncertain Income Tax Positions	14	6	2
Goodwill impairment	26	—	—
Other, net	(17)	(30)	5
Total income tax expense (benefit)	$ 178	$ 649	$ 607

The tax effects of the principal temporary differences between financial reporting and income tax reporting at December 31 are as follows:

	2023	2022
Net operating loss carryforward	$ 166	$ 103
Credit carryforward	20	19
Operating lease liabilities	177	149
Accruals	139	96
Pension	88	99
Inventories	34	34
Research and experimental expenditures	124	94
Other	36	18
Deferred taxes assets—total	784	612
Property, plant and equipment	(1,257)	(1,301)
Intangibles	(240)	(272)
Operating lease right-of-use asset	(168)	(149)
Turnaround costs	(61)	(49)
Consolidated partnerships	(213)	(234)
Equity method investments	(228)	(237)
Other	(47)	(42)
Deferred tax liabilities—total	(2,214)	(2,284)
Valuation allowance	(118)	(47)
Total net deferred tax liabilities	$ (1,548)	$ (1,719)
Balance sheet classifications		
Noncurrent deferred tax asset	$ 12	$ 16
Noncurrent deferred tax liability	(1,560)	(1,735)
Total net deferred tax liabilities	$ (1,548)	$ (1,719)

At December 31, 2023, the Company had federal, foreign and state net operating loss carryforwards ("NOLs") of approximately $15, $432 and $952, respectively. The federal NOL and certain foreign and state NOLs do not expire, while certain other foreign and state NOLs expire in varying amounts between 2024 and 2043. The federal NOL and certain state NOLs are subject to limitations on an annual basis. At December 31, 2023, the Company had various federal and state credit carryforwards of $2 and $18, respectively, which either do not expire or expire in varying amounts between 2028 and 2033. Management believes the Company will realize the benefit of a portion of the net operating loss and credit carryforwards before they expire, but to the extent that the full benefit may not be realized, a valuation allowance has been recorded. The valuation allowance increased by $71 primarily due to the impairment of long-lived assets of Westlake Epoxy's base resin business in the Netherlands, which generated deferred tax assets including net operating loss carryforwards that are not expected to be realized.

The Company has recognized a liability for uncertain income tax positions of $38 as of December 31, 2023. The Company does not believe it is likely that any material amounts will be paid in 2024. The ultimate resolution and timing of payment for remaining matters continues to be uncertain.

The Company files income tax returns in the U.S. federal jurisdiction, various states and foreign jurisdictions. The Company is no longer subject to examinations by tax authorities before the year 2014.

On October 8, 2021, the Organization for Economic Co-operation and Development (the "OECD")/G20 Inclusive Framework on Base Erosion and Profit Shifting released a statement indicating that its members had agreed to a Two – Pillar Solution to Address the Tax Challenges Arising from the Digitalisation of the Economy. Pillar One aims to reallocate a taxpayer's residual profits to the market jurisdictions with which the taxpayer has a nexus. Pillar Two aims to establish a minimum global tax rate of 15%, assessed through a top-up tax imposed on a country-by-country basis. Pillar One targets multinational companies with global annual revenue exceeding €20 billion and profit-to-revenue ratio of more than 10%. Based on the current threshold requirements, the Company does not expect to be subject to Pillar One. On December 20, 2021, the OECD released the Pillar Two model rules providing a framework for implementing a 15% minimum tax, also referred to as the Global Anti-Base Erosion ("GloBE") rules, on earnings of multinational companies with consolidated annual revenue exceeding €750 million.

On December 12, 2022, European Union (EU) member states agreed to adopt the 15% minimum tax under the Pillar Two model rules to be enacted into the member states' domestic tax laws by December 31, 2023, with an effective date beginning in 2024. As of the targeted enactment deadline, only a handful of EU member states did not comply. Outside of the EU, several other jurisdictions that the Company operates in have enacted legislation consistent with the GloBE rules, while other foreign countries continue to debate adoption and timing to adopt. The Company's global footprint includes operations within the EU, as well as other non-EU jurisdictions that have enacted GloBE related legislation, such as Japan, South Korea, Vietnam, the UK and Switzerland. At this time, the Company is evaluating what effect, if any, Pillar Two or GloBE will have on the Company's consolidated financial statements. The Company will continue to closely monitor Pillar Two developments and evaluate the potential impact to the Company as more foreign countries enact legislation, and as new information and guidance becomes available. The impacts of Pillar Two, if any, will begin to be recorded in 2024 as the rules become effective.

18. Earnings and Dividends per Share

The Company has unvested restricted stock units outstanding that are considered participating securities and, therefore, computes basic and diluted earnings per share under the two-class method. Basic earnings per share for the periods are based upon the weighted average number of shares of common stock outstanding during the periods. Diluted earnings per share include the effects of certain stock options and performance stock units.

	Year Ended December 31,		
	2023	2022	2021
Net income attributable to Westlake Corporation	$ 479	$ 2,247	$ 2,015
Less:			
Net income attributable to participating securities	3	12	10
Net income attributable to common shareholders	$ 476	$ 2,235	$ 2,005

The following table reconciles the denominator for the basic and diluted earnings per share computations shown in the consolidated statements of operations:

	Year Ended December 31,		
	2023	2022	2021
Weighted average common shares—basic	127,806,317	127,970,445	128,002,911
Plus incremental shares from:			
Assumed exercise of options and vesting of performance stock units	792,124	875,117	695,071
Weighted average common shares—diluted	128,598,441	128,845,562	128,697,982
Earnings per common share attributable to Westlake Corporation:			
Basic	$ 3.73	$ 17.46	$ 15.66
Diluted	$ 3.70	$ 17.34	$ 15.58

Excluded from the computation of diluted earnings per share for the years ended December 31, 2023, 2022 and 2021 are options to purchase 263,131, 315,864 and 461,618 shares of common stock, respectively. These options were outstanding during the periods reported but were excluded because the effect of including them would have been antidilutive.

Dividends per Share

Dividends per common share for the years ended December 31, 2023, 2022 and 2021 were as follows:

	Year Ended December 31,		
	2023	2022	2021
Dividends per common share	$ 1.7140	$ 1.3090	$ 1.1350

19. Supplemental Information

Other Assets, Net

Other assets, net were $651 and $617 at December 31, 2023 and 2022, respectively. Deferred turnaround costs, net of accumulated amortization, included in other assets, net were $391 and $359 at December 31, 2023 and 2022, respectively.

Accrued and Other Liabilities

Accrued and other liabilities were $1,614 and $1,409 at December 31, 2023 and 2022, respectively. Accrued rebates and accrued litigation reserves, which are components of accrued and other liabilities, were $217 and $297 at December 31, 2023 and $227 and $90 at December 31, 2022, respectively. No other component of accrued and other liabilities was more than five percent of total current liabilities. Accrued liabilities with related parties were $41 and $44 at December 31, 2023 and 2022, respectively.

Non-cash Investing Activity

Capital expenditure related liabilities, included in accounts payable and accrued and other liabilities, were $184, $171, and $156 at December 31, 2023, 2022, and 2021, respectively.

Restructuring, Transaction and Integration-related Costs

For the years ended December 31, 2023, 2022 and 2021, the restructuring, transaction and integration-related costs of $28, $33 and $21, respectively. The costs in 2023 are restructuring and integration costs primarily related to plant closures resulting from the Company's manufacturing footprint optimization efforts. The 2022 and 2021 expenses primarily consist of integration-related consulting fees, restructuring expenses and costs associated with the Company's acquisitions in previous years.

Other Income, Net

For the year ended December 31, 2023, other income, net included interest income, insurance recoveries and income from unconsolidated subsidiaries of $104, $28 and $19, respectively. For the year ended December 31, 2022, other income, net included interest income, income from pension and post-retirement plans and income from unconsolidated subsidiaries of $24, $26 and $15, respectively. For the year ended December 31, 2021, other income, net included interest income, income from pension and post-retirement plans and income from unconsolidated subsidiaries of $7, $23 and $13, respectively.

Operating Lease Supplemental Cash Flow

Supplemental cash flow information related to leases was as follows:

	Year Ended December 31,		
	2023	2022	2021
Operating cash flows from operating leases [1]	$ 139	$ 134	$ 114
Right-of-use assets obtained in exchange for operating lease obligations	240	187	215

(1) Includes cash paid for amounts included in the measurement of operating lease liabilities recorded in the consolidated balance sheets.

Cash Flow Information

	Year Ended December 31,		
	2023	2022	2021
Cash paid for:			
Interest paid, net of interest capitalized	$ 159	$ 172	$ 130
Income taxes paid	421	570	466

20. Related Party and Affiliate Transactions

The Company and Lotte have a joint venture, LACC, to operate an ethylene facility with2.2 billion pounds per year of ethylene production capacity. See Note 9 for details of the Company's transactions with LACC.

The Company leases office space for management and administrative services from an affiliate of the Company's principal stockholder. For each of the years ended December 31, 2023, 2022 and 2021, the Company incurred lease payments of approximately $3.

Cypress Interstate Pipeline L.L.C., a natural gas liquids pipeline joint venture in which the Company owns a50% equity interest, transports natural gas liquid feedstocks to the Company's Lake Charles complex through its pipeline. The Company accounts for its investments in Cypress Interstate Pipeline L.L.C. under the equity method of accounting. The investment in Cypress Interstate Pipeline L.L.C. at December 31, 2023 and 2022 was $8 and $7, respectively. For the years ended December 31, 2023, 2022 and 2021, the Company incurred pipeline lease service fees of approximately $20, $16 and $14, respectively, payable to this joint venture for usage of the pipeline. The amounts due to this joint venture were $1 at December 31, 2023 and 2022.

The Company owns an approximately20% equity interest in both YNCORIS GmbH & Co. KG (formerly known as InfraServ Knapsack GmbH & Co. KG) and InfraServ Gendorf GmbH & Co. KG (collectively "Infraserv"). The Company accounts for its investments in Infraserv under the equity method of accounting. The Company has service agreements with these entities, including contracts to provide electricity, technical and leasing services to certain of the Company's production facilities in Germany. The investment in Infraserv was $53 and $53 at December 31, 2023 and 2022, respectively. For the years ended December 31, 2023, 2022 and 2021, the Company incurred charges aggregating approximately $176, $188 and $174, respectively, for these services. The amounts accrued for these related parties were approximately34 and $37 at December 31, 2023 and 2022, respectively.

In conjunction with the Westlake Epoxy acquisition, the Company acquired49.99% equity interest in Westlake UV Coatings (Shanghai) Co., Ltd. (formerly known as Hexion UV Coatings Co., LTD). The investment in Westlake UV Coatings (Shanghai) Co., Ltd. was $7 at December 31, 2023. The Company accounts for its investments in Westlake UV Coatings (Shanghai) Co., Ltd. under the equity method of accounting.

Dividends received from equity method investments were $18, $16 and $15 for the years ended December 31, 2023, 2022 and 2021, respectively.

One of the Company's directors serves as Chairman and Chief Executive Officer of American Air Liquide, Inc. and Executive Vice President of the Air Liquide Group ("Air Liquide"). The Company purchased oxygen, nitrogen and utilities and leased cylinders from various affiliates of American Air Liquide, Inc. aggregating approximately $43, $43 and $39 for the years ended December 31, 2023, 2022 and 2021, respectively. The Company also sold certain utilities to Air Liquide aggregating approximately $14, $11 and $8 during the years ended December 31, 2023, 2022 and 2021, respectively. The amounts payable to Air Liquide were $3 and $4 at December 31, 2023 and 2022, respectively, and the amounts receivable from Air Liquide were $1 and $2 at December 31, 2023 and 2022, respectively.

21. Westlake Chemical Partners LP

In 2014, the Company formed Westlake Partners to operate, acquire and develop ethylene production facilities and related assets. Also in 2014, Westlake Partners completed its initial public offering of 12,937,500 common units. Most recently, on March 29, 2019, Westlake Partners purchased an additional4.5% newly issued limited partner interests in OpCo and completed a private placement of 2,940,818 common units. TTWF LP, the Company's principal stockholder and a related party, acquired 1,401,869 units out of the 2,940,818 common units issued in the private placement. At December 31, 2023, Westlake Partners had a22.8% limited partner interest in OpCo, and the Company retained a 77.2% limited partner interest in OpCo and a significant interest in Westlake Partners through the Company's ownership of Westlake Partners' general partner, 40.1% of the limited partner interests (consisting of 14,122,230 common units) and incentive distribution rights.

On October 4, 2018, Westlake Partners and Westlake Partners GP, the general partner of Westlake Partners, entered into an Equity Distribution Agreement with UBS Securities LLC, Barclays Capital Inc., Citigroup Global Markets Inc., Deutsche Bank Securities Inc., RBC Capital Markets, LLC, Merrill Lynch, Pierce, Fenner & Smith Incorporated and Wells Fargo Securities, LLC to offer and sell Westlake Partners' common units, from time to time, up to an aggregate offering amount of $50. This Equity Distribution Agreement was amended on February 28, 2020 to reference a new shelf registration for utilization under this agreement. No common units had been issued under this program as of December 31, 2023.

22. Commitments and Contingencies

The Company is involved in a number of legal and regulatory matters, principally environmental in nature, that are incidental to the normal conduct of its business, including lawsuits, investigations and claims. The outcome of these matters are inherently unpredictable. The Company believes that, in the aggregate, the outcome of all known legal and regulatory matters will not have a material adverse effect on its consolidated financial statements; however, under certain circumstances, if required to recognize costs in a specific period, when combined with other factors, outcomes with respect to such matters may be material to the Company's consolidated statements of operations in such period. The Company's assessment of the potential impact of environmental matters, in particular, is subject to uncertainty due to the complex, ongoing and evolving process of investigation and remediation of such environmental matters, and the potential for technological and regulatory developments. In addition, the impact of evolving claims and programs, such as natural resource damage claims, industrial site reuse initiatives and state remediation programs creates further uncertainty of the ultimate resolution of these matters. The Company anticipates that the resolution of many legal and regulatory matters, and in particular environmental matters, will occur over an extended period of time.

Caustic Soda Antitrust. The Company and other caustic soda producers were named as defendants in multiple purported class action civil lawsuits filed since March 2019 in the U.S. District Court for the Western District of New York. The lawsuits allege the defendants conspired to fix, raise, maintain and stabilize the price of caustic soda, restrict domestic (U.S.) supply of caustic soda and allocate caustic soda customers. The other defendants named in the lawsuits are Olin Corporation, K.A. Steel Chemicals (a wholly-owned subsidiary of Olin), Occidental Chemical Corporation d/b/a OxyChem, Shintech Incorporated and Formosa Plastics Corporation, U.S.A. Each of the lawsuits is filed on behalf of the respective named plaintiff or plaintiffs and a putative class comprised of either direct purchasers or indirect purchasers of caustic soda in the U.S. The plaintiffs in the putative class for such direct purchasers seek $861 in single damages from the defendants, in addition to treble damages and attorney's fees. The plaintiffs in the putative class for such indirect purchasers seek approximately $500 in single damages from the defendants, in addition to treble damages (if permitted under applicable state law) and injunctive relief. The defendants' joint motion to dismiss the direct purchaser lawsuits was denied. The defendants' joint motion to dismiss the indirect purchaser lawsuits was granted in part and denied in part. Both groups of cases have proceeded to discovery. On October 16, 2023, the Company entered into a settlement agreement whereby the Company agreed, subject to the terms and conditions of the settlement agreement, to pay $19 to the direct purchaser plaintiffs and in which the Company admits to no violations of law and will be released from liability and dismissed from the direct purchaser lawsuits with prejudice upon receipt of final court approval of the settlement. Preliminary court approval of the settlement, which remains subject to certification of the class, has been requested and is pending. If the court grants such preliminary approval, notice of the settlement will be disseminated to the settlement class, and the settlement class members will, for a limited time period, have an opportunity to opt-out of or object to the settlement. The Company reserved $19 in connection with the litigation in the third quarter of 2023. Beginning in October 2020, similar class action proceedings were also filed in Canada before the Superior Court of Québec as well as before the Federal Court. These proceedings seek the certification or authorization of a class action on behalf of all residents of Canada who purchased caustic soda (including, in one of the cases, those who merely purchased products containing caustic soda) from October 1, 2015 through the present or such date deemed appropriate by the court. On December 10, 2021, the Superior Court of Québec stayed its proceedings until after a final certification decision is released in the Federal Court proceedings. At this time, the Company is not able to estimate the impact, if any, that these lawsuits could have on the Company's consolidated financial statements either in the current period or in future periods.

Ethylene Antitrust. The Company and other ethylene consumers were named as defendants in a civil lawsuit filed by Shell Chemical Europe B.V. ("SCE") in March 2023 in the District Court of Amsterdam, the Netherlands, following the European Commission Decision AT.40410 – Ethylene from July 14, 2020. SCE is a producer of ethylene in the European market and the lawsuit alleges the defendants conspired to lower the purchase price for ethylene and ethylene derivatives by manipulating the monthly contract price. SCE initially sought declaratory relief. In October 2023, SCE amended its claim and is now seeking a judgment establishing that the Company and the co-defendants are jointly and severally liable for alleged damages in the amount of approximately €1,026 million with interest compounding daily from January 1, 2020 and continuing until payment in full, as provided by Dutch law, resulting from artificially lowered prices for ethylene and ethylene derivatives during the specified period. SCE is also seeking reimbursement of costs related to the proceeding plus statutory interest. The Company and other ethylene consumers were also named as defendants in a civil lawsuit filed by Stichting Ethylene Claims ("Stichting") in November 2023 in the District Court of Amsterdam, the Netherlands, following the same European Commission decision. Stichting is a foundation under Dutch Law that claims to represent various parties asserting injury by the same alleged conduct of defendants, seeking a declaratory judgment establishing that the Company and other defendants are jointly and severally liable for an unspecified amount of damages. At this time, the Company is not able to estimate the impact, if any, that the SCE lawsuit and/or the Stichting lawsuit could have on the Company's consolidated financial statements either in the current period or in future periods.

Triad Hunter. In April 2018, Triad Hunter, LLC ("Triad Hunter") filed suit against the Company and certain of its subsidiaries in the Court of Common Pleas in Monroe County, Ohio seeking injunctive relief and alleging negligence and trespass at the Natrium Plant with respect to Triad Hunter's well drilling activities in Ohio. On October 27, 2022, the jury returned a verdict finding that the Company had committed trespass and was negligent in conducting salt mining operations at the Natrium Plant. Upon receipt of the jury verdict, the Company reserved approximately $70 for the damages awarded to Triad Hunter. The court subsequently denied Triad Hunter's request to enjoin further solution mining from one of the brine fields at the Natrium Plant. On September 12, 2023, final judgment was entered in the amount of $70 with interest accruing at the rate of 5% from the date the judgment was rendered, as provided by law. The Company appealed the verdict and sought a stay of execution pending appeal on October 30, 2023.

Petro 2 Flash Fire and EDC Storage Tank Explosion. Petro 2 Flash Fire and EDC Storage Tank Explosion. In September 2021, shortly after the turnaround on Westlake Chemical OpCo LP's Petro 2 ethylene facility commenced, there was a flash fire at the quench tower of the Petro 2 facility. In January 2022, an ethylene dichloride storage tank exploded at one of the Company's Lake Charles, Louisiana complexes. Certain employees and contractors working on the sites at the time of the events were injured during these events and multiple lawsuits were filed against us. Final settlements totaling approximately $382 in the aggregate were reached with all of the plaintiffs to fully resolve the lawsuits, but payment by the Company and the insurance carriers has not yet been completed. Certain of the Company's excess insurers have taken the coverage position that there has been insufficient exhaustion of the retention levels beneath their respective attachment points to obligate them to cover these events. The practical effect of this coverage position is that approximately $150 of the $382 total final settlement amount is currently not being covered by the insurers and, as such, approximately $150 has been included in cost of sales in the fourth quarter of 2023. The Company disagrees with the insurers' coverage position, and is pursuing recovery of the approximately $150 and any other amount the Company may be entitled to. The Company has reserved for the full final settlement amount and recorded an insurance receivable for $43 and a legal accrual of $183 as of December 31, 2023.

Environmental Contingencies. As of December 31, 2023 and 2022, the Company had reserves for environmental contingencies totaling approximately $58 and $55, respectively, most of which was classified as noncurrent liabilities. The Company's assessment of the potential impact of these environmental contingencies is subject to considerable uncertainty due to the complex, ongoing and evolving process of investigation and remediation, if necessary, of such environmental contingencies, and the potential for technological and regulatory developments.

Calvert City Proceedings. For several years, the Environmental Protection Agency (the "EPA") has been conducting remedial investigation and feasibility studies at the Company's Calvert City, Kentucky facility pursuant to the Comprehensive Environmental Response, Compensation, and Liability Act of 1980 ("CERCLA"). As the current owner of the Calvert City facility, the Company was named by the EPA as a potentially responsible party ("PRP") along with Goodrich Corporation ("Goodrich") and its successor-in-interest, Avient Corporation (formerly known as PolyOne Corporation, "Avient"). On November 30, 2017, the EPA published a draft Proposed Plan, incorporating by reference an August 2015 draft Remedial Investigation ("RI") report, an October 2017 draft Feasibility Study ("FS") report and a Technical Impracticability Waiver document dated December 19, 2017. On June 18, 2018, the EPA published an amendment to its Proposed Plan. The amended Proposed Plan describes a final remedy for the onshore portion of the site comprised of a containment wall, targeted treatment and supplemental hydraulic containment. The amended Proposed Plan also describes an interim approach to address the contamination under the river that would include recovery of any mobile contaminants by an extraction well along with further study of the extent of the contamination and potential treatment options. The EPA's estimated cost of implementation is $107, with an estimated $1 to $3 in annual operation and maintenance ("O&M") costs. In September 2018, the EPA published the Record of Decision ("ROD") for the site, formally selecting the preferred final and interim remedies outlined in the amended Proposed Plan. In October 2018, the EPA issued Special Notice letters to the PRPs for the remedial design phase of work under the ROD. In April 2019, the PRPs and the EPA entered into an administrative settlement agreement and order on consent for remedial design. In October 2019, the PRPs received special notice letters for the remedial action phase of work at the site. The Company, jointly with the other PRPs, submitted a good faith offer response in December 2019. On September 17, 2020, the EPA and the Department of Justice filed a proposed consent decree for the remedial action with the U.S. District Court for the Western District of Kentucky. On November 16, 2020, the Department of Justice filed a motion to approve and enter the consent decree. On January 28, 2021, the Court granted the unopposed motion to enter the consent decree, which became effective the same day. The Company's allocation of liability for remedial and O&M costs at the Calvert City site, if any, is governed by a series of agreements between the Company, Goodrich and Avient. These agreements and the associated litigation are described below.

In connection with the 1990 and 1997 acquisitions of the Goodrich chemical manufacturing complex in Calvert City, Goodrich agreed to indemnify the Company for any liabilities related to preexisting contamination at the complex. For its part, the Company agreed to indemnify Goodrich for post-closing contamination caused by the Company's operations. The soil and groundwater at the complex, which does not include the Company's nearby PVC facility, had been extensively contaminated by Goodrich's operations. In 1993, Goodrich spun off the predecessor of Avient, and that predecessor assumed Goodrich's indemnification obligations relating to preexisting contamination. In 2003, litigation arose among the Company, Goodrich and Avient with respect to the allocation of the cost of remediating contamination at the site. The parties settled this litigation in December 2007 and the case was dismissed. In the settlement, the parties agreed that, among other things: (1) Avient would pay 100% of the costs (with specified exceptions), net of recoveries or credits from third parties, incurred with respect to environmental issues at the Calvert City site from August 1, 2007 forward; and (2) either the Company or Avient might, from time to time in the future (but not more than once every five years), institute an arbitration proceeding to adjust that percentage. In May 2017, Avient filed a demand for arbitration. In this proceeding, Avient sought to readjust the percentage allocation of future costs and to recover approximately $11 from the Company in reimbursement of previously paid remediation costs. The Company's cross demand for arbitration seeking unreimbursed remediation costs incurred during the relevant period was dismissed from the proceedings when Avient paid such costs in full at the beginning of the arbitration hearing.

On July 10, 2018, Avient sued the Company in the U.S. District Court for the Western District of Kentucky and sought to invalidate the arbitration provisions in the parties' 2007 settlement agreement and enjoin the arbitration it had initiated in 2017. On July 30, 2018, the district court refused to enjoin the arbitration and, on January 15, 2019, the court granted the Company's motion to dismiss Avient's suit. On February 13, 2019, Avient appealed those decisions to the U.S. Court of Appeals for the Sixth Circuit. The court of appeals issued an opinion and final order on September 6, 2019, affirming the district court.

The arbitration hearing began in August 2018 and concluded in December 2018. On May 22, 2019, the arbitration panel issued its final award. It determined that Avient was responsible for 100% of the allocable costs at issue in the proceeding and that Avient would remain responsible for100% of the costs to operate the existing groundwater remedy at the Calvert City site. In August 2019, Avient filed a motion to vacate before the U.S. District Court for the Western District of Kentucky, seeking to invalidate the final award under the Federal Arbitration Act. On February 11, 2020, the U.S. District Court for the Western District of Kentucky denied Avient's motion to vacate and affirmed the arbitration final award. Avient did not file a notice of appeal before the March 10, 2020 deadline to contest the court's decision. Accordingly, the final award was affirmed, and the arbitration proceeding is fully and finally resolved.

In March 2022, the Company filed a demand for arbitration seeking reimbursement for certain allocable costs incurred during the applicable period since May 2017, and which Avient has failed to pay or disputed as not subject to indemnity under the 1990 and 1997 agreements. In April 2022, Avient filed a complaint in the federal district court for the Western District of Kentucky disputing the enforceability of the 2007 settlement agreement and seeking to enjoin arbitration. Avient claims that the allocable costs at issue are up to $22, for which Avient claims the Company is totally liable. On September 30, 2023, the court issued an order denying without prejudice the parties' cross motions for summary judgment and set a 30-day deadline for the parties to refile and provide additional briefing on specific aspects of the arguments before the court. The parties each refiled and provided additional briefing by the deadline and the motions are pending before the court. The Company disputes these claims and at this time, the Company believes it is unlikely that any remediation costs allocable to it would result in material expenditures in any individual reporting period.

Sulphur Brine Dome. Westlake owns and operates salt solution-mining caverns at the Sulphur Brine Dome in Sulphur, Louisiana. The Louisiana Department of Natural Resources issued Compliance Order No. IMD 2022-027 and several supplements to that order, the latest in October 2023, in response to pressure anomaly events in two of the Company's brine caverns. The brine caverns were not active, operating wells but under ongoing, post-operational monitoring requirements. The compliance order and supplements thereto have required us to undertake various activities related to response planning, monitoring, investigation and mitigation of potential impacts in the event of future cavern pressure anomalies or failures. Since December 2022 continuous brine injection has maintained cavern pressure while the Company continues to pursue active monitoring, studies, groundwater monitoring, modeling and other activities under the compliance order and supplements. In September 2023, the Office of Conservation issued an emergency declaration as a conservative step and to ensure that the full suite of powers and resources are available to the government in its response and management of the evolving circumstances at the Sulphur Dome. The capital costs and expenditures required to comply with the compliance order and supplements have been and will continue to be incurred. In response to the most recent supplement to the compliance order, the Company reserved approximately $13 in connection with groundwater monitoring wells and monitoring required by the supplement. At this time, the Company is unable to estimate the impact, if any, that other ongoing expenditures or future injunctive relief ordered by the government could have on the Company's consolidated financial statements either in the current period or in future periods.

Environmental Remediation: Reasonably Possible Matters. The Company's assessment of the potential impact of environmental contingencies is subject to considerable uncertainty due to the complex, ongoing and evolving process of investigation and remediation, if necessary, of such environmental contingencies, and the potential for technological and regulatory developments. As such, in addition to the amounts currently reserved, the Company may be subject to reasonably possible loss contingencies related to environmental matters in the range of $98 to $165.

Other Commitments

The Company has various unconditional purchase obligations, primarily to purchase goods and services, including commitments to purchase various utilities, feedstock, nitrogen, oxygen, product storage and pipeline usage. At December 31, 2023, unrecorded unconditional total purchase obligations were $6,345, which included approximately $983 in 2024, $1,016 in 2025, $936 in 2026, $825 in 2027, $628 in 2028, and $1,957 thereafter.

23. Segment and Geographic Information

Segment Information

The Company has two principal operating segments, Performance and Essential Materials and Housing and Infrastructure Products. These segments are strategic business units that offer a variety of different materials and products. The Company manages each segment separately as each business requires different technology and marketing strategies.

The Company's Performance and Essential Materials segment manufactures and markets polyethylene, styrene monomer, ethylene co-products, PVC, VCM, ethylene dichloride ("EDC"), chlor-alkali (chlorine and caustic soda), chlorinated derivative products and epoxy resins. The Company's ethylene production is used in the Company's polyethylene, styrene and VCM operations. In addition, the Company sells ethylene and ethylene co-products, primarily propylene, crude butadiene, pyrolysis gasoline and hydrogen, to external customers. The Company's primary North American manufacturing facilities are located in its Calvert City, Kentucky; Lake Charles, Plaquemine and Geismar, Louisiana; Longview and Deer Park, Texas; Lakeland, Florida and Argo, Illinois sites. The Company's primary European facilities are located in Germany and the Netherlands. The Company produces ethylene and polyethylene at its facilities in Lake Charles, Louisiana; Calvert City, Kentucky and Longview, Texas. The Company produces chlorine, caustic soda, VCM, EDC, PVC, hydrogen and chlorinated derivative materials at its facilities in Lake Charles, Plaquemine and Geismar, Louisiana; Calvert City, Kentucky; Natrium, West Virginia; Longview, Washington; Beauharnois, Quebec; Aberdeen, Mississippi and in Germany. Epoxy resins primarily comprise of Epoxy Specialty Resins and Base Epoxy Resins and Intermediaries. Epoxy Specialty Resins are produced at manufacturing facilities in Duisburg and Esslingen in Germany; Argo and Lakeland in the United States; one plant in Spain and one plant in South Korea. Base Epoxy Resins and Intermediaries are produced at Company's facilities plants in Pernis, the Netherlands and Deer Park, United States. In addition, the Company also has other manufacturing facilities and product development facilities in North America, Europe and Asia.

No single customer accounted for more than 10% of sales in the Performance and Essential Materials segment for the years ended December 31, 2023, 2022 or 2021.

The Housing and Infrastructure Products segment manufactures and markets products including residential siding, trim and mouldings, stone, roofing, windows, outdoor living products, PVC pipe and fittings and PVC compounds. As of December 31, 2023, the Company owned or leased 70 manufacturing facilities in North America, Europe and Asia. The Company's North American PVC facilities within the Performance and Essential Materials segment supply most of the PVC required for the building products and pipes and fittings plants. The raw materials for stone, roofing and accessories, windows, shutters and specialty tool products are externally purchased. PVC required for the PVC compounds plants is either internally sourced from Company's North American or Asian facilities or externally purchased at market prices based on the location of the plants.

No single customer accounted for more than 10% of sales in the Housing and Infrastructure Products segment for the years ended December 31, 2023, 2022 or 2021.

The accounting policies of the individual segments are the same as those described in Note 1.

WESTLAKE CORPORATION

NOTES TO CONSOLIDATED FINANCIAL STATEMENTS—(Continued)
(in millions of dollars, except share amounts and per share data)

	Year Ended December 31,		
	2023	2022	2021
Net external sales			
Performance and Essential Materials			
Performance Materials	$ 4,656	$ 6,964	$ 5,997
Essential Materials	3,680	4,044	2,673
Total Performance and Essential Materials	8,336	11,008	8,670
Housing and Infrastructure Products			
Housing Products	3,494	3,864	2,334
Infrastructure Products	718	922	774
Total Housing and Infrastructure Products	4,212	4,786	3,108
	$ 12,548	$ 15,794	$ 11,778
Intersegment sales			
Performance and Essential Materials	$ 408	$ 908	$ 798
Housing and Infrastructure Products	—	—	—
	$ 408	$ 908	$ 798
Income (loss) from operations			
Performance and Essential Materials	$ 59	$ 2,416	$ 2,549
Housing and Infrastructure Products	710	675	356
Corporate and other	(40)	(41)	(105)
	$ 729	$ 3,050	$ 2,800
Depreciation and amortization			
Performance and Essential Materials	$ 881	$ 784	$ 665
Housing and Infrastructure Products	207	263	168
Corporate and other	9	9	7
	$ 1,097	$ 1,056	$ 840
Other income, net			
Performance and Essential Materials	$ 25	$ 37	$ 33
Housing and Infrastructure Products	32	17	10
Corporate and other	79	19	10
	$ 136	$ 73	$ 53
Provision for (benefit from) income taxes			
Performance and Essential Materials	$ (63)	$ 435	$ 542
Housing and Infrastructure Products	194	185	80
Corporate and other	47	29	(15)
	$ 178	$ 649	$ 607
Capital expenditures			
Performance and Essential Materials	$ 857	$ 913	$ 567
Housing and Infrastructure Products	164	187	88
Corporate and other	13	8	3
	$ 1,034	$ 1,108	$ 658

A reconciliation of total segment income from operations to consolidated income before income taxes is as follows:

	Year Ended December 31,		
	2023	2022	2021
Income from operations for reportable segments	$ 729	$ 3,050	$ 2,800
Interest expense	(165)	(177)	(176)
Other income, net	136	73	53
Income before income taxes	$ 700	$ 2,946	$ 2,677

	December 31, 2023	December 31, 2022
Total assets		
Performance and Essential Materials	$ 13,538	$ 13,978
Housing and Infrastructure Products	4,888	5,022
Corporate and other	2,609	1,550
	$ 21,035	$ 20,550

Geographic Information

	Year Ended December 31,		
	2023	2022	2021
Net sales to external customers[1]			
United States	$ 8,955	$ 10,899	$ 8,157
Foreign			
Canada	808	1,051	980
Germany	602	875	628
China	220	262	216
Mexico	217	228	149
Brazil	215	353	141
France	138	172	100
Italy	135	200	181
Other	1,258	1,754	1,226
	$ 12,548	$ 15,794	$ 11,778

	December 31, 2023	December 31, 2022
Property, plant and equipment, net		
United States	$ 7,395	$ 7,155
Foreign		
Germany	790	749
Other	334	573
	$ 8,519	$ 8,477

(1) Net sales are attributed to countries based on location of customer.

Item 9. Changes in and Disagreements with Accountants on Accounting and Financial Disclosure

None.

Item 9A. Controls and Procedures

Disclosure, Controls and Procedures

We carried out an evaluation, under the supervision and with the participation of our management, including our President and Chief Executive Officer (our principal executive officer) and our Executive Vice President and Chief Financial Officer (our principal financial officer), of the effectiveness of our disclosure controls and procedures pursuant to Rules 13a-15 and 15d-15 under the Securities Exchange Act of 1934 as of the end of the period covered by this Form 10-K. Based upon that evaluation, our President and Chief Executive Officer and our Executive Vice President and Chief Financial Officer concluded that our disclosure controls and procedures are effective as of December 31, 2023 to provide reasonable assurance that information required to be disclosed by us in the reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC's rules and forms, and is accumulated and communicated to management as appropriate to allow timely decisions regarding required disclosure.

Internal Control Over Financial Reporting

Westlake's management's report on internal control over financial reporting appears on page 57 of this Annual Report on Form 10-K. In addition, PricewaterhouseCoopers LLP, the independent registered public accounting firm that audited the financial statements included in this Annual Report on Form 10-K, has also audited the effectiveness of internal control over financial reporting as of December 31, 2023, as stated in their report that appears on page 58 of this Annual Report on Form 10-K.

Changes in Internal Control Over Financial Reporting

There were no changes in our internal control over financial reporting that occurred during the quarter ended December 31, 2023 that have materially affected, or are reasonably likely to materially affect, our internal control over financial reporting.

Item 9B. Other Information

Rule 10b5-1 Trading Arrangements. During the three months ended December 31, 2023, no director or officer of the Companyadopted or terminated a "Rule 10b5-1 trading arrangement" or "non-Rule 10b5-1 trading arrangement," as each term is defined in Item 408 of Regulation S-K.

Item 9C. Disclosure Regarding Foreign Jurisdictions that Prevent Inspections

Not applicable.

PART III

Except as noted below, the information required by Items 10, 11, 12, 13 and 14 is incorporated by reference to the Proxy Statement, which will be filed with the SEC pursuant to Regulation 14A under the Exchange Act within 120 days of December 31, 2023.

Item 10. Directors, Executive Officers and Corporate Governance.

The information required by Item 401(b) of Regulation S-K with respect to our executive officers is set forth in Part I of this Form 10-K.

Item 11. Executive Compensation.

Item 12. Security Ownership of Certain Beneficial Owners and Management and Related Stockholder Matters.

Equity Compensation Plan Information

Securities authorized for issuance under equity compensation plans are as follows:

Plan Category	Number of securities to be issued upon exercise of outstanding options, warrants and rights (a)	Weighted-average exercise price of outstanding options, warrants and rights (b)	Number of securities remaining available for future issuance under equity compensation plans (excluding securities reflected in column (a)) (c)
Equity compensation plans approved by security holders	1,912,438 (1)	$ 87.37 (2)	4,403,389
Equity compensation plans not approved by security holders	N/A	N/A	N/A
Total	1,912,438	$ 87.37 (2)	4,403,389

(1) Includes shares reserved for issuance pursuant to restricted stock units, stock options and performance stock units.

(2) Price applies only to the stock options included in column (a). Exercise price is not applicable to the other awards included in column (a).

Other information regarding our equity compensation plans is set forth in the section entitled "Executive Compensation" in our Proxy Statement, which information is incorporated herein by reference.

Item 13. Certain Relationships and Related Transactions, and Director Independence.

Item 14. Principal Accountant Fees and Services.

PART IV

Item 15. Exhibits and Financial Statement Schedules

(a)(1) The financial statements listed in the Index to Consolidated Financial Statements in Item 8 of this Form 10-K are filed as part of this Form 10-K.

(a)(2) All schedules are omitted because the information is not applicable, not required, or has been furnished in the Consolidated Financial Statements or Notes thereto in Item 8 of this Form 10-K.

(a)(3) Exhibits

Exhibit No.	Exhibit Index
3.1†	Restated Certificate of Incorporation of Westlake as filed with the Delaware Secretary of State onFebruary 20, 2024.
3.2	Amended and Restated Bylaws of Westlake (incorporated by reference to Westlake's Current Report on Form 8-K, filed on February 18, 2022, File No. 001-32260).
4.1†	Description of Securities Registered Pursuant to Section 12 of the Securities Exchange Act of 1934
4.2	Indenture dated as of January 1, 2006 by and among Westlake, the potential subsidiary guarantors listed therein and JPMorgan Chase Bank, National Association, as trustee (incorporated by reference to Westlake's Current Report on Form 8-K, filed on January 13, 2006, File No. 1-32260).
4.3	Seventh Supplemental Indenture, dated as of February 12, 2013, among the Company, the Subsidiary Guarantors (as defined therein) and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.16 to Westlake's Annual Report on Form 10-K for the year ended December 31, 2012, filed on February 22, 2013, File No. 1-32260).
4.4	Eighth Supplemental Indenture (including the form of the Notes), dated as of August 10, 2016, among Westlake Corporation, the Guarantors (as defined therein) and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.2 to Westlake's Current Report on Form 8-K, filed on August 10, 2016, File No. 001-32260).
4.5	Tenth Supplemental Indenture (including the form of the Notes), dated as of November 29, 2017, among Westlake Corporation, the Subsidiary Guarantors (as defined therein) and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.2 to Westlake's Current Report on Form 8-K, filed on November 28, 2017, File No. 001-32260).
4.6	Eleventh Supplemental Indenture (including the form of the Notes), dated as of November 28, 2017, among Westlake Corporation, the Subsidiary Guarantors (as defined therein) and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.3 to Westlake's Current Report on Form 8-K, filed on November 28, 2017, File No. 001-32260).
4.7	Twelfth Supplemental Indenture (including the form of the Notes), dated as of July 17, 2019, between Westlake Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.2 to Westlake's Current Report on Form 8-K filed on July 17, 2019, File No. 1-32260).
4.8	Thirteenth Supplemental Indenture (including the form of Notes), dated as of June 12, 2020, between Westlake Corporation and The Bank of New Mellon Trust Company, N.A., as trustee (incorporated by reference to Westlake's Current Report on Form 8-K filed on June 12, 2020, File No. 1-32260).
4.9	Fourteenth Supplemental Indenture (including the Notes), dated as of August 19, 2021, between Westlake Corporation and The Bank of New York Mellon Trust Company, N.A., as trustee (incorporated by reference to Exhibit 4.2 to Westlake's Current Report on Form 8-K, filed on August 19, 2021, File No. 001-32260).
4.10	Paying Agency Agreement dated as of July 17, 2019, between Westlake Corporation and The Bank of New York Mellon, London Branch, as paying agent (incorporated by reference to Exhibit 4.4 to Westlake's Current Report on Form 8-K, filed on July 17, 2019, File No. 1-32260).
	Westlake and its subsidiaries are party to other long-term debt instruments not filed herewith under which the total amount of securities authorized does not exceed 10% of the total assets of Westlake and its subsidiaries on a consolidated basis. Pursuant to paragraph 4(iii)(A) of Item 601(b) of Regulation S-K, Westlake agrees to furnish a copy of such instruments to the SEC upon request.

Exhibit No.	Exhibit Index
10.1	Amended and Restated Loan Agreement, dated as of November 1, 2017, by and between the Louisiana Local Government Environmental Facilities and Community Development Authority and Westlake Corporation (incorporated by reference to Exhibit 4.6 to Westlake's Current Report on Form 8-K, filed on November 28, 2017, File No. 001-32260).
10.2	Amended and Restated Senior Unsecured Revolving Credit Agreement by and among Westlake Chemical OpCo LP, Westlake Polymers LLC and the lenders party thereto, dated as of June 1, 2017 (incorporated by reference to Exhibit 10.9 to Westlake Chemical Partners LP's Annual Report on Form 10-K for the year ended December 31, 2019, filed on February 28, 2020, File No. 001-36567).
10.3	First Amendment to Amended and Restated Senior Unsecured Revolving Credit Agreement (incorporated by reference to Exhibit 10.1 to Westlake Chemical Partners LP's Quarterly Report on Form 10-Q for the quarter ended September 30, 2018, filed on November 6, 2018, File No. 001-36567)
10.4	Second Amendment to Amended and Restated Senior Unsecured Revolving Credit Agreement by and among Westlake Chemical OpCo LP, Westlake Polymers LLC, and the lenders party thereto, dated as of July 12, 2022 (incorporated by reference to Exhibit 10.1 to Westlake Chemical Partners LP's Current Report on Form 8-K, filed on July 15, 2022, File No. 001-36567).
10.5	Senior Unsecured Revolving Credit Agreement by and among Westlake Chemical Partners LP and Westlake Chemical Finance Corporation, dated as of April 29, 2015 (incorporated by reference to Exhibit 10.1 to Westlake Chemical Partners LP's Current Report on Form 8-K filed on April 30, 2015, File No. 001-36567).
10.6	First Amendment to Senior Unsecured Revolving Credit Agreement by and between Westlake Chemical Partners LP, as borrower, and Westlake Chemical Finance Corporation, as lender, dated as of August 1, 2017 (incorporated by reference to Exhibit 10.1 to Westlake Chemical Partners LP's Quarterly Report on Form 10-Q for the quarter ended June 30, 2017, File No. 001-36567).
10.7	Second Amendment to Senior Unsecured Revolving Credit Agreement by and between Westlake Chemical Partners LP, as borrower, and Westlake Chemical Finance Corporation, as lender, dated as of November 28, 2017 (incorporated herein by reference to Exhibit 10.12 to Westlake's Annual Report on Form 10-K for the year ended December 31, 2017, filed on February 21, 2018, File No. 001-32260).
10.8	Third Amendment to Senior Unsecured Revolving Credit Agreement by and between Westlake Chemical Partners LP, as borrower, and Westlake Chemical Finance Corporation, as lender, dated as of March 19, 2020 (incorporated herein by reference to Exhibit 10.1 to Westlake Chemical Partners LP's Quarterly Report on Form 10-Q for the quarter ended March 31, 2020, filed on May 6, 2020, File No. 001-36567).
10.9	Fourth Amendment to Senior Unsecured Revolving Credit Agreement by and between Westlake Chemical Partners LP, as borrower, and Westlake Chemical Finance Corporation, as lender, dated as of July 12, 2022 (incorporated herein by reference to Exhibit 10.2 to Westlake Chemical Partners LP's Current Report on Form 8-K, filed on July 15, 2022, File No. 001-36567).
10.10	Credit Agreement dated as of June 9, 2022, by and among Westlake Corporation, the lenders from time to time party thereto, the issuing banks party thereto and JPMorgan Chase Bank, National Association, as Administrative Agent, relating to a $1.5 billion senior unsecured revolving credit facility (incorporated by reference to Exhibit 10.1 to Westlake's Current Report on Form 8-K, filed on June 09, 2022, File No. 001-32260).
10.11	Form of Registration Rights Agreement between Westlake and TTWF LP (incorporated by reference to Westlake's Registration Statement on Form S-1/A, filed on July 2, 2004).
10.12+	Westlake Corporation 2013 Omnibus Incentive Plan (as amended and restated as of May 19, 2017) (incorporated by reference to Appendix B to Westlake's Definitive Proxy Statement on Schedule 14A filed on April 7, 2017, File No.1-32260).
10.13+	Westlake Corporation 2013 Omnibus Incentive Plan (as amended and restated effective May 11, 2023) (incorporated by reference to Appendix B to Westlake's Definitive Proxy Statement on Schedule 14A filed on March 31, 2023, File No.1-32260).
10.14+	Westlake Corporation Amended and Restated Annual Incentive Plan as of May 11, 2023 (incorporated by reference to Westlake's Quarterly Report on Form 10-Q for the quarter ended June 30, 2023, filed on August 3, 2023, File No. 001-32260).
10.15	Investment Management Agreement among Westlake Corporation, Westlake Chemical OpCo LP and Westlake Chemical Partners LP, dated as of August 1, 2017 (incorporated herein by reference to Exhibit 10.1 to Westlake's Quarterly Report on Form 10-Q for the quarter ended September 30, 2017, filed on November 7, 2017, File No. 001-32260).

Exhibit No.	Exhibit Index
10.16+	Form of Restricted Stock Unit Award Letter for 2021 Executive Officer Awards (incorporated by reference to Exhibit 10.20 to Westlake's Annual Report on Form 10-K for the year ended December 31, 2020 filed on February 24, 2021, File No. 001-32260).
10.17+	Form of Performance Stock Unit Award Letter for 2021 Executive Officer Awards (incorporated by reference to Exhibit 10.21 to Westlake's Annual Report on Form 10-K for the year ended December 31, 2020 filed on February 24, 2021, File No. 001-32260).
10.18+	Form of Restricted Stock Unit Award Letter for 2023 Executive Officer Awards (incorporated by reference to Exhibit 10.3 to Westlake's Quarterly Report on Form 10-Q for the quarter ended June 30, 2023 filed on August 3, 2023, File No. 001-32260).
10.19+†	Form of Stock Option Award Letter for 2024 Executive Officer Awards.
10.20+†	Form of Performance Stock Unit Award Letter for 2024 Executive Officer Awards.
21†	Subsidiaries of Westlake.
23.1†	Consent of PricewaterhouseCoopers LLP.
31.1†	Rule 13a-14(a) / 15d-14(a) Certification (Principal Executive Officer).
31.2†	Rule 13a-14(a) / 15d-14(a) Certification (Principal Financial Officer).
32.1††	Section 1350 Certification (Principal Executive Officer and Principal Financial Officer).
97†	Westlake Corporation Policy for Recovery of Erroneously Awarded Compensation, effective as of October 2, 2023
101.INS†	XBRL Instance Document-The instance document does not appear in the interactive data file because its XBRL tags are embedded within the Inline XBRL document.
101.SCH†	XBRL Taxonomy Extension Schema Document.
101.CAL†	XBRL Taxonomy Extension Calculation Linkbase Document.
101.DEF†	XBRL Taxonomy Extension Definition Linkbase Document.
101.LAB†	XBRL Taxonomy Extension Label Linkbase Document.
101.PRE†	XBRL Taxonomy Extension Presentation Linkbase Document.
104	Cover Page Interactive Data File-The cover page interactive data file does not appear in the Interactive Data File because its XBRL tags are embedded within the Inline XBRL document and contained in Exhibit 101.

† Filed herewith.
†† Furnished herewith.
+ Management contract, compensatory plan or arrangement.
* On February 18, 2022, Westlake Chemical Corporation changed its corporate name to Westlake Corporation. Accordingly, filings made prior to such date were made under the name Westlake Chemical Corporation.

Item 16. Form 10-K Summary.

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) of the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

WESTLAKE CORPORATION

Date: February 21, 2024

/S/ ALBERT CHAO

Albert Chao, President and Chief Executive Officer

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/S/ ALBERT CHAO **Albert Chao**	President, Chief Executive Officer and Director (Principal Executive Officer)	February 21, 2024
/S/ M. STEVEN BENDER **M. Steven Bender**	Executive Vice President and Chief Financial Officer (Principal Financial Officer)	February 21, 2024
/S/ JOHNATHAN S. ZOELLER **Johnathan S. Zoeller**	Vice President and Chief Accounting Officer (Principal Accounting Officer)	February 21, 2024
/S/ JAMES CHAO **James Chao**	Chairman of the Board of Directors	February 21, 2024
/S/ CATHERINE T. CHAO **Catherine T. Chao**	Director	February 21, 2024
/S/ DAVID T. CHAO **David T. Chao**	Director	February 21, 2024
/S/ JOHN T. CHAO **John T. Chao**	Director	February 21, 2024
/S/ MICHAEL J. GRAFF **Michael J. Graff**	Director	February 21, 2024
/S/ MARIUS A. HAAS **Marius A. Haas**	Director	February 21, 2024
/S/ KIMBERLY S. LUBEL **Kimberly S. Lubel**	Director	February 21, 2024
/S/ MARK A. MCCOLLUM **Mark A. McCollum**	Director	February 21, 2024
/S/ R. BRUCE NORTHCUTT **R. Bruce Northcutt**	Director	February 21, 2024
/S/ CAROLYN C. SABAT **Carolyn C. Sabat**	Director	February 21, 2024
/S/ JEFFREY W. SHEETS **Jeffrey W. Sheets**	Director	February 21, 2024